

Chapter 1

The Lowry Method for Protein Quantitation

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Introduction

The most accurate method of determining protein concentration is probably acid hydrolysis followed by amino acid analysis. Most other methods are sensitive to the amino acid composition of the protein and absolute concentrations cannot be obtained. The procedure of Lowry et al. (1) is no exception, but its sensitivity is moderately constant from protein to protein, and it has been so widely used that Lowry protein estimations are a completely acceptable alternative to a rigorous absolute determination in almost all circumstances where protein mixtures or crude extracts are involved.

Materials

1. *Complex-forming reagent*: prepare immediately before use by mixing the following 3 stock solutions A, B, and C in the proportion 100:1:1, respectively.

Solution A: 2% (w/v) Na_2CO_3 in distilled water

Solution B: 1% (w/v) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ in distilled water

Solution C: 2% (w/v) sodium potassium tartrate in distilled water

2. *2N NaOH*
3. *Folin reagent* (commercially available): Use at 1N concentration.
4. *Standards*. Use a stock solution of standard protein (e.g., bovine serum albumin fraction V) containing 4 mg/mL protein in distilled water stored frozen at -20°C . Prepare standards by diluting the stock solution with distilled water as follows:

Stock solution	μL	0	1	25	250	625	1250	2500	6250	12500	25000	50000
Water	μL	500	499	498	494	488	475	438	375	250	125	0
Protein concentration	$\mu\text{g/mL}$	0	10	20	50	100	200	500	1000	2000		

Method

1. To 0.1 mL of sample or standard, add 0.1 mL of 2N NaOH. Hydrolyze at 100°C for 10 min in a heating block or a boiling water bath.
2. Cool the hydrolyzate to room temperature and add 1 mL of freshly mixed complex-forming reagent. Let the solution stand at room temperature for 10 min.
3. Add 0.1 mL of Folin reagent, using a Vortex mixer, and let the mixture stand at room temperature for 30–60 min (do not exceed 60 min).
4. Read the absorbance at 750 nm if the protein concentration was below $500 \mu\text{g/mL}$ or at 550 nm if the protein concentration was between 100 and $2000 \mu\text{g/mL}$.
5. Plot a standard curve of absorbance as a function of initial protein concentration and use it to determine the unknown protein concentrations.

Notes

1. If the sample is available as a precipitate, then dissolve the precipitate in 2N NaOH and hydrolyze as in step 1. Carry 0.2 mL aliquots of the hydrolyzate forward to step 2.
2. Whole cells or other complex samples may need pretreatment, as described for the Burton assay for DNA (See Vol. 2). For example, the PCA/ethanol precipitate from extraction I may be used directly for the Lowry assay or the pellets remaining after the PCA hydrolysis (step 3 of the Burton assay) may be used for Lowry. In this latter case, both DNA and protein concentrations may be obtained from the same sample.
3. Rapid mixing as the Folin reagent is added is important for reproducibility.
4. A set of standards is needed with each group of assays, preferably in duplicate. Duplicate or triplicate unknowns are recommended.

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Chapter 2

Determination of Protein Molecular Weights by Gel Permeation High Pressure Liquid Chromatography

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Introduction

An essential part of the characterization of any protein is the determination of its molecular weight. The method of choice for these determinations, because of its simplicity and rapidity, has most frequently been sodium dodecyl sulfate (SDS) gel electrophoresis (*See* Chapter 6). The alternative method of gel filtration under denaturing conditions (1,2) has not been so widely used, in part as a result of the longer times required for a single run. How-

ever, recent developments in gel filtration supports for high pressure liquid chromatography (HPLC) have made more rapid separations possible, and thus gel permeation HPLC is becoming a widely used technique for molecular weight determinations (3,4). Gel permeation HPLC, in addition to taking less time than SDS gel electrophoresis, allows easier quantitation and recovery of separated proteins, and the resolution is better than that achieved by gel filtration with conventional materials.

The volume accessible to a protein in gel filtration supports depends on both its size and shape. Thus in order to determine molecular weight the sample protein must have the same shape as the proteins used for calibration. In the presence of denaturants, such as 6M guanidine hydrochloride, all proteins in their reduced state adopt a linear random coil conformation whose molecular radius is proportional to molecular weight (5). Under these conditions the molecular weight of a protein can be expressed in terms of its elution volume from the column (1-3).

Denaturation causes an increase in the intrinsic viscosity of the protein, and hence an increase in its molecular dimensions. Thus, under denaturing conditions the molecular weight exclusion limits of gel filtration matrices are lower than those in the absence of denaturant. Of the column supports used for gel permeation HPLC under denaturing conditions those of the TSK-G3000 SW type are suitable for proteins of less than 70,000 molecular weight (mw), whereas the TSK-G4000 SW type can be used for proteins up to 160,000 mw.

Gel permeation HPLC in guanidine hydrochloride has proved to be a reliable and convenient method for accurate molecular weights determinations (3,4). A wide range of sensitivities can be covered, from femtomolar to nanomolar amounts. Because of the high absorbance of guanidine hydrochloride at < 220 nm, the eluate cannot be monitored at this wavelength, therefore proteins in the nanomolar range are detected by absorbance at 280 nm. The sensitivity of detection can be increased by incorporating radioactive label into the protein, for instance by reduction and [^{14}C]-carboxymethylation of cysteine residues

Gel permeation HPLC is suitable for preparative purposes as well as analytical. Up to approximately 50 nmol can be separated on the standard size columns (7.5×600 mm); amounts greater than this reduce the resolution of the separation. Larger amounts can be separated either on preparative columns, or by successive runs with 50 nmol aliquots. For preparative purposes, it may be advantageous to use buffers other than guanidine hydrochloride, such as urea/formic acid (6), nondenaturing buffers (4,7,8), or volatile buffers (8,9). However, the resolution of separation in these buffers is frequently not as good as that in guanidine hydrochloride, and inaccurate estimates of molecular weight will arise from any protein-protein or protein-column interactions that may occur.

Materials

1. 6M guanidine hydrochloride (GdHCl) in 0.1M potassium dihydrogen phosphate. The pH of the buffer does not require adjustment and should be approximately 4.5. GdHCl of the highest grade available must be used. Filter the buffer through a 0.45 micron filter and thoroughly degas (either by applying a vacuum for 5–10 min until bubbling ceases, or by bubbling helium through the buffer for at least 5 min). Also filter and degas solutions of water and methanol (HPLC grade) for washing the column after use.
2. HPLC equipment, including an absorbance detector for 280 nm and a chart recorder.
3. Column for gel permeation HPLC, e.g., TSK-G type made by Toyo Soda and sold by several manufacturers.
4. 0.5M Tris HCl, pH 8.5, 6M GdHCl
5. 10 mM Tris HCl, pH 8.5; 6M GdHCl
6. 1M solution of dithiothreitol. Store at -20°C in aliquots; after thawing, use immediately and do not refreeze.
7. [^{14}C]-iodoacetamide (40–60 mCi/mmol) dissolved in 0.5M Tris HCl, pH 8.5 and 6M GdHCl, and then stored at -20°C . Iodoacetamide (nonlabeled), stored at 4°C in the dark (if any yellow color is present, then recrystallize from heptanol)

8. 2,4-Dinitrophenyl (DNP)-lysine and Blue Dextran 2000, approximately 5 mg/mL solutions of each in 6M GdHCl, 0.1M KH_2PO_4

Method

1. Reduce and carboxymethylate the protein standards and the sample protein as described in Chapter 5, but dialyze first against 10 mM Tris HCl (pH 8.5) and 6M GdHCl, then against 6M GdHCl in 0.1M KH_2PO_4 . Suitable protein standards are listed in Table 1.
2. Flush the column at 1 mL/min with water for at least 30 min, followed by 6M GdHCl, 0.1M KH_2PO_4 until the absorbance at 280 nm drops and becomes stable.
3. Zero the chart recorder and reduce the pump flow rate to 0.5 mL/min.
4. Determine the void volume (V_o) and total available volume (V_t) of the column by injecting a sample containing 90 μL Blue Dextran and 10 μL DNP-lysine solu-

TABLE 1
Protein Standards for Gel Permeation HPLC in Guanidine Hydrochloride

Column type	Protein	Molecular weight
TSK-G3000SW	Bovine serum albumin	66,300
	Ovalbumin	43,000
	α -Chymotrypsinogen	25,700
	Lysosyme	14,400
	Trypsin inhibitor (bovine)	6,500
	Insulin, B chain	3,420
	Insulin, A chain	2,380
TSK-G4000SW	RNA polymerase (<i>E. coli</i>) β and β' subunits	160,000
	Phosphorylase 6	92,500
	Bovine serum albumin	66,300
	Ovalbumin	43,000
	RNA polymerase (<i>E. coli</i>), α subunit	39,000
	Trypsin inhibitor (soybean)	21,000
	Lysosyme	14,400

tions (peaks of a suitable height will be obtained at 280 nm on the 0–0.08 absorbance range). Measure the distance on the chart recorder paper between the sample application and the location of the first and last peaks. V_o and V_t can then be calculated from the distances obtained for the first and last peaks respectively. After the DNP-lysine has been eluted, inject a mix of standard proteins, using amounts equivalent to the sample protein. Adjust the sensitivity range to obtain suitable peak heights.

5. For each protein, calculate the elution volume (V_e) from the distance on the chart recorder between the sample application and the protein peak. The distribution coefficient (K_d) can then be calculated using the following equation:

$$K_d = \left(\frac{V_e - V_o}{V_t - V_o} \right)$$

A plot of molecular weight (M) to the power 0.555 versus K_d to the power 0.333 will give a linear relationship. In Fig. 1 an elution profile and $M^{0.555}$ vs $K_d^{0.333}$ plot are shown for a range of standard proteins on a TSK-G3000 SW

6. Inject the sample, after all standard proteins have eluted from the column. Calculate K_d and determine the molecular weight of the sample protein from the $M^{0.555}$ vs $K_d^{0.333}$ plot.
7. Do not leave the column and HPLC equipment in GdHCl overnight, but flush with water for at least 30 min at a flow rate of 1 mL/min. If the column is not to be used the following day, then flush through with methanol for 30 min at 1 mL/min.

Notes

1. Reduction and carboxymethylation ensures that the protein adopts a linear random coil conformation in 6M GdHCl, and thus the elution volume will be proportional to molecular weight (and hence $M^{0.555}$ vs $K_d^{0.333}$ will be linear). This process also has the additional advantage of allowing the introduction of

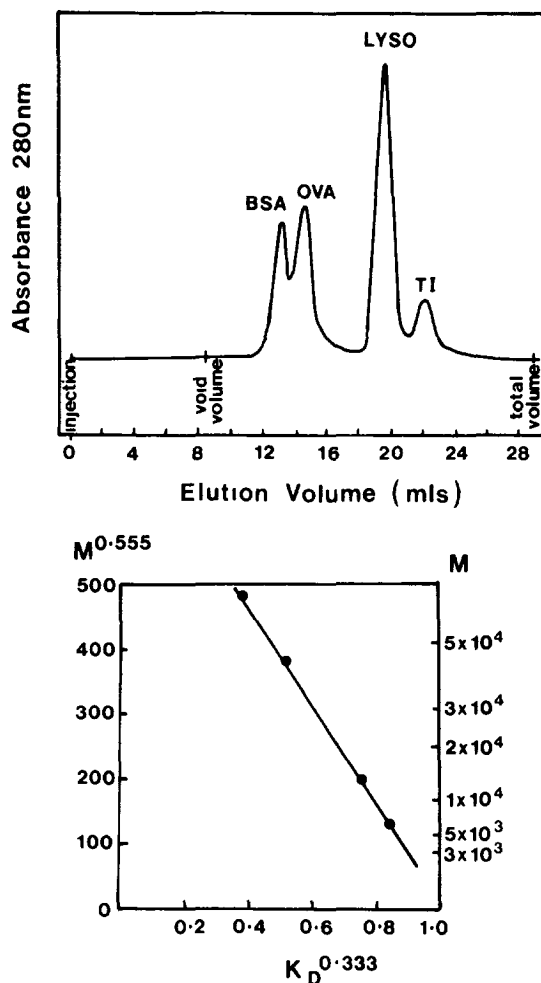


Fig 1 Typical elution profile obtained from TSK-G3000 SW column in 6M guanidine hydrochloride and the calibration curve for molecular weight determinations (Top) 40 μ L of a mixture containing bovine serum albumin (BSA), ovalbumin (OVA), lysosyme (LYSO), and bovine trypsin inhibitor (TI) was injected onto a TSK-G3000 SW column (Pharmacia), equilibrated in 6M guanidine hydrochloride, 0.1M KH_2PO_4 with a flow rate of 0.5 mL/min. The eluate was monitored at 280 nm. The void volume and total volume of the column were previously determined with Blue Dextran 2000 and 2,4-dinitrophenyl-lysine (Bottom) The data were plotted according to the equation given in the text to give the calibration curve for molecular weight (M) determination

- ^{14}C -label into the protein, thus enabling more sensitive detection than absorbance at 280 nm. If ^{14}C -label is not required, the $[\text{}^{14}\text{C}]$ -iodoacetamide may be omitted from the carboxymethylation procedure.
2. On TSK-G4000 SW columns, Blue Dextran 2000 may give two peaks, one at the void volume and one after the void volume. Thus ensure that V_o is determined from the first peak. For TSK-G3000 SW columns, $^3\text{H}_2\text{O}$ gives more accurate determinations of V_t than does DNP-lysine. However, small inaccuracies in either V_o or V_t only effect the slope of the $M^{0.555}$ vs $K_d^{0.333}$ plot, and do not effect the estimation of molecular weight.
 3. The chloride ions present in GdHCl will, when under high pressure, corrode stainless steel. It is therefore vital that GdHCl is not left in the HPLC equipment for long periods—hence the necessity to flush out the equipment with water each day.
 4. To preserve the life of the column, all samples should be centrifuged before injecting into the system. If possible a guard column should be attached in front of the analytical column.

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Chapter 3

Immunoaffinity Purification of Protein Antigens

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Introduction

The unique high specificity of antibodies, both polyclonal and monoclonal, makes them extremely valuable tools for rapid, selective purification of antigens. In principle, the antibody immobilized on a column support is used to selectively adsorb antigen from a mixture containing many other proteins (1,2). The other proteins, for which the antibody has no affinity, may be washed away, and the purified antigen then eluted from the immunoadsorbent. In order to dissociate the antigen from its high affinity antibody, the conditions for elution are necessarily extreme (1-3) and thus must be carefully chosen to permit isolation of active protein.

Although the principles involved are relatively simple, in practice many pitfalls may be encountered. How-

ever, by adopting a systematic approach to the development and optimization of the purification protocol, these pitfalls can usually be avoided.

Materials

1. Monoclonal antibody, partially purified from ascites fluid or hybridoma supernatant by ammonium sulfate precipitation, ion-exchange chromatography (DEAE-cellulose), or affinity chromatography with either DEAE-Affi-Gel Blue (BioRad) or protein-A Sepharose (see Chapters 28–31).
2. Affi-Gel 10 (BioRad), an *N*-hydroxysuccinimide ester of derivatized crosslinked agarose gel beads.
3. 0.1M HEPES, pH 7.5.
4. 1M ethanolamine HCl, pH 8.
5. 10 mM sodium phosphate, 0.15M sodium chloride, pH 7.4 (PBS).
6. 10 mM sodium phosphate, 1M sodium chloride, pH 7.4 (high salt buffer).
7. 50 mM sodium citrate pH 3.
8. 1M Tris.
9. Stock solution of sodium azide (20%).

Method

Preparation of Immunoadsorbent

1. Dialyze the antibody solution (0.5–5.0 mg/mL) against HEPES buffer to remove all spurious primary amines such as Tris and free amino acids.
2. Wash the Affi-Gel with 3 bed-volumes of isopropanol, followed by 3 bed-volumes of ice-cold deionized distilled water. This washing procedure must be completed within 20 min and is best performed by filtration in a small Buchner funnel.
3. Add the antibody solution to the washed gel (1 mg antibody/mL of gel) and agitate gently (e.g., by rotating

end-over-end) overnight at 4°C. Do not use magnetic stirrers since these may fragment the gel beads.

4. Add 0.1 mL ethanolamine/mL gel, and agitate for 1 hour further to block all unreacted ester groups.
5. Assay the supernatant for unbound antibody; usually less than 10% of the antibody remains unbound. Since the *N*-hydroxy-succinimide released during the coupling adsorbs at 280 nm above pH 2, dilute an aliquot of the supernatant in 0.1M HCl and measure the protein absorbance at 280 nm.
6. Wash the immunoabsorbent with HEPES buffer until all reactants are removed and the absorbance at 280 nm is zero.
7. Precycle the immunoabsorbent with the buffers to be used for the purification procedure by washing with 5 bed-volumes of high salt buffer followed by 5 volumes of sodium citrate buffer.
8. Wash the immunoabsorbent with PBS to return the pH to neutrality and store at 4°C in the presence of 0.02% sodium azide.

Immunoaffinity Purification

1. Incubate the antigen containing extract (in PBS) with the immunoabsorbent by rotating end-over-end for 2 h at 4°C.
2. Remove the supernatant by filtration and wash the immunoabsorbent with at least 10 bed-volumes of high salt buffer followed by 5 volumes of PBS.
3. Elute the antigen batchwise as follows. Gently agitate the immunoabsorbent with 1 bed-volume of sodium citrate buffer for 10 min at 4°C. Remove the eluate by filtration. Repeat with three further applications of sodium citrate buffer. Alternatively the immunoabsorbent may be packed into a column and eluted with sodium citrate buffer at a flow rate of 1 mL/min, or less.
4. Adjust the pH of the eluate to neutrality with 1M Tris.
5. Wash the immunoabsorbent with 5 bed-volumes of PBS and store at 4°C in PBS, 0.02% sodium azide.

Notes

1. Since the most favorable conditions for purification may differ for each antigen and antibody, small-scale trials should be made to systematically optimize the various stages of the purification. In order to do this, a suitable assay for the antigen must be available (e.g., by enzymatic activity, radioimmunoassay, or SDS gel electrophoresis). The amounts of antigen and total protein are therefore estimated in the extract before and after immunoadsorption, in the wash buffers and in the eluate.

The various stages that can be optimized are discussed below

- (a) Antibody-to-Coupling-Gel Ratio. Usually 1–10 mg of antibody is coupled per mL of gel. High concentrations may lead to reduced efficiency of binding because of steric hindrance, or to such strong binding of the antigen that it cannot be readily eluted.
- (b) Coupling Matrix. Cyanogen bromide (CNBr)-activated Sepharose (available commercially from Pharmacia) can also be used to couple antibody. With both Affi-Gel and CNBr-activated Sepharose, the orientation of the antibody is random, and therefore the efficiency of the antigen–antibody interaction may be decreased. To overcome this, the antibody may be coupled using chemical crosslinkers, such as dimethyl pimelimidate, to either protein A-Sepharose or anti-immunoglobulin crosslinked to Affi-Gel or CNBr-activated Sepharose.
- (c) Optimum Conditions for Antigen Binding. Buffers other than PBS may be more suitable for particular antigen purifications; optimum conditions are usually within the pH range 6–9 and in the presence of approximately 0.15M salt. The time of incubation of the antigen with immunoadsorbent may also require optimization.
- (d) Reduction of Nonspecific Binding. Nonspecific binding of proteins to the immunoadsorbent may be minimized by addition to the antigen-containing solution of a nonantigen protein (e.g.,

bovine serum albumin), a non-ionic detergent (e.g., Triton X-100), or an organic solvent (e.g., ethylene glycol). Altering the pH or increasing the ionic strength of the antigen solution may also decrease nonspecific binding. Alternatively, a non-immune antibody matrix (coupled in the same manner as the immunoabsorbent) can be used to pre-adsorb the extract before incubation with the immunoaffinity matrix

- (e) Antigen-to-Antibody Ratio. Optimal elution and minimal nonspecific adsorption are usually obtained if the coupled antibody is fully saturated with antigen.
- (f) Washing Procedures. To remove nonspecifically adsorbed protein, the immunoabsorbent should be washed with several column volumes of buffer. Optimal removal is attained by altering either the ionic strength (e.g., 1M salt, or distilled water), or pH or by inclusion of organic solvent (e.g., ethylene glycol). Obviously under the conditions chosen the antigen must not be eluted from the immunoabsorbent.
- (g) Elution Conditions. The conditions used to elute immunoabsorbents are extreme (Table 1) and may cause denaturation and loss of activity of the antigen. Thus a study of the stability of the antigen under various potential eluting conditions must be made to determine the minimum and maximum conditions that the protein activity will withstand. These conditions can then be tested for elution of the antigen from the immunoabsorbent. Combinations, such as denaturants at low pH, may be required to disrupt antigen-antibody interactions of very high affinity. Specific elution may be possible for some immunoabsorbents, e.g., if the immunodeterminant is carbohydrate, the antigen may be eluted with the carbohydrate or an analogue.
- (h) Batch Procedure vs Column. The procedure described in the previous section uses a batch technique throughout; alternatively, the immunoabsorbent may be poured into a column at any stage during the purification. When the antigen extract is applied to the immunoabsorbent packed in a col-

TABLE 1
Conditions for Elution from Immunoabsorbents

Low pH (<3)	50 mM glycine HCl, pH 2-3 50 mM sodium citrate, pH 2.5-3 1M propionic acid
High pH (<10)	0.15M ammonium hydroxide, pH 11 0.15M NaCl 50 mM diethylamine, pH 11.5 0.5% sodium deoxycholate
Denaturants	4M guanidine hydrochloride 8M urea
Chaotropic ions (up to 3 M)	3M sodium isothiocyanate
($\text{CCl}_3\text{COO}^- \gg \text{SCN}^- >$ $\text{CF}_3\text{COO}^- > \text{ClO}_4^- > \text{I}^- >$ Cl^-)	3M sodium iodide
Polarity reducing	Ethylene glycol (50%) Dioxan (10%)
Low ionic strength	Distilled water

umn, the flow rate should be low, < 20 mL/h. Recycling the flow-through may allow further binding of antigen. The direction of flow should be reversed for elution to prevent strongly bound antigens from interacting with the immunoabsorbent throughout the column before elution. If the affinity of the antibody for antigen is high, elution may be helped by stopping the flow for 10 min or longer when the column has been equilibrated in elution buffer.

2. Monospecific polyclonal antiserum, obtained by immunization with purified protein, may be used instead of monoclonal antibody. The serum should be purified before use by one of the techniques suggested for monoclonal antibody purification. (See Chapter 31).
3. Immunoaffinity chromatography has been successfully used to purify several membrane proteins. Because of the hydrophobic nature of these proteins, all solutions used throughout the purification must contain deter-

gent (e.g., Triton X-100, NP40, or sodium deoxycholate). Sodium deoxycholate is extremely effective for solubilizing membrane proteins, however, it is incompatible with buffers of high ionic strength, or pH of less than 8.0 since a viscous gel is formed. One detergent may be exchanged for another during washing while the antigen is bound to the immunoabsorbent. Detergents may reduce the affinity of antibody for antigen; thus, pilot studies must be made to determine the optimum conditions for solubilization and the antigen-antibody interaction.

4. If the antigen does not bind to the immunoabsorbent despite optimization of the extract conditions, then either the affinity of the antibody is too low or the coupling procedure has caused denaturation of the antibody. The former cause is intrinsic to the antibody and cannot be remedied. Denaturation upon coupling may be circumvented by use of a different coupling support, or by partially blocking some of the coupling groups prior to addition of antibody (thus the gel is first incubated with coupling buffer alone for a few hours, and then with the antibody). Alternatively the antibody may be noncovalently bound to immobilized protein A or anti-immunoglobulin. With this type of immunoabsorbent, antibody and antigen will be eluted together and therefore a further purification step will be required.
5. Another problem that may be encountered is that the antigen sometimes cannot be eluted from the immunoabsorbent. This may be the result of a very high affinity of the antibody for antigen. Since Fab fragments frequently have lower affinities than the antibodies from which they are derived, an immunoabsorbent made with Fab fragments may allow elution of the antigen. Alternatively the antigen may elute from the immunoabsorbent, but in an inactive form and therefore would not be detected. To test whether this is the case, SDS gel electrophoresis should be used to analyze both the eluate, and an aliquot of the immunoabsorbent (boiled in 2% SDS for 2 min) for protein of the appropriate molecular weight.
6. The life of the immunoabsorbent will depend on how frequently it is used and the conditions required to

elute the antigen. The immunoabsorbent should be returned to normal conditions as soon as possible after elution, and should be stored at 4°C (not frozen) in the presence of a bacterial inhibitor. Centrifuging the antigen-containing extract to remove particulate matter will also preserve the life of the immunoabsorbent. Also if proteases are present in the extract, inhibitors such as phenylmethyl sulfonyl fluoride (2 mM), iodoacetamide (2.5 mM), and EDTA (5 mM), should be added to protect both the antigen and antibody from proteolysis

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Chapter 4

Electrophoretic and Chromatographic Separation of Peptides on Paper

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Introduction

Electrophoresis and chromatography on paper have proved extremely useful techniques for the separation of small peptides prior to amino acid sequencing. Polarity determines the rate of migration of a peptide during chromatography, whereas both charge and size are the main determinants during electrophoresis. The two techniques therefore are complimentary, and when used in conjunction can resolve most peptide mixtures into their pure components.

Ideally the mixture to be separated should contain less than 20 peptides, since only a limited amount of a mixture can be loaded per chromatogram. These techniques therefore are frequently used for final separation

following an initial purification by either gel filtration or ion-exchange chromatography. On Whatman 3 MM paper up to 0.5 mg of peptide mixture can be loaded per centimeter (fivefold less on Whatman No. 1); heavier loadings result in distortion and streaking of the peptide bands with a concomitant decrease in resolution. Optimum resolution is achieved with peptides of less than 20 amino acids, thus these techniques are usually used for separation of peptides obtained by enzymatic digestion (e.g., by trypsin, or *Staphylococcus aureus* V8 protease) rather than those obtained by chemical cleavage (e.g., by cyanogen bromide). Peptides larger than 20 amino acids remain at the origin in both electrophoresis and chromatography, and in addition their presence can cause streaking of smaller peptides during electrophoresis. The peptide mixture must be free from salts that cause severe streaking of the peptide bands, thus the initial purification steps, or the enzymatic digestion are best carried out in volatile buffers (e.g., ammonium bicarbonate, acetic acid, pyridine acetate).

Several stains can be used to visualize the peptides; some react with amino groups and therefore stain all peptides, e.g., ninhydrin, fluorescamine (1), whereas others are specific for peptides containing a particular amino acid, e.g., phenanthrenequinone for arginine (2) or the Ehrlich test for tryptophan. Some stains are compatible with others, thus allowing one chromatogram to be stained both for specific amino acids, and for all peptides (3). All these stains are destructive, and make the peptides unsuitable for amino acid sequencing. Thus for preparative purposes narrow strips from the edges of the chromatogram are stained and the corresponding peptides marked and eluted from the unstained portion. Alternatively the whole chromatogram is stained with a weak solution of fluorescamine (1), which reacts with only a small percentage of each peptide, but is sufficiently sensitive to allow their detection.

Trial runs with small aliquots of the peptide mixture are carried out in order to determine the optimum conditions for separation. By judicious choice of the solvent mixture and the duration of the separation, all the peptides in a mixture can frequently be resolved by either chromatography or electrophoresis. More complex mix-

tures will require separation by one technique followed by the other.

Materials

1. Whatman chromatography paper No. 1 and 3 MM, sheet size 46×57 cm.
2. Disposable plastic gloves. These must always be worn when handling the paper, since fingerprints will be stained by the sensitive stains used.
3. Glacial acetic acid, pyridine and *n*-butanol, all A.R grade.
4. Glass chromatography tanks for descending chromatography (at least $51 \times 20 \times 56$ cm) complete with troughs. High-voltage flat-bed electrophoresis apparatus fitted with safety cutoffs and cooling plates to take a maximum sheet size of 61×28 cm, and a high voltage power supply, to give at least 200 mA at 3 kV.
5. Dressmaking pinking shears, sewing machine, and cotton.
6. Electric fan heater or hair dryer.
7. Microsyringe with blunt-ended needle. The most useful sizes are 10 and 50 μ L.
8. Ninhydrin spray. Aerosols of ninhydrin solution are commercially available; otherwise a solution (0.25% in acetone) may be prepared and used with a universal aerosol spray.
9. Oven (110°C) for developing ninhydrin-sprayed chromatograms.
10. Flat-bottomed specimen tubes with approximately 10 mL capacity. These are for collecting eluted peptides and should be siliconized with Repelcote.

N.B.: *n*-butanol, pyridine, ninhydrin solution, and Repelcote are all toxic and should therefore be used in a fume hood.

Method

Paper Chromatography

1. Place a folded sheet of Whatman 3 MM in the bottom of the chromatography tank and saturate this with chromatography solvent (*see* Table 1). Fill the trough with

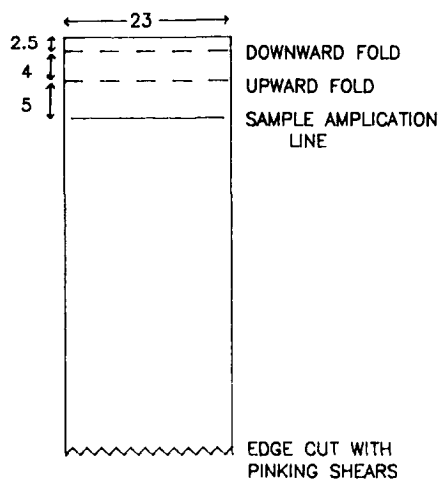


Fig 1 (a) Folding and marking of paper for chromatography.

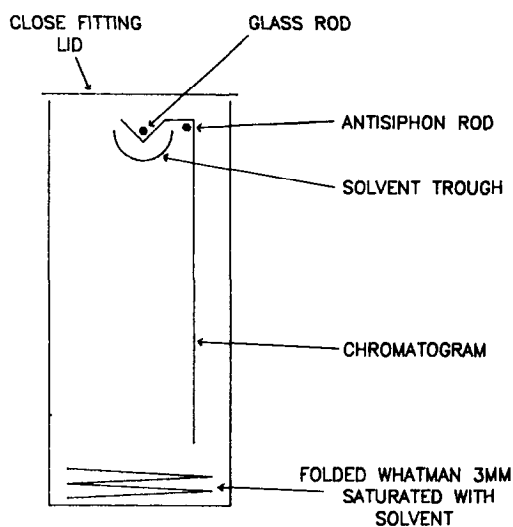


Fig 1 (b) Positioning of paper in chromatography tank.

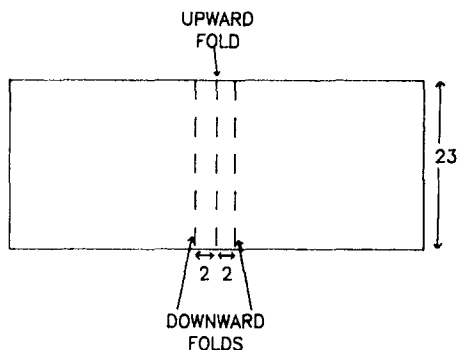


Fig. 1 (c) Folding of paper for electrophoresis. All dimensions are given in centimeters.

solvent, close the lid, and allow the atmosphere within the tank to equilibrate. Meanwhile, prepare the chromatogram.

2. Cut a sheet of Whatman 3 MM in half lengthwise and fold as shown in Fig. 1a. Draw a pencil line 2 cm from the second fold to within 1 cm of either edge—this becomes the guideline for loading the sample. The bottom edge of the sheet should be cut with pinking shears to ensure that the solvent runs evenly across the width of the chromatogram.
3. Dissolve the peptide mixture in aqueous solution (e.g., water, 0.1M acetic acid, or 0.1M ammonium bicarbonate) and centrifuge to remove any insoluble material. Using a microsyringe, load the sample (0.1–0.5 mg/cm) on the pencil line. The area of application should not touch the bench, the folds are usually sufficient to ensure this does not happen. Do not allow the width of the application strip to exceed 0.3 cm; this may require repeated applications, drying between each with a warm air stream from a fan heater or hair dryer.
4. When the sample is dry, place the chromatogram in the tank with the downward fold in the trough and held in place with a glass rod; the upward fold should be over the antisiphon rod (see Fig. 1b).

5. Leave the chromatogram to develop until optimum separation is achieved. The time required, usually between 1 and 5 d, should be determined previously by trial separations of aliquots of the mixture (see notes).

Paper Electrophoresis

1. Cut a sheet of Whatman 3 MM in half lengthwise and fold as shown in Fig. 1c. The position of the upward fold should previously be determined by trial runs on aliquots of the mixture (see notes), and for most peptide mixtures will be nearer to the anode.
2. Load the peptide mixture along the upward fold as described for paper chromatography.
3. When the sample is dry, evenly wet the electrophoretogram with the appropriate buffer (*see* Table 1). This is best done using a 10 mL pipet. First, run the pipet along the two downward folds, allowing a small amount of solvent to absorb onto the paper, which will then wet the sample application line by capillary action (do not wet the application line directly). The remainder of the paper is then wetted by passing the pipet backwards and forwards across the width of the paper while allowing the buffer to slowly flow out of the pipet (do not overwet). Blot the paper (excluding the sample application line) between two sheets of Whatman 3 MM and place it in the electrophoresis apparatus.
4. Fill the electrode troughs with buffer and make wicks the same width as the electrophoretogram by folding Whatman 3 MM paper. Wet the wicks with buffer, and blot with Whatman 3 MM. Place the wicks in the electrode troughs and overlap them onto the electrophoretogram by approximately 1 cm.

N.B.: Excessive wetting of either the electrophoretogram or wicks leads to condensation, and subsequent short circuiting of the current.

5. Ensure that the water for the cooling plates is turned on, and turn on the power supply. Electrophoresis at 1–2 kV for 1–2 h is usually sufficient for the separation of most peptide mixtures; the exact conditions should be determined previously by trial runs.

Detection of Peptides and Their Elution

1. When the chromatographic or electrophoretic separations have finished, hang the paper to dry in a warm air stream from a fan heater or hair-dryer
2. Cut 2-cm-wide strips from each long edge of the chromatogram and mark each so that they can later be lined up with the remainder of the sheet.
3. Attach these strips to a sheet of Whatman No. 1 with paper clips, thus keeping them straight and allowing several side strips to be stained simultaneously. Spray the strips evenly with ninhydrin solution (do not overwet since this may cause the peptide bands to streak). After allowing the strips to dry for a few minutes, place them in a 110°C oven for 5 min. The peptides should be clearly visible as blue bands; if not, the spraying and heating process may be repeated.
4. Match the strips with the chromatogram and draw lines across to connect the corresponding peptide bands on the two strips.
5. The peptides are eluted from the unstained paper by descending chromatography with 0.1M acetic acid. To do this, cut a sheet of Whatman No. 1 in quarters lengthwise. Sew a quarter sheet along each edge of the chromatogram and cut across the full width for each peptide band. One piece of the Whatman No. 1 is folded as given for chromatography and the other cut into a point (if the peptide band is wider than 3 cm, it is advisable to cut two points). These strips are placed in a chromatography tank with 0.1M acetic acid in the trough. Siliconized tubes are placed under each point to collect the eluted peptides (if a large tank is used, the tubes can be placed on a glass shelf supported at the correct height in the tank). Leave for 24 h before removing the tubes and drying the eluted samples *in vacuo* over sodium hydroxide and phosphorus pentoxide.

Notes

1. The optimum conditions for peptide separation within one chromatogram or electrophoretogram are deter-

mined by preliminary trial separations with aliquots of the mixture. Approximately 0.1–0.5 mg of mixture is loaded across a 1 cm strip; thus one-half sheet of Whatman 3 MM can be used for 11 trial separations, each mixture being separated from the next by 1 cm. For chromatography it is only necessary to determine the optimum time for separations, while for electrophoresis the time, voltage, and position for sample application must be determined. Some commonly used chromatography solvents and electrophoresis buffers are shown in Table 1

2. Other stains besides ninhydrin can be used for the detection of peptides (*see* Table 2 and Chapter 21). Unfortunately the use of stained side strips does not detect irregularities in the peptide bands across the width of

TABLE 1
Useful Solvents for Paper Chromatography and Electrophoresis

Method	Composition, v/v	
Chromatography	<i>n</i> -Butanol:acetic acid	(upper phase only, lower phase in bottom of tank)
	water	
	.pyridine	
	30 9 24:20	
Electrophoresis	<i>n</i> -Butanol:acetic	pH 3 5
	acid:water:pyridine	
	15 3 12:10	
	<i>n</i> -Butanol:acetic acid:water	
	4 1.5	
Electrophoresis	Pyridine:acetic acid:water	pH 4 8
	1 10:289	
	Pyridine:acetic acid:water	
	3.3 395	
	Pyridine:glacial acetic acid:water	
Electrophoresis	25 1 225	pH 6 5
	Formic acid:acetic acid:water	
	52:29:919	

the chromatogram. This may be overcome by spraying the chromatogram lightly with a weak solution of fluorescamine (0.001–0.0005% in acetone), after an initial treatment with 3% pyridine in acetone (1). Only a small portion of the free amino groups react under these conditions, allowing visualization of the peptides by fluorescence under UV light (366 nm) without rendering all the peptides unsuitable for sequencing. Fluorescamine (0.01% in acetone) is also considerably more sensitive than ninhydrin, detecting as little as 0.1 nmol/cm², compared to 5 nmol/cm², and thus can also be used for staining side strips.

3. The main problem encountered during paper electrophoresis or chromatography is uneven migration and/or streaking of the peptide bands. Several factors can cause the occurrence of this phenomenon: (a) The presence of nonvolatile salts in the sample; (b) peptides too large (ideally < 20 amino acids); (c) too much sample applied (not more than 0.5 mg/cm); (d) excessive wetness during electrophoresis; (e) insufficient cooling during electrophoresis; (f) an uneven temperature gradient in the chromatography tank (e.g., because of exposure to draft or to direct sunlight); (g) excessive wetting with staining agent.
4. It is essential that pure solvents are used since impurities such as aldehydes may lead to loss of aromatic acids. The addition of a trace of β -mercaptoethanol to the solvent prevents autoxidation of methionine and cysteine derivatives.
5. Cleanliness is essential to avoid contamination of the peptides with either proteases or microorganisms. Thus all surfaces should be cleaned regularly with 70% ethanol to remove microorganisms and dust. Chromatograms must be handled at all times with either disposable plastic gloves or forceps.
6. The purity of eluted peptides can be assessed by trial runs under different conditions as outlined previously. Ultimately *N*-terminal analysis (see Chapter 23) indicates whether the peptide is sufficiently pure for amino acid sequencing.
7. Yields from paper depend on the properties of the peptide, and are usually between 60 and 80%, but some-

TABLE 2
Alternative Stains for the Detection of Peptides^a

Stain	Specificity	Color	Sensitivity
<i>Ninhydrin-collidine</i> (4) 600 mL ethanol + 200 mL glacial acetic acid + 80 mL collidine + 1 g ninhydrin Store 4°C Develop 5–10 min 80°C	Free amino groups	Various (pink, yellow, green, blue purple) Colors fade, store stained papers –20°C	5 nmol cm ^{–2}
<i>Iodine–starch</i> (5) Spray with iodine solution (0.5% in chloroform) Leave room temperature to dry Spray with starch solution (1% in water)	Peptide bond	Blue black	>5 nmol cm ^{–2}
<i>Ehrlich reagent</i> (3,4) 2% (w/v) <i>p</i> -dimethylaminobenzaldehyde in acetone, immediately before spraying mix 9 vol with 1 vol conc HCl Develop for a few minutes room temperature Can be used after ninhydrin, but will bleach ninhydrin colors	Tryptophan	Blue purple	5 nmol cm ^{–2}
<i>Phenanthrene-1,2-quinone</i> (2) 0.02% Phenanthrene-1,2-quinone in absolute ethanol (A), store in dark 4°C for months 10% (w/v) NaOH in 60% (v/v) aqueous ethanol, (B), store 1 month	Arginine	Greenish-white fluorescence against dark background	0.1 nmol cm ^{–2}

Mix equal parts A and B prior to spraying
 Dry at room temperature, 20 min
 Visualize with UV (254 or 366 nm)

Pauly test (3,4)

1 g sulfanilamide + 10mL 12M HCl + 90 mL water, (A), 5 g NaNO₂ in 100 mL water, (B), 50 mL saturated Na₂CO₃ + 50 mL water, (C)
 Store all solutions 4°C Mix 5 mL A + 5 mL B
 Add 40 mL *n*-butanol after 1 min Shake 1 min, leave to settle, use butanol layer to spray or dip chromatogram 5 min at room temperature
 Spray with C for imidizoles other than histidine

Histidine	Cherry red	5 nmol cm ⁻²
Tyrosine	Dull brown	

Nitrosonaphthol Test (3)

0.1% (w/v) α-nitroso-β-naphthol in acetone (A), acetone conc HNO₃ (9:1, v/v) (B), freshly prepared
 Dip or spray with A. Dry
 Dip or spray with B Dry 5–10 min
 Heat gently with hot air

Tyrosine	Rose on yellow, background color fades	10 nmol cm ⁻²
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^aSee also Chapter 21

times as low as 20%. If smaller amounts of peptides are fractionated Whatman No. 1 may be used and up to 0.1 mg/cm loaded (when wet, Whatman No. 1 tears easily and therefore should be handled with care).

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Chapter 5

Peptide Mapping by Reverse-Phase High Pressure Liquid Chromatography

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Introduction

Reverse-phase high pressure liquid chromatography (HPLC) has proved to be an extremely versatile technique for rapid separation of peptides (reviewed in ref. 1). One of its uses is for peptide mapping, or "fingerprinting" (2-5) as an alternative procedure to the conventional two-dimensional separations on paper or thin-layer supports (See Chapter 21). Although the map obtained is one dimensional, the excellent resolving power of reverse-phase HPLC enables separation of the majority of peptides within a mixture. HPLC offers the advantages of high reproducibility, easy quantitation, and rapid analysis time, and is also suitable for automation.

A wide range of sensitivities (from subfemto- to nanomolar amounts) can be covered by use of an appropriate method for detection of the peptides. For nanomolar quantities the eluate is monitored at two or more wavelengths, thus detecting all peptides (at < 220 nm), and only those containing aromatic amino acids (tyrosine and tryptophan at 280 nm; phenylalanine at 255 nm). The sensitivity of detection is increased to picomolar levels by either fluorescence detection of tyrosine and tryptophan (6) or postcolumn fluorescent derivatization with fluorescamine (3,7) or *o*-phthalaldehyde (6,7). By incorporation of radioactive label into proteins, as little as subfemtomolar amounts may be analyzed. To minimize peptide losses at these low levels, unlabeled carrier protein (e.g., bovine serum albumin) should be added prior to proteolysis. Proteins can be labeled biosynthetically with amino acids (e.g., ^{35}S -methionine, ^{35}S -cysteine, ^3H -lysine) thus allowing positive identification of precursors produced during *in vivo* pulse labeling or *in vitro* translation (4). Alternatively, the label can be introduced postsynthetically by modification with radioactive reagents (e.g., ^{125}I -iodination, ^{14}C -carboxymethylation). Radioactive labeling can also be useful for peptide maps on larger amounts of protein, thus allowing the identification of peptides containing specific amino acids (e.g., cysteine or methionine).

For reverse-phase HPLC of peptides, the most suitable column supports are the cyanopropyl-(CN-), octyl-(C₈-), or octadecyl-(C₁₈-) silica types (the latter two usually give better separation than the CN-silica types). Several buffers have been used for peptide separation, such as orthophosphoric acid (2), pyridine-acetic acid (3), triethylammonium phosphate (7), and trifluoroacetic acid (TFA) (8). Peptides are eluted with increasing concentrations of organic solvents, most frequently acetonitrile or *n*-propanol. In this laboratory, TFA/acetonitrile is routinely used, since it is both volatile, and allows high sensitivity detection by absorbance at < 220 nm.

Materials

1. 0.1% TFA (buffer A) and 0.1% TFA/60% acetonitrile (buffer B). TFA and acetonitrile should be HPLC or

sequencing grade, and the water used must be free of organic contaminants (most laboratory distilled water requires further purification, e.g., by use of a Millipore Milli-Q system). Filter the buffers through a 0.45 μm filter to remove all particulate matter and degas (either by applying a vacuum for 5–10 min until all bubbling ceases, or by bubbling helium through for at least 5 min).

2. HPLC equipment, including a solvent programmer and two detectors, one at 280 nm to monitor tryptophan- and tyrosine-containing peptides and the other within the range 205–215 nm to monitor all peptides.
3. Column for reverse-phase HPLC; either octyl-(C_8 -) or octadecyl-(C_{18} -) silica type.
4. 0.5M Tris HCl, pH 8.5–6M guanidine hydrochloride (GdHCl). The highest grade of GdHCl should be used.
5. 10 mM Tris HCl, pH 8.5–2M GdHCl.
6. 0.1M ammonium bicarbonate.
7. Stock solution of *N*-tosyl-L-phenylalanyl chloromethyl ketone- (TPCK-) treated trypsin, 10 mg/mL in 0.1 mM HCl. This solution can be stored at -20°C for months without loss of activity.
8. 1M stock solution of dithiothreitol (DTT). Store at -20°C in aliquots, after thawing, use immediately and do not refreeze.
9. [^{14}C]-iodoacetamide (40–60 mCi/mmol) dissolved in 0.5M Tris HCl, pH 8.5, 6M GdHCl and stored at -20°C . Iodoacetamide (nonlabeled), if any yellow color is present the iodoacetamide must be recrystallized from heptanol. Store the solid at 4°C in the dark.

Methods

Full Reduction and Carboxymethylation

1. Dissolve the protein in 0.5M Tris HCl, pH 8.5–6M GdHCl at 1–10 mg/mL.
2. Make 10 mM in DTT and incubate at 37°C for at least 2 h.
3. Cool on ice and add 50 μCi [^{14}C]-iodoacetamide. Incubate in the dark for 30 min on ice.

4. Add iodoacetamide to 20 mM final concentration and incubate for a further 2 h in the dark and on ice.
5. Dialyze against 10 mM Tris HCl, pH 8.5–2M GdHCl for at least 3 h and then twice against 0.1M ammonium bicarbonate.

These dialysis steps remove both the unreacted iodoacetamide (to ensure that methionine residues are not modified) and the GdHCl prior to proteolysis.

Trypsin Digestion

1. Add TPCK-treated trypsin (2% by weight of the protein) and incubate for 24 h at 37°C.
2. Stop the digestion either by injecting onto the HPLC column immediately, or by adding soyabean trypsin inhibitor or *N*-tosyl-L-lysyl chloromethyl ketone (TLCK) in a slight molar excess over the protease.

Reverse-Phase HPLC

1. Set a flow rate of 1 mL/min through the column and program a gradient of 0–100% buffer B over 60 min.
2. Start the gradient and monitor at both wavelengths on the highest sensitivity. If peaks are observed during the run then repeat this process until they disappear (see Notes).
3. To check the system inject a tryptic digest of a standard protein e.g., cytochrome c or bovine serum albumin. The injection volume can be up to 2 mL, but a better resolution is obtained with less than 200 μ L. Up to approximately 50 nmol of protein can be injected per run on a 25 cm $\times$ 4.6 mm column. Ideally the amount of standard protein should correspond approximately to that of the protein of interest so that the sensitivities of the two detectors can be adjusted to give the optimum peak height.
4. At 5 min after the injection, start the gradient. When the gradient has been completed, maintain 100% buffer B for a further 10 min. The majority of peptides will elute between 0 and 40% acetonitrile (i.e., 0 and 67% buffer B), and should be sharp discrete peaks (see Fig. 1).

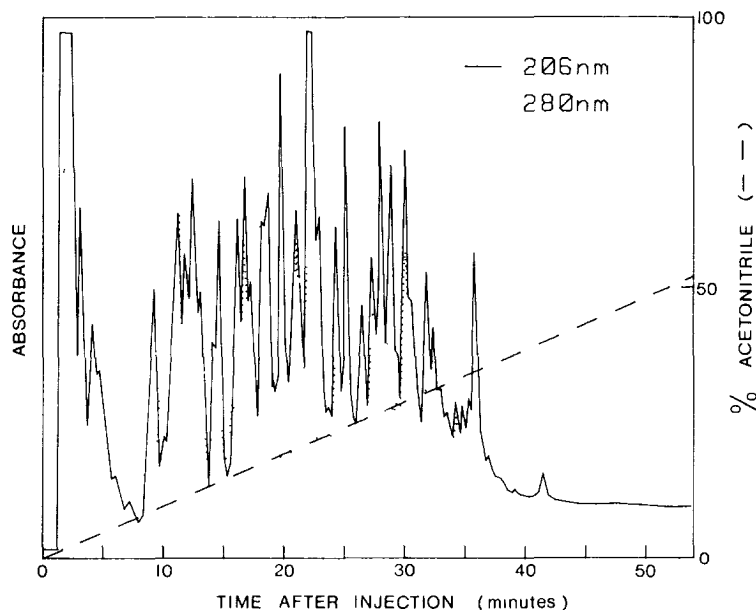


Fig. 1 Tryptic peptide map of bovine serum albumin. 500 μ g of bovine serum albumin was digested for 24 h at 37°C with TPCK-trypsin (10 μ g). The resultant digest was analyzed on a SynChropak C-18 column (SynChrom Inc.) in 0.1% TFA with an acetonitrile gradient of 0–60% in 60 min. The eluate was monitored at 280 and 206 nm.

5. Allow the column to re-equilibrate at 0% buffer B for 10 min prior to injection of the next digest.
6. Repeat steps 3–5 for each protein digest.

Notes

1. Reduction of disulfide bonds followed by carboxymethylation of cysteine residues prevents the formation of intermolecular disulfide bonds either before or after proteolysis. Under the alkaline conditions used, the iodoacetamide reacts most rapidly with the cysteine residues, then with the thiol groups of the DTT; thus, with a molar ratio of 1:2 of DTT to iodoacetamide, alkylation of methionine residues does not occur. The reaction may be optimized if the moles

of disulfide bonds in the protein are known; use a 2 mM excess of DTT over disulfide bonds, and a 1.1-fold molar excess of iodoacetamide over the total thiol groups in the solution.

2. If the protein is insoluble in ammonium bicarbonate, try the addition of urea to 2M to aid solubilization. Urea will not effect the digestion, but when injected onto the HPLC column will give a large UV absorbing peak at the void volume; therefore, allow the absorbance to return to baseline level before starting the acetonitrile gradient.
3. At high sensitivities an increase in the absorbance at the lower wavelength may be observed as the proportion of buffer B increases. This results from absorbance by acetonitrile, and may be counteracted by addition of more TFA to buffer A (0.01–0.03%). Peaks may also be observed during the blank run that remain even after several acetonitrile gradients have been passed through the column. These peaks are caused by impurities in the buffers. Provided the peaks are not too large, they can be subtracted from the peptide maps obtained; alternatively, the buffers will require further purification.
4. To preserve the life of the column, all samples must be centrifuged before injection. Also, if possible, a guard column (packed with a similar support material to that of the main column) should be used.
5. Poorly resolved and broad peaks are normally a result of overloading the column with peptides; alternatively, the column may need to be replaced.
6. Peptide mapping by HPLC covers a large range of sensitivity, from very small amounts of radioactively labeled protein up to approximately 50 nmol. Protein may be radioactively labeled *in vivo* (e.g., ^{35}S -methionine, ^3H -lysine) or *in vitro* (e.g., ^{125}I -iodination or ^{32}P -phosphorylation). When the amount of radioactively labeled protein is small ($< 5\text{ }\mu\text{g}$), a cold carrier protein (e.g., bovine serum albumin) should be added to minimize losses; this has the additional advantage of acting as an internal control on the reproducibility of the separation.

7. In the procedure described here, trypsin is used to digest the protein. This method of peptide mapping is also suitable for use with other proteases that generate small peptides, e.g., *Staphylococcus aureus* V8 protease.

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Chapter 6

SDS Polyacrylamide Gel Electrophoresis of Proteins

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Introduction

Probably the most widely used of techniques for analyzing mixtures of proteins is SDS polyacrylamide gel electrophoresis. In this technique, proteins are reacted with the anionic detergent, sodium dodecylsulfate (SDS, or sodium lauryl sulfate) to form negatively charged complexes. The amount of SDS bound by a protein, and so the charge on the complex, is roughly proportional to its size. Commonly, about 1.4 g SDS is bound per 1 g protein, although there are exceptions to this rule. The proteins are generally denatured and solubilized by their binding of SDS, and the complex forms a prolate ellipsoid or rod of a length roughly proportionate to the protein's molecular weight. Thus, proteins of either acidic or basic *pI* form

negatively charged complexes that can be separated on the bases of differences in charges and sizes by electrophoresis through a sieve-like matrix of polyacrylamide gel.

This is the basis of the SDS gel system, but it owes its popularity to its excellent powers of resolution that derive from the use of a "stacking gel." This system employs the principles of isotachopheresis, which effectively concentrates samples from large volumes (within reason) into very small zones, that then leads to better separation of the different species. The system is set up by making a "stacking gel" on top of the "separating gel," which is of a different pH. The sample is introduced to the system at the stacking gel. With an electric field applied, ions move towards the electrodes, but at the pH prevailing in the stacking gel, the protein-SDS complexes have mobilities intermediate between the Cl^- ions (present throughout the system) and glycinate ions (present in the reservoir buffer). The Cl^- ions have the greatest mobility. The following larger ions concentrate into narrow zones in the stacking gel, but are not effectively separated there. When the moving zones reach the separating gel, their respective mobilities change in the pH prevailing there and the glycinate ion front overtakes the protein-SDS complex zones to leave them in a uniformly buffered electric field to separate from each other according to size and charge. More detailed treatments of the theory of isotachopheresis and electrophoresis generally are available in the literature (e.g., 1).

The system of buffers used in the gel system described below is that of Laemmli (2), and is used in a polyacrylamide gel of slab shape. This form allows simultaneous electrophoresis of more than one sample, and thus is ideal for comparative purposes.

Materials

1. The apparatus required is available commercially or can be made in the workshop, but generally conforms to the design by Studier (3). The gel is prepared and run in a narrow chamber formed by two glass plates separated by spacers of perspex or other suitable material,

as shown in Fig. 1. The spacers are longer than the glass plates (so that they are easy to remove), are about 1 cm wide, and are of any thickness (say, 0.5–1 mm for a thin gel). The sample wells into which samples are loaded are formed by a template "comb" that extends across the top of the gel and is of the same thickness as the spacers. Typically, the "teeth" on this comb will be 1 cm long, 2–10 mm wide, and separated by about 3 mm. The chamber is sealed with white petroleum jelly (Vaseline). A dc power supply is also required.

- 2 Stock solutions. Chemicals should be analytical reagent (Analar) grade and water should be distilled. Stock solutions should all be filtered. Cold solutions should be warmed to room temperature before use.

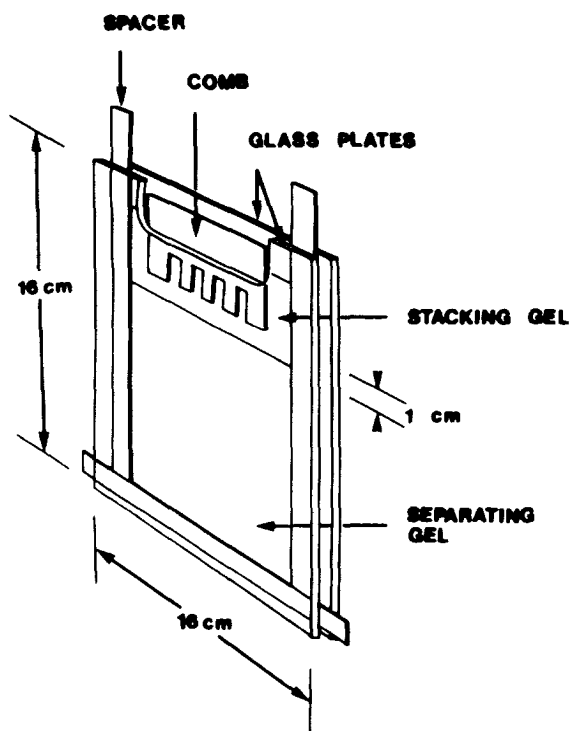


Fig. 1. The construction of a slab gel, showing the positions of the glass plates, the spacers, and the comb.

(i) *Stock acrylamide solution* (total acrylamide content, %T = 30% w/v, ratio of crosslinking agent to acrylamide monomer, %C = 2.7% w/w)

Acrylamide	73 g
Bis acrylamide	2 g

Dissolve the above and make up to 250 mL in water. This stock solution is stable for weeks in brown glass, at 4°C.

(ii) *Stock separating gel buffer*

SDS	1.0 g
"Tris" buffer	
[2-amino-2-(hydroxymethyl)-propane-1,3-diol]	45.5 g

Dissolve the above in less than 250 mL of water, adjust the pH to 8.8 with HCl, and make the volume to 250 mL. This stock solution is stable for months at 4°C.

(iii) *Stock ammonium persulfate.*

Ammonium persulfate	1.0 g
---------------------	-------

Dissolve in 10 mL of water. This stock solution is stable for weeks in brown glass at 4°C.

(iv) *Stock stacking gel buffer.*

SDS	1.0 g
"Tris" buffer	15.1 g

Dissolve the above in less than 250 mL of water, adjust the pH to 6.8 with HCl, and make up to 250 mL. Check the pH before use. This stock solution is stable for months at 4°C.

(v) *Reservoir buffer* (0.192M glycine, 0.025M Tris, 0.1% w/v SDS).

Glycine	28.8 g
"Tris" buffer	6.0 g
SDS	2.0 g

Dissolve the above and make to 2 L in water. The solution should be at about pH 8.3 without adjustment. This solution is readily made fresh each time.

(vi) *Stock (double strength) sample solvent:*

SDS	0.92 g
β -Mercaptoethanol	2 mL
Glycerol	4.0 g
"Tris" buffer	0.3 g
Bromophenol Blue	2 mL
(0.1% w/v solution in water)	

Dissolve the above in less than 20 mL of water, adjust the pH to 6.8 with HCl, and make to 20 mL. Check the pH before use. Exposed to oxygen in the air, the reducing power of the β -mercaptoethanol wanes with time. Periodically (after a few weeks) add extra agent or renew the solution. This stock solution is stable for weeks at 4°C.

(vu) Protein stain.

Coomassie Brilliant Blue R250	0.25 g
Methanol	125 mL
Glacial acetic acid	25 mL
Water	100 mL

Dissolve the Coomassie dye in the methanol component first, then add the acid and water. If dissolved in a different order, the dye's staining behavior may differ. The stain is best used when freshly made. For best results do not reuse the stain—its efficacy declines with use. If this dye is not available, use the equivalent dye PAGE blue 83.

(viii) Destaining solution.

Methanol	100 mL
Glacial acetic acid	100 mL
Water	800 mL

Mix thoroughly. Use when freshly made.

Method

- 1 Thoroughly clean and dry the glass plates and three spacers, then assemble them as shown in Fig. 1, with the spacers set 1–2 mm in from the edges of the glass

plates. Hold the construction together with bulldog clips. White petroleum jelly (melted in a boiling water bath) is then applied around the edges of the spacers to hold them in place and seal the chamber. Clamp the chamber in an upright, level position

2. A sufficient volume of separating gel mixture (say 30 mL for a chamber of about $14 \times 14 \times 0.1$ cm) is prepared as follows. Mix the following:

Stock acrylamide solution	15 mL
Distilled water	7.5 mL

Degas on a water pump, and then add.

Stock separating gel buffer	7.5 ml
Stock ammonium persulfate solution	45 μ L
<i>N, N, N', N'</i> -tetramethylethylenediamine (TEMED)	15 μ L

Mix gently and use immediately (because polymerization starts when the TEMED is added). The degassing stage removes oxygen, which inhibits polymerization by virtue of mopping up free radicals, and also discourages bubble formation when pouring the gel.

3. Carefully pipet or pour the freshly mixed solution into the chamber without generating air bubbles. Pour to a level about 1 cm below where the bottom of the well-forming comb will come when it is in position. Carefully overlayer the acrylamide solution with butan-2-ol without mixing (to eliminate oxygen and generate a flat top to the gel). Leave the mixture until it is set (0.5–1.5 h).
4. Prepare stacking gel (5 mL) as follows. Mix the following:

Stock acrylamide solution	0.75 mL
Distilled water	3 mL

Degas on a water pump, then add.

Stock stacking gel buffer	1.25 mL
Stock ammonium persulfate solution	15 μ L
TEMED	5 μ L

Mix gently and use immediately.

Pour off the butan-2-ol from the polymerized separating gel, wash the gel top with water and then a little stacking gel mixture, and fill the gap remaining in the chamber with the stacking gel mixture. Insert the comb and allow the gel to stand until set (about 0.5–1 h).

5. When the stacking gel has polymerized, remove the comb without distorting the shapes of the well. Remove the clips holding the plates together, and install the gel in the apparatus. Fill apparatus with reservoir buffer. The reservoir buffer can be circulated between anode and cathode reservoirs, to equalize their pH values. The buffer can also be cooled (by circulating it through a cooling coil in ice), so that heat evolved during electrophoresis is dissipated and does not affect the size or shape of protein zones (or bands) in the gel. Push out the bottom spacer from the gel and remove bubbles from both the top and underneath of the gel, for they could partially insulate the gel and distort electrophoresis. Check the electrical circuit by turning on the power (dc) briefly, with the cathode at the stacking gel end of the gel (i.e., the top). Use the gel immediately.
6. While the gel is polymerizing (or before making the gel), prepare samples for electrophoresis. A dry sample may be dissolved directly in single-strength sample solvent (i.e., the stock solution diluted twofold with water) or dissolved in water and diluted with one volume of stock double-strength sample solvent. The concentration of sample in the solution should be such as to give a sufficient amount of protein in a volume not greater than the size of the sample well. Some proteins may react adequately with SDS within a few minutes at room temperature, but as a general practice, heat sample solutions in boiling water for 2 min. Cool the sample solution before loading it. The bromophenol blue dye indicates when the sample solution is acidic by turning yellow. If this happens, add a little NaOH, enough to just turn the color blue.
7. Load the gel. Take up the required volume of sample solution in a microsyringe or pipet and carefully inject

it into a sample well through the reservoir buffer. The amount of sample loaded depends upon the method of its detection (see below). Having loaded all samples without delay, start electrophoresis by turning on power (dc). On a gel of about 0.5–1 mm thickness and about 14 cm length, an applied voltage of about 150 V gives a current of about 20 mA or so (falling during electrophoresis if constant voltage is employed). The bromophenol blue dye front takes about 3 h to reach the bottom of the gel. Greater voltage speeds up electrophoresis, but generates more heat in the gel.

8. At the end of electrophoresis (say, when the dye front reaches the bottom of the gel), protein bands in the gel may be visualized by staining. Remove the gel from between the glass plates and immerse it in the protein stain immediately (although delay of an hour or so is not noticeably detrimental in a gel of 15%T). The gel is left there with gentle agitation until the dye has penetrated the gel (about 1.5 h for 15%T gels of 0.5–1 mm thickness). Dye that is not bound to protein is removed by transferring the gel to destaining solution. After about 24 h, with gentle agitation and several changes of destaining agent, the gel background becomes colorless and leaves protein bands colored blue, purple, or red. Coomassie Brilliant Blue R250 and PAGE blue 83 each visibly stain as little as 0.1–1 μg of protein in a band of about 1 cm width.

Notes

- 1 The reducing agent in the sample solvent reduces intermolecular disulphide bridges and so destroys quarternary structure and separates subunits, and also oxidizes intramolecular disulfide bonds to ensure maximal reaction with SDS. The glycerol is present to increase the density of the sample, to aid the loading of it onto the gel. The bromophenol blue dye also aids loading of the sample, by making it visible, and indicates the position of the front of electrophoresis in the gel. The dye also indicates when the sample solution is acidic by turning yellow.
- 2 The polymerization of acrylamide and bisacrylamide is

initiated by the addition of TEMED and persulfate. The persulfate activates the TEMED and leaves it with an unpaired electron. This radical reacts with an acrylamide monomer to produce a new radical that reacts with another monomer, and so on to build up a polymer. The bis acrylamide is incorporated into polymer chains this way and so forms crosslinks between them.

3. The gel system described is suitable for electrophoresis of proteins in the M_r range of 10,000–100,000. Smaller proteins move at the front or form diffuse, fast-moving bands, whereas larger proteins hardly enter the gel, if at all. Electrophoresis of larger proteins requires gels of larger pore size, which are made by dilution of the stock acrylamide solution (reduction of %T) or by adjustment of %C (the smallest pore size is at 5%C, whatever the %T). The minimum %T is about 3%, useful for separation of proteins of molecular weights of several millions. Such low %T gels are extremely weak and may require strengthening by the inclusion of agarose to 0.5% w/v. Smaller pore gels, for electrophoresis of small proteins, are prepared by increasing %T and adjustment of %C. Such adjustment of %T and %C may be found empirically to improve resolution of closely migrating species.

A combination of large and small pore gels, suitable for electrophoresis of mixtures of proteins of wide-ranging sizes, can be made in a gradient gel, prepared with use of a gradient-making apparatus when pouring the separating gel (see Chapter 7).

4. Since proteins (or rather, their complexes with SDS) are resolved largely on the basis of differences in their sizes, electrophoretic mobility in SDS gels may be used to estimate the molecular weight of a protein by comparison with proteins of known size [as described in (1)]. However, it should be remembered that some proteins have anomalous SDS-binding properties, and hence anomalous mobilities in SDS gels.
5. If necessary, the gel may be stored for 24 h (preferably in the cold) either as the separating gel only, under a buffer of stock-separating gel buffer diluted fourfold in water, or together with the stacking gel with the comb

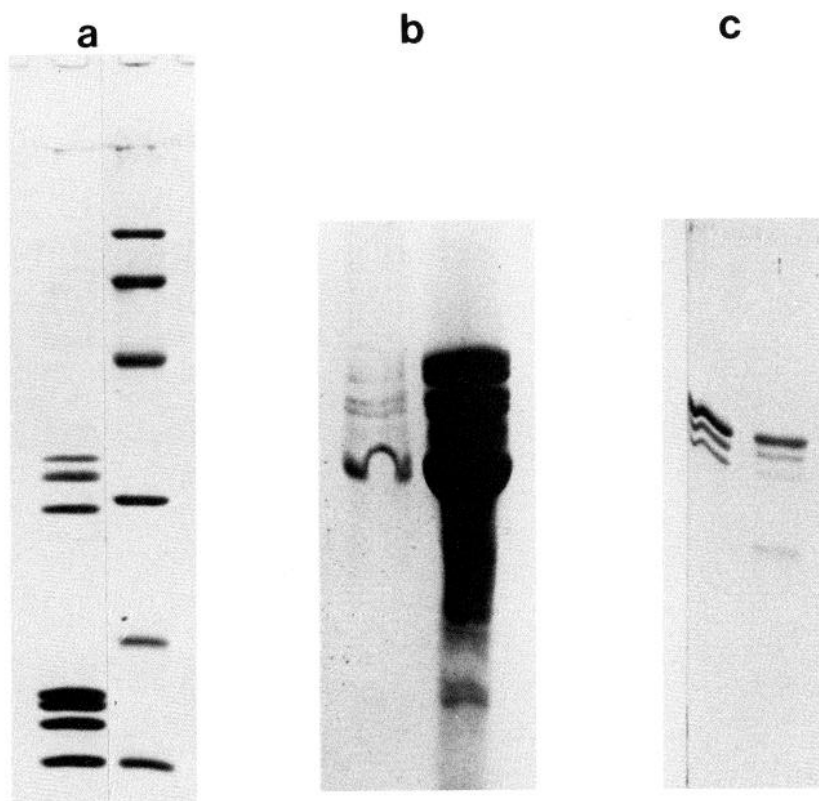


Fig. 2. Examples of proteins electrophoresed on SDS polyacrylamide (15%T) gels and stained with Coomassie Brilliant Blue R 250 as described in the text. Electrophoresis was from top to bottom. (a) Good electrophoresis. Sample, left, 15- μ g loading of histone proteins from chicken erythrocyte nuclei. Sample right, a 5- μ g loading (total) of molecular weight marker proteins (obtained from Pharmacia). The M_s are, from top to bottom: phosphorylase b, 94,000; albumin, 67,000; ovalbumin, 43,000; carbonic anhydrase, 30,000; trypsin inhibitor, 20,100; α -lactalbumin, 14,400. (b) Examples of artifacts. Sample, left, the fastest (bottom) band has distorted as it encountered a region of high polyacrylamide density (which arose during very rapid gel polymerization). Sample, right, the effect on protein overloading of increasing a band's size (the sample proteins are as in sample, left). Extreme overloading may also cause narrowing of faster-migrating bands, as has happened here to some extent (cf. fast bands' widths with widths of bands in sample, left). (c) Example of an artifact. Sample, left, the "end well effect" of distortion of the sample loaded into the very end well, not seen in samples in other wells (e.g., sample, right).

Table 1
Some Problems That May Arise During the Preparation and Use of
SDS Gels

	Fault	Cause	Remedy
(i)	Failure or a decreased rate of gel polymerization	(a) Oxygen is present (b) Stock solutions (especially acrylamide and persulfate) are aged	(a) Degas the solutions (b) Renew the stock solutions
(ii)	Formation of a sticky top to the gel	Penetration of the gel by butan-2-ol	Overlay the gel solution with butan-2-ol without mixing them. Do not leave butan-2-ol to stand on a polymerized gel
(iii)	Poor sample wells		
(a)	The wells are distorted or broken	(a) The stacking gel resists the removal of the comb	(a) Remove the comb carefully or use a gel of lower %T
(b)	The wells contain a loose webbing of polyacrylamide	(b) The comb fits loosely	(b) Replace the comb with a tighter-fitting one
(iv)	Unsatisfactory staining		
(a)	The staining is weak.	(a) The dye is bound inefficiently	(a) Use a more concentrated dye solution, a longer staining time, or a more sensitive stain. The stain solution should contain organic solvent (e.g., methanol), which strips the SDS from the protein to which the dye may then bind

(continued)

Table 1 (*continued*)

	Fault	Cause	Remedy
(b)	The staining is uneven	(b) The dye penetration or destaining is uneven	(b) Agitate the gel during staining and destaining. Increase the staining/destaining time.
(c)	Stained bands become decolorized	(c) The dye has been removed from the protein	(c) Restain the gel. Reduce the destaining time or use a dye that stains proteins indelibly, e.g., Procion Navy MXRB (see Chapter 14).
(d)	The gel is marked nonspecifically by the dye	(d) Solid dye is present in the staining solution	(d) Ensure full dissolution of the dye, or filter the solution before using it.
(v)	Contaminants are apparent	(a) The apparatus and/or stock solutions are contaminated	(a) Clean or renew them as required.
		(b) Nonproteinaceous material in the sample (e.g., nucleic acid) has been stained	(b) Try another stain that will not stain the contaminants.
		(c) Samples have cross-contaminated each other because of their overloading or their sideways seepage between the gel layers	(c) Do not overfill sample wells. Ensure good adherence of the gel layers to each other by thorough washing of the polymerized gel before application of subsequent layers.

(continued)

Table 1 (*continued*)

	Fault	Cause	Remedy
(vi)	Protein bands are not sufficiently resolved	(a) Insufficient electrophoresis	(a) Prolong the run
		(b) The separating gel's pore size is incorrect	(b) Alter the %T and/or %C of the separating gel
(vii)	There are small changes in standard proteins' electrophoretic mobilities from time to time	(a) The amounts loaded differ greatly	(a) Keep the loadings roughly similar in size each time
		(b) The constituents of the gel vary in quality from batch to batch or with age	(b) Use one batch of a chemical for as long as possible Replace aged stock solutions and reagents
(viii)	Distortion of bands		
(a)	Bands have become smeared or streaked	(a) Proteins in the sample are insoluble or remain aggregated in the sample solvent	(a) Use fresh sample solvent and/or extra SDS and reducing agent in it (especially for concentrated sample solutions)
		There is insoluble matter or a bubble in the gel that has interfered with protein band migration	Filter the stock solutions before use and remove any bubbles from the gel mixtures
		The pore size of the gel is inconsistent	Ensure that the gel solutions are well mixed and that polymerization is not very

(continued)

Table 1 (*continued*)

	Fault	Cause	Remedy
			rapid (to slow it down, reduce the amount of persulfate added)
(b)	Protein migration has been uneven (bands are bent)	Part of the gel has been insulated	(b) Remove any bubbles adhering to the gel before electrophoresis
		Electrical leakage	Ensure that the side spacers are in place
		Cooling of the gel is uneven (allowing one part of the gel to run more quickly than another)	Improve the cooling of the gel, or reduce the heating by reducing the voltage or ionic strength of buffers
		The band and/or its neighbors are overloaded	Repeat the electrophoresis, but with smaller loadings. Leave gaps (i.e., unloaded sample wells) between neighboring heavily loaded samples. If necessary, alter the system [e.g. see (vi)] and so the relative mobilities of bands, so that they do not interfere with each other
		The sample well used was at the very end of the row of	Avoid using the end wells

(continued)

Table 1 (*continued*)

Fault		Cause	Remedy
		wells (the "end well effect")	
(c)	Bands are not of uniform thickness	(c) The sample was loaded unevenly	(c) Check that the sample well bottoms are straight and horizontal [see (iii)]

left in place to prevent drying out

6. Proteins dissolved in sample solvent are stable for many weeks if kept frozen (at -10°C or below), although repeated freezing and thawing causes protein degradation
7. The result of electrophoresis in SDS gels ideally has protein(s) as thin, straight band(s) that are well-resolved from other bands. This may not always be so, however. Some faults and their remedies are given in Table 1. Some examples are shown in Fig. 2.
8. Be wary of the dangers of electric shock and of fire, and of the neurotoxic acrylamide monomer.

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Chapter 7

Gradient SDS Polyacrylamide Gel Electrophoresis

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Introduction

The preparation of fixed-concentration polyacrylamide gels has been described in Chapter 6. However, the use of polyacrylamide gels that have a gradient of increasing acrylamide concentration (and hence decreasing pore size) can sometimes have advantages over fixed-concentration acrylamide gels. During electrophoresis in gradient gels, proteins migrate until the decreasing pore size impedes further progress. Once the "pore limit" is reached, the protein banding pattern does not change appreciably with time, although migration does not cease completely. There are three main advantages of gradient gels over linear gels

1. The advancing edge of the migrating protein zone is retarded more than the trailing edge, thus resulting in a sharpening of the protein bands.
2. The gradient in pore size increases the range of molecular weights that can be fractionated in a single gel run.
3. Proteins with close molecular weight values are more likely to separate in a gradient gel than a linear gel.

The usual limits of gradient gels are 3–30% acrylamide in linear or concave gradients. The choice of range will of course depend on the size of proteins being fractionated. The system described here is for a 5–20% linear gradient using SDS polyacrylamide gel electrophoresis. The theory of SDS polyacrylamide gel electrophoresis has been described in Chapter 6.

Materials

1. Stock acrylamide solution (30% acrylamide, 0.8% bis-acrylamide). Dissolve 75 g of acrylamide and 2.0 g of *N,N'*-methylene bisacrylamide in about 150 ml of water. Filter and make the volume to 250 ml. Store at 4°C. The solution is stable for months.
2. Buffers. (a) 1.875M Tris-HCl, pH 8.8
(b) 0.6M Tris-HCl, pH 6.8 } Store at 4°C
3. Ammonium persulfate solution (5%, w/v). Make fresh as required.
4. SDS solution (10% w/v). Stable at room temperature. In cold conditions, the SDS can come out of solution, but may be redissolved by warming.
5. *N,N,N',N'*-Tetramethylene diamine (TEMED).
6. Gradient forming apparatus (see Fig. 1). Reservoirs with dimensions of 2.5 cm id and 5.0 cm height are suitable. The two reservoirs of the gradient former should be linked by flexible tubing to allow them to be moved independently. This is necessary since although equal volumes are placed in each reservoir, the solutions differ in their densities and the relative positions of A and B have to be adjusted to balance the two solutions when the connecting clamp is opened.

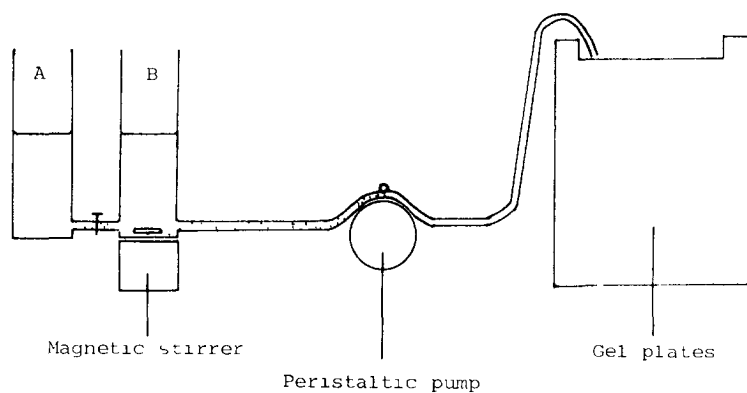


Fig. 1 Diagram of an apparatus for forming gradient gels

Method

1. Prepare the following solutions, A and B.

	Solution A, mL	Solution B, mL
Tris, pH 8.8	3.0	3.0
Water	9.3	0.6
Stock acrylamide, 30%	2.5	10.0
10% SDS	0.15	0.15
Ammonium persulfate (5%)	0.05	0.05
Sucrose	—	2.2 g (equivalent to 1.2 mL volume)

2. Degas each solution under vacuum for about 30 s and then, when you are ready to form the gradient, add TEMED (10 μ L) to each solution.
3. Once the TEMED is added and mixed in, pour solutions A and B into the appropriate reservoirs (see Fig. 1).
4. *With the stirrer stirring*, fractionally open the connection between A and B and adjust the relative heights of A and B such that there is no flow of liquid between the two reservoirs (easily seen because of the difference in densities). Do not worry if there is some mixing between reservoirs—this is inevitable.

5. When the levels are balanced, completely open the connection between A and B, turn the pump on, and fill the gel apparatus by running the gel solution down one edge of the gel slab. Surprisingly, very little mixing within the gradient occurs using this method. A pump speed of about 5 mL/min is suitable. If a pump is not available, the gradient may be run into the gel under gravity.
6. When the level of the gel reaches about 3 cm from the top of the gel slab, connect the pump to distilled water and overlay the gel with 3–4 mm of water.
7. The gradient gel is now left to set for 30 min. Remember to rinse out the gradient former before the gel sets in it.
8. Prepare a stacking gel by mixing the following:

Tris pH 6.8	1.0 mL
Stock acrylamide	1.35 mL
Water	7.5 mL
10% SDS	0.1 mL
Ammonium persulfate (5%)	0.05 mL

9. Degas this mixture under vacuum for 30 s and then add TEMED (10 μ L).
10. Pour off the water overlaying the gel and wash the gel surface with about 2 mL of stacking gel solution and then discard this solution.
11. The gel slab is now filled to the top of the plates with stacking gel solution and the well-forming comb placed in position (*see* Chapter 6).
12. When the stacking gel has set ($\sim$ 15 min), carefully remove the comb. The gel is now ready for running. The conditions of running and sample preparation are exactly as described for SDS gel electrophoresis in Chapter 6.

Notes

1. The total volume of liquid in reservoirs A and B should be chosen such that it approximates to the volume available between the gel plates. However, allowance

must be made for some liquid remaining in the reservoirs and tubing.

2. As well as a gradient in acrylamide concentration, a density gradient of sucrose (glycerol could also be used) is included to minimize mixing by convectional disturbances caused by heat evolved during polymerization. Some workers avoid this problem by also including a gradient of ammonium persulfate to ensure that polymerization occurs first at the top of the gel, progressing to the bottom. However, we have not found this to be necessary in our laboratory.

Chapter 8

Acetic Acid–Urea Polyacrylamide Gel Electrophoresis of Proteins

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Introduction

In SDS polyacrylamide gel electrophoresis, proteins are separated essentially on the basis of their sizes, by the sieving effect of the polyacrylamide gel matrix (*see* Chapter 6). In the absence of SDS, the proteins would still be subject to the sieving effect of the gel matrix, but their charges would vary according to their amino acid content. This is because the charge on a protein at any particular pH is the sum of the charges prevailing on the side chain groups of its constituent amino acid residues, and the free amino and carboxyl groups at its termini (although these are relatively trivial in anything other than a very small peptide).

Thus, in an ionic detergent-free gel electrophoretic system, both the molecular size and charge act as bases for effective protein separation. The pH prevailing in such a system might be anything, but is commonly about pH 3. Since the pK_a values of the side chain carboxyl groups of aspartic and glutamic acids are about 3.8 and 4.2, respectively, even these amino acids will contribute little to the negative charge on a protein at this pH. Thus at pH 3, all proteins are likely to be positively charged and to travel towards the cathode in an electric field.

In such an acid-polyacrylamide gel electrophoresis system, two proteins of similar size but different charge may be separated from each other. Since SDS gels may be unable to achieve this end, these two electrophoresis systems usefully complement each other for analysis of small amounts of proteins. Proteins that might be usefully studied in the acid-gel system are minor primary structure variants (of slightly different charge), or modified forms of the same protein. Thus, a protein that has had some threonine or serine side chains phosphorylated, or lysine side chains acetylated, will be more acidic (or less basic) than the unmodified form of the same protein, and so will have a different electrophoretic mobility in the appropriate acid-gel system (for instance, see the acetylated derivatives of H4 in Fig. 1).

Commonly, the hydrogen bond-breaking agent urea is added to the simple acid-gel electrophoresis system in amounts traversing its entire range of solubility. This denaturant increases the frictional coefficient of proteins and so alters their electrophoretic mobilities. This has often proved useful in obtaining optimal resolution of proteins of interest and so urea is included in the system described below, which uses 2.5M urea. The system is buffered to about pH 3 with acetic acid, and is similar to the system described by Panyim and Chalkley (1).

Materials

1. The apparatus required for running slab gels is available commercially or may be made in the workshop, but is usually of the type described by Studier (2). The gel is cast and used in a chamber formed between two glass plates, as are SDS gels (for further details, see Chapter 6). A dc power supply is required.

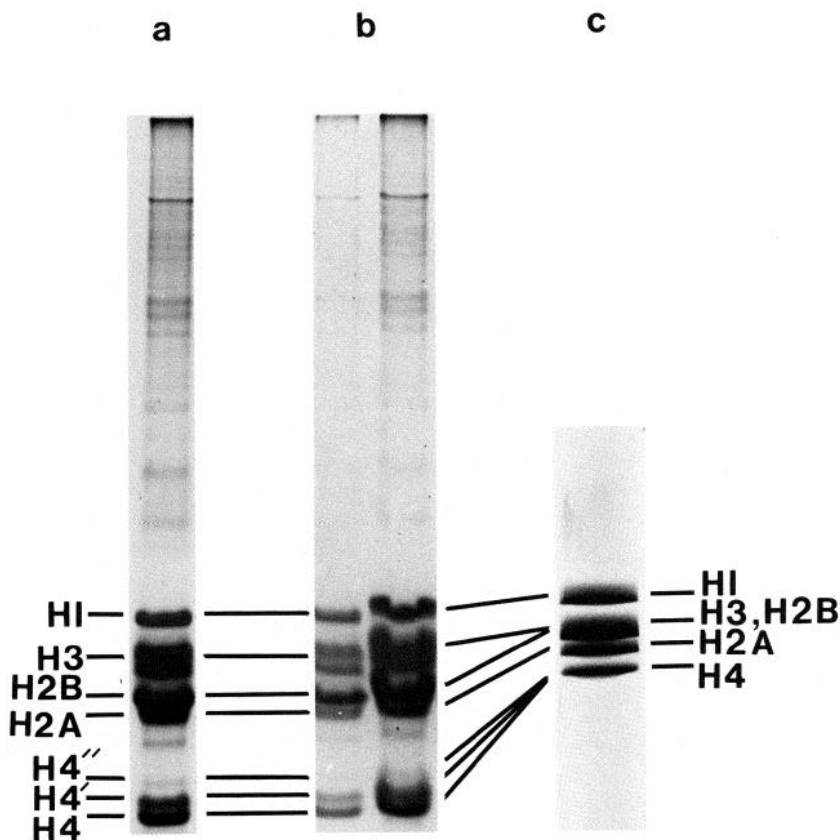


Fig. 1. Examples of electrophoresis on acetic acid (0.9M, pH 3)-polyacrylamide (20%T, 1.5%C) gels. (a) Slab gel containing 2.5M urea, stained with Coomassie Brilliant Blue R250, as described in the text. Sample: 8 μ g of a mouse liver nuclei extract. The histones are identified. The H3 band probably also contains another protein, H1^o. Note the mono- and the faint band of diacetylated forms of H4 (H4' and H4'', resp.) migrating behind the non-acetylated H4. (b) Slab gel containing 2.5M urea, stained with Procion Navy MXRB as described in Chapter 14. Sample: mouse liver nuclei extract. Left, loading 8 μ g; right, loading 24 μ g. Note the different sensitivity of the stain [cf. (a)] and the distortion of bands that occurs with heavier loading. (c) Rod gel (5 mm diameter) containing *no* urea, stained with Procion Navy MXRB as in (10). Sample: 50 μ g of pig thymus histones. Note alteration of H2B mobility relative to other histones because of the omission of urea.

- 2 Stock solutions. Use analytical grade (Analar) reagents and distilled water. Filter stock solutions and warm to room temperature before use.

- (i) *Stock acrylamide solution* (total acrylamide content, %T = 30% w/v, ratio of crosslinking agent to acrylamide monomer, %C = 1.5% w/w).

Acrylamide	73.8 g
Bis acrylamide	1.1 g

Dissolve the above and make to 250 mL in water. Filter the solution before use. This stock solution is stable for weeks in brown glass at 4°C.

- (ii) *Stock ammonium persulfate*

Ammonium persulfate 1 g

Dissolve in 10 mL of water. This stock solution is stable for weeks in brown glass at 4°C.

- (iii) *Reservoir buffer, pH 3* (0.9M acetic acid).

Acetic acid (glacial) 51.5 mL

Make up to 1L with water. Can be stored, but is readily made fresh each time.

- (iv) *Sample solvent*

HCl (1M)	1 mL
β-mercaptoethanol	0.5 mL
Urea	5.4 g
Pyronin Y (0.4% w/v solution in water)	0.5 mL

Add 4.5 mL distilled water to the above and fully dissolve the urea. The final volume is 10 mL. Although it is probably best to use this solution when it is fresh, it may be stored frozen at -20°C for weeks without apparent adverse effect.

- (v) *Protein stain*

Coomassie Brilliant Blue R250	0.25 g
Methanol	125 mL
Acetic acid (glacial)	25 mL
Water	100 mL

Dissolve the dye in the methanol, then add the acid and water. If dissolved up in a different order the dye's staining behavior may differ. The stain is best used when freshly made. For best results do not reuse the stain—its efficacy declines with use. If this dye is not available use the equivalent dye PAGE blue 83.

(vi) *Destaining solution:*

Methanol	100 mL
Acetic acid (glacial)	100 mL
Water	800 mL

Mix thoroughly. Make when required and use fresh.

Method

1. Assemble clean glass plates and spacers into the form of a chamber (as described for SDS gels, Chapter 6), and seal it with molten white petroleum jelly. Clamp it in an upright, level position.
2. Prepare a sufficient volume of separating gel mixture (30 mL for a chamber of about $14 \times 14 \times 0.1$ cm allows for some wastage), as follows. Mix the following:

Stock acrylamide solution	20 mL
Urea	4.5 g
Glacial acetic acid	1.57 mL
<i>N,N,N',N'</i> -tetramethyl- ethylenediamine (TEMED)	100 μ L

Make to 29.45 mL with distilled water, degas on a water pump, then add stock ammonium persulfate solution (0.55 mL).

Mix gently and use immediately because polymerization starts when the persulfate is added. The degassing removes oxygen, which can inhibit polymerization (by virtue of mopping up free radicals), and also discourages bubble formation when pouring the gel.

3. Carefully pipet or pour freshly made solution into the chamber and remove any bubbles present. Pour to a level about 0.5 cm below where the bottom of the well-forming template ("comb") will come when it is in position. Carefully overlay the acrylamide solution with butan-2-ol, without mixing. This insulates the solution

from oxygen and generates a flat top to the gel. The acrylamide should polymerize in an hour or so at room temperature.

4. Prepare the upper gel layer (5 mL) as follows. Mix the following:

Stock acrylamide solution	1.25 mL
Urea	0.75 g
Glacial acetic acid	78.2 μ L
TEMED	25 μ L

Make to 4.75 mL with water, degas on water pump, then add stock ammonium persulfate solution (0.24 mL). Mix gently and use immediately. Pour off the butan-2-ol from the polymerized separating gel, wash the gel top with water and then a little of the upper gel mixture, and then fill the gap remaining in the chamber with upper gel mixture. Insert the well-forming comb and allow to stand until set (about 0.5–1 h).

5. When the upper gel has set, remove the comb without breaking or distorting the sample wells. Install the gel in the apparatus and fill the reservoirs with buffer (0.9M acetic acid). The gel may be run at room temperature without buffer circulation or cooling. Cooling tends to slow up the rate of electrophoresis and so the advantage of the decreased rate of protein diffusion (which causes band widening) is counteracted by increased diffusion occurring during the longer time required to complete the run. Push out the bottom spacer from the chamber and, with the cathode at the bottom end of the gel, turn on the dc power supply, to give about 180 V. Continue this electrophoresis without added samples ("pre-electrophoresis") at a constant voltage until the current falls to a steady level (say, from 25 to 5 mA for a 0.5–1 mm-thick slab of this sort), or alternatively, at constant current until the voltage increases to a steady level. This process may take about 5 h, but may be done conveniently overnight. The pre-electrophoresed gel may be stored under fresh 0.9M acetic acid for at least several days at room temperature.
6. While the gel is polymerizing, prepare samples for electrophoresis. Dry samples will be dissolved directly in sample solvent or aqueous solutions may be diluted

with not less than one volume of sample solvent. The concentration of protein in the solution should be as great as possible so that the volume of solution loaded onto the gel is as small as possible.

7. Prepare to load the gel. Use fresh reservoir buffer in the apparatus. Take up the required volume of sample solution in a microsyringe or pipet and carefully inject it into a sample well, loading it through the reservoir buffer without mixing. The amount of sample loaded depends partly on the sensitivity of the method of detection (see below), but in any case it is often found that good, even, and straight bands are obtained most frequently when samples are lightly loaded, say about 1 $\mu\text{g}/1$ cm wide band in a 0.5–1 mm thick gel. Having loaded the samples without delay, start electrophoresis at 180 V. The pyronin Y will take 4–5 h to migrate to the bottom of a thin, 14 cm-long gel of the type described here. Decreasing the voltage will prolong the run, whereas an increased voltage will generate more heat, which may not benefit the appearance of the protein bands.
8. At the end of electrophoresis, remove the gel and immerse it in protein stain for at least several hours with gentle agitation. The dye that is then not bound to the protein bands may be removed by washing the gel in several changes of destaining solution. After destaining, protein bands are seen to be colored blue to purplish-red. Excessive destaining will decolorize these protein bands, but they may be restained. Coomassie Brilliant Blue R250 and PAGE Blue 83 each visibly stain as little as 0.1–1 μg of protein in a 1 cm-wide band.

Notes

1. The purpose of the upper, weak polyacrylamide, gel is to provide a medium in which sample wells can be formed and from which the well-forming comb can be readily removed (20%T gel tends to break when the comb is removed). Since the upper gel contains weaker acid than does the separating gel, its pH is slightly higher. However, the purpose of this design

is that the upper gel has lower conductivity than the rest of the system and when the electric field is applied this has a small band-sharpening effect similar to that produced by the "stacking gel" used in SDS gels (*see* Chapter 6).

2. A more typical stacking gel for 0.9M acetic acid/urea gels has been described by Spiker (3). It has been found in this laboratory that although bands may have an improved appearance, they may instead become smeared.
3. Addition of TEMED and persulfate to the gel mixture initiates its polymerization. This occurs by their interaction and formation of a TEMED radical that reacts with an acrylamide monomer. This in turn produces a radical that reacts with another acrylamide monomer, or occasionally one half of a bis acrylamide molecule. Incorporation of bis acrylamide molecules into different chains forms crosslinks between them.
4. The 9M urea in the sample solvent has two functions. First, to disrupt aggregates, and second, to increase the density of the solution (which aids in the loading of the sample beneath the less dense reservoir buffer). The β -mercaptoethanol that is also present reduces inter- and intramolecular disulfide bonds and so, together with the urea, destroys higher-order protein structures. The pyronin Y dye is present to aid sample application by making it visible, and also to mark the approximate position of the front of electrophoresis, near which it runs.
5. The "pre-electrophoresis" treatment of the gel before addition of the samples removes persulfate and other ions that would otherwise slow up the rate of sample electrophoresis and also spoil the resolution of protein bands.
6. As mentioned above, protein bands in this system are often slightly misshapen, and this problem is exacerbated by heavy loading (*see* Fig 1) Thus, slab gels may give unsatisfactory resolution for quantification purposes (*see* Chapter 14). As an alternative, gels may be made in the forms of rods by casting them in tubes of glass. The result has the proteins as discs running through the rod (e.g., *see* Fig 1) These discs suffer less from distortion than do bands on slabs. The

separating gel for rod gels is prepared as for slab gels, and polymerized in glass tubes that have been siliconized before-hand. No upper gel is added, but samples are applied directly onto the separating gel. The gel may be extruded from the glass tube, as described in (4), or removed after cracking the glass in a vice or with a hammer. The apparatus required for running rod gels is available commercially or may be made to a simple design (*see ref. 4*).

7. The 20%T gel described is suitable for electrophoresis of smaller proteins (say, of M_r below 50,000). For larger proteins a gel of larger pore size is more convenient by virtue of allowing greater mobility. The pore size may be increased by reduction of %T by simple dilution of the stock acrylamide solution, or by adjustment of %C (which gives the smallest pores when at 5%). It may be found that adjustment of the gel's pore size in this way may also improve the resolution of proteins of interest.
8. If the resolution of the proteins of interest is not good enough, then alteration of pH of the system may be beneficial, by virtue of altering their respective charges by titration of the side chain groups (e.g., *see ref. 1*). The literature describes polyacrylamide gel systems of various pH values, e.g., pH 4.5 (5,6); pH 7.1 (7), pH 8.9 (8).
9. The concentration of urea in the gel may be altered, to alter the relative mobilities of proteins in the system. The effects of altering the urea concentration have to be determined empirically. An example is shown in Fig. 1, in which it may be seen that in the absence of urea (as in 1M urea) histone proteins H2B and H3 comigrate, whereas in 2.5M urea they are resolved from each other (*see also ref. 1*).
10. In addition to urea, the non-ionic detergent Triton X 100 (or Triton DF-16) may be added to the gel (9). This agent binds to proteins in proportion to their hydrophobicity, and alters their electrophoretic mobilities accordingly. This technique has proved useful in the study of proteins that differ slightly in their hydrophobic character and that are not separated by ordinary acid/urea or SDS gel electrophoresis.

11. The acid-urea system described may be adapted for separation of nondenatured proteins, which can be detected in the gel by their enzymatic properties and prepared (undenatured) from the gel. For this purpose, denaturants (urea and Triton) are omitted and instead of the sample solvent given above, with its urea and reducing agent, the 0.9M acetic acid, 30% w/v sucrose sample solvent of Panyim and Chalkley (1) may be used.
12. Coomassie Brilliant Blue R250 may not fully penetrate to and stain the center of bands of concentrated protein, especially in thicker slabs and rod gels. Procion Navy MXRB does not suffer this defect, although it is several-fold less sensitive (*see Fig. 1*). The method of staining with Procion Navy described by Goodwin et al. (10) is suitable for rod gels, but the heated destaining process is inconvenient for slabs. However, the method described in Chapter 14 gives an equivalent result and is suitable for all forms of gel.
13. Various problems may arise to spoil the otherwise perfect gel. Failure of the gel to polymerize, distortion of sample wells and other problems that are general to the process of polyacrylamide gel electrophoresis are dealt with in Chapter 6. The main failing with the present system in particular is the common, uneven shape of the protein bands especially in conditions of heavy loading (e.g., *see Fig. 1*). As mentioned above, use of a stacking gel may alleviate this problem, but generally speaking light loading (about 1 μ g/band or less) and small sample volumes go a long way to promote good results. Avoid using the sample wells at the very ends of the row of wells, for samples in these often suffer distortion during the electrophoresis.
14. Be wary of the dangers of electric shock and of fire, and of the neurotoxic acrylamide monomer.

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Chapter 9

The In Vivo Isotopic Labeling of Proteins for Polyacrylamide Gel Electrophoresis

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Introduction

Autoradiography (*see* Chapter 17 and Vol. 2) offers a convenient, quick, and cheap means of quantifying proteins separated by gel electrophoresis

For metabolic experiments proteins must be labeled with a radioactive isotope *in vivo* prior to isolation and subsequent electrophoretic analysis. The isotope chosen, of course, must correspond to the question to be investigated, but they are generally ^3H , ^{14}C , ^{32}P , and ^{35}S , which are all beta-emitters. The energies and half lives are:

0.018 MeV (12.43 yr), 0.159 MeV (5600 yr), 1.706 MeV (14.3 d), 0.167 MeV (87.4 d), respectively. ^3H is a popular isotope for biological use, potentially available at high specific activity, but its low energy precludes its autoradiography directly and it needs to be either treated for fluorography or quantified by gel slicing and scintillation counting. Thus the higher energy ^{14}C - and ^{35}S -labeled amino acids are commonly used as protein labels since they can be directly detected by autoradiography. However, the low specific activity of ^{14}C labeled amino acids often prevents their use, with the result that ^{35}S -methionine or ^{35}S -cysteine labeling, either alone or in combination, are more commonly used. Nevertheless, it is worth noting that, in a recent paper, Bravo and Celis (1) were able to detect 28% more proteins by labeling with a mixture of 16 ^{14}C -amino acids than could be detected with ^{35}S -methionine and that such labeling also allows the long term storage of data on gels. Unfortunately, the cost in isotopes is often prohibitive. In this paper methods for labeling both cells in culture and tissues in vitro with ^{35}S -methionine, ^3H or ^{14}C -amino acids, and ^{32}P for phosphoproteins are described, with particular emphasis on sample preparation for two-dimensional polyacrylamide gel electrophoresis (Chapter 10). But these techniques may easily be modified to encompass other uses, such as the isotopes ^{59}Fe , ^{125}I , or ^{131}I for specific studies of particular proteins.

Materials

1. Tissue culture medium: I use a medium rich in amino acids, alpha—minimal essential medium, but lacking in methionine and stored at 4°C . Any defined tissue culture medium may be used.
2. Dialyzed fetal calf serum (DFCS): serum is serially dialyzed against two changes of phosphate-buffered saline to remove amino acids and stored at -20°C . Do not store this at 4°C , since endogenous proteolytic activity will result in a relatively high concentration of amino acids.
3. Ca^{2+} and Mg^{2+} -free phosphate-buffered saline (PBS) (0.14M NaCl, 2.7 mM KCl, 1.5 mM KH_2PO_4 , 8.1 mM Na_2HPO_4).

4. 0.1% (w/v) trypsin in PBS citrate (PBS containing 20 mM sodium citrate).
5. Lysis buffer: 9.5M urea, 2% v/v Nonidet, 2% v/v ampholines (1.6%, pH range 5–7; 0.4%, pH range 3.5–10), 5% v/v beta-mercaptoethanol. Stored frozen at -20°C in 0.5 mL aliquots (*see* Chapter 10).
6. Isotopes: ^{35}S -Methionine, ^{32}P -orthophosphate, and ^3H -amino acids are used as supplied. ^{14}C -amino acids are lyophilized and resuspended in a medium lacking amino acids at 500 $\mu\text{Ci/mL}$.

Method

Cells in Monolayers

1. Cells are plated directly into microwells at about 2000 cells/well in 0.25 mL of growth medium. They are left to attach for at least 5 h, but preferably overnight.
2. The media is removed and the cells washed with 0.5 mL medium lacking methionine and replaced with 0.1 mL medium supplemented with 10% v/v dialyzed fetal calf serum, lacking methionine, but containing 100 μCi ^{35}S -methionine and 1 mg/L unlabeled methionine.
3. The cells are labeled for 20 h at 37°C (wrap the microtiter plates in cling film to prevent evaporation).
4. Following incubation, the medium is removed, the cells washed twice with PBS to remove serum proteins and, if a total cell extract for two-dimensional gel electrophoresis is required, the cells are lysed with 20 μL lysis buffer. The samples may be stored in small vials or in the microtiter wells at -70°C , or loaded directly onto isoelectric focusing gels (*see* Chapter 10).
5. Alternatively, cells may be trypsinized and processed according to the analytical technique required.
6. Cells may also be adequately labeled for up to about 4 h in a medium completely lacking unlabeled methionine. But under these circumstances, equilibrium labeling may not be achieved.
7. It is difficult to maintain tight physiological control of cells growing in microwells and thus in some experiments where better control of cell physiology is required, cells growing in 25 mL flasks may be labeled in

2 mL of medium containing 10% DFCS and methionine at 1 mg/L containing 100–200 μCi of ^{35}S -methionine. Bottles are occasionally agitated to prevent desiccation of the cells. After labeling, the cells are washed twice with PBS and may be collected either by scraping with a rubber policeman, or trypsinized and processed as before

Cells Growing in Suspension

1. Cells growing in suspension are collected by centrifugation, washed in methionine-free medium, and regained by centrifugation
2. The cells are resuspended at about 25% of their saturation density in 15 mL Falcon plastic snap-cap tubes previously gassed with a 5% CO_2 –95% air mixture in medium containing 10% DFCS lacking methionine, but supplemented with 400 $\mu\text{Ci/mL}$ ^{35}S -methionine and agitated with a small magnetic flea.
3. Cells are labeled for 1–4 h at 37°C , collected by centrifugation, washed twice with PBS, and processed as before.

Tissues

1. The tissue to be labeled is excised, blotted briefly onto filter paper, and chopped into small pieces (approximately 2 mm diameter).
2. These pieces are placed into glass scintillation vials, gassed with a 5% CO_2 –95% air mixture with 1 mL of tissue culture medium containing 10% dialyzed fetal calf serum, but lacking methionine supplemented with 200 $\mu\text{Ci/mL}$ ^{35}S -methionine.
3. The vials are placed in a shaking water bath at 37°C and labeling is performed for 1 h; thereafter, the sample is processed as appropriate.

Notes

1. Accurate protein synthetic rates over short periods may be determined concurrently in parallel flasks by measuring the rate of incorporation of a mixture of

three ^{14}C -labeled essential amino acids into acid insoluble counts per cell.

2. The handling of cells to be labeled with ^3H or ^{14}C -amino acids or ^{32}P -orthophosphate, is identical. The ^3H or ^{14}C -amino acids are exposed to cells in 0.1 mL of medium at 500 $\mu\text{Ci/mL}$ in medium lacking the appropriate amino acids or to ^{32}P -orthophosphate at 2 mCi/mL in medium lacking phosphate.
3. To determine the number of counts incorporated, a small aliquot (1–5 μL) should be removed from the final sample, added to 0.5 mL of water containing 10 μg of bovine serum albumin, and precipitated with 0.5 mL of ice-cold 20% (w/v) TCA. The precipitate is allowed to develop for 10 min and then collected onto Whatman glass fiber discs with four washings of 5 mL of 5% (w/v) TCA. The discs are finally washed with ethanol, dried, and either digested overnight with a tissue solubilizer for ^3H samples, and counted in a compatible scintillant, or counted directly in a scintillation counter. ^{32}P may alternatively be determined by Cerenkov counting.
4. It must be remembered that mammalian cells show large changes in the synthesis of proteins when growth conditions are changed. Thus, even amino acid deprivation results in a reduction of the initiation rate of ribosomes onto mRNA, with the result of preferential synthesis of proteins whose mRNAs have high intrinsic rates of initiation. It is also worth repeating that accurate protein synthesis measurements may be achieved only when very small amounts of radioactive precursors are added to reduced volumes of the same conditioned growth medium containing large amounts of unlabeled precursors that has been removed from the growing culture (*see* ref. 2 for a full discussion). These above conditions are clearly not met when fresh medium, often lacking methionine, is used to label proteins for gel electrophoresis. Considerable care should therefore be exercised in standardizing growth and labeling conditions for any experiments involving comparison of different samples. Often a compromise has to be effected between high levels of incorporation and the physiological constraints of maintaining precursor pool sizes

5. The only other problems that may be encountered, providing care is taken over sterility, pH, and temperature regulation, is the toxicity of isotopes. This is rarely a problem for a day's labeling, but could potentially be so if proteins are labeled for longer. Fibroblasts will survive in 0.5 mCi ^{32}P /mL in medium containing 0.2 mM phosphate and in mutant isolation, 1.5 dpm of ^3H -amino acid incorporation per cell is considered lethal, but only after a period in the cold for accumulation of radioactive damage. Thus, toxicity will not be a problem using the above labeling procedures.
6. The procedures given above will result in ^{35}S -methionine-labeled proteins with specific activities of around 10^5 cpm/ μg of protein. But obviously the optimal conditions and resultant specific activity of proteins will depend on the cell type and growth conditions. Nevertheless, the methods are applicable to a wide range of cells in culture and tissues. It is also worth noting that, following slab gel electrophoresis, a 1 mm² spot containing 3 dpm of ^{35}S -methionine may be readily detected (>0.01 OD above background) after 1 wk.

Acknowledgments

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Chapter 10

Two-Dimensional Polyacrylamide Gel Electrophoresis of Proteins

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Introduction

Since O'Farrell (1) introduced the improved technique for high resolution two-dimensional polyacrylamide gel electrophoresis (2-D PAGE), it has become one of the most powerful tools for the separation and quantification of proteins from complex mixtures. The principal reason for this is that the method employs separation of denatured proteins according to two different parameters, molecular weight and isoelectric point. Consequently, it has sufficient resolution to separate individual proteins as discrete

spots on the gel. Each parameter may also be varied and therefore, with the modification of non-equilibrium pH-gradient electrophoresis (NEPHGE) to analyze basic proteins (2), almost any polypeptide may be investigated. Thus to date, the O'Farrell 2-D gel system has no serious rivals, with the possible exception of the Kaltschmidt and Wittmann (3) gel system for analyzing ribosomal proteins. Ribosomal proteins, however, may be adequately separated with NEPHGE.

It has been estimated from mRNA complexity studies that individual mammalian cells may contain 10,000 polypeptides ranging in abundance from 10^9 copies/cell to a few hundred. However, by 2-D PAGE analysis only about 1800 individual proteins have so far been detected (4), even after long exposures. This is usually interpreted as the inability to detect minor proteins, but Duncan and McConkey (5) have argued that in fact 2000 is close to the number of proteins in a cell and that the remaining rare mRNAs are rarely, if ever translated. If this is the case, then 2-D PAGE represents an even more powerful technique than previously expected for investigating changes in cellular physiology. However, these investigations are still limited because, of these two thousand proteins, only a few have been positively identified, although protein maps of individual cell types and sera have been published (4). Nevertheless, with subcellular fractionation, antibody detection techniques, and the use of peptide mapping, which has been performed on proteins representing as little as 0.01% of the total protein, it is to be expected that a substantial increase in the number of proteins identified will soon occur, enabling rapid advances in the study of cell biochemistry.

Because of its resolution, the 2-D PAGE technique has been applied to a great number of biological problems ranging from the analysis of proteins in different tissues, under various hormonal states and at different stages of development, to the analysis of cells in culture, and finally to the analysis of polypeptides within a single cell. For these studies of different cellular states, the only technique that may rival the 2-D PAGE method is that of Rot curve analysis of mRNA populations, which in terms of the ability to detect differences (although not in the identification of

proteins) has a similar degree of resolution (5). Other powerful applications of the 2-D PAGE system include the detection of proteins that contain single amino acid substitutions, which confer a change in isoelectric point on the protein. This has resulted in definitive identification of mis-sense mutations within proteins (1) and the visualization of mistranslated proteins (6).

The future promises not only further investigations of different metabolic states, but also definitive protein maps, stored as computer matrixes, of cells and sera from a variety of pathological states. These, with the development of specialized equipment to analyze thousands of gels a year in a single laboratory, may enable rapid clinical diagnosis of diseases by comparison of clinical samples to the computerized 2-D PAGE record. Similarly, computer data bases will enable searches for information about the distribution and occurrence of different proteins in hundreds of cell types analyzed at different stages of development, hormonal stimulation, and transformation. However, as yet we await the standardization of these techniques of protein nomenclature, and of rapid, reliable computer analyses (7).

Materials

One-Dimensional Isoelectric Focusing

1 Lysis buffer.

- 9.5M urea
- 2% v/v Nonidet P-40 (NP-40)
- 2% v/v Ampholines (1.6%, pH range 5-7, 0.4%, pH range 3-10)
- 5% v/v β -mercaptoethanol

This buffer may be stored frozen at -20°C for long periods of time in 0.5 mL aliquots. But do not continually freeze and thaw it. Use an aliquot once and discard the remainder.

2. 30% Acrylamide stock solution

- 28.38% w/v acrylamide
- 1.62% w/v N,N' -methylene-bisacrylamide

All acrylamide solutions are light sensitive and should be stored in the dark at 4°C. It is usable for at least 1 month. Acrylamide is a potent neurotoxin, so caution should be exercised to protect against contact with the dust. It should *never* be mouth-pipeted.

3. 10% w/v Nonidet in water
4. Anode electrode solution.

0.01M phosphoric acid

This should be made fresh from a 1M stock (63.64 mL phosphoric acid to 1 L water)

5. Cathode electrode solution:

0.02M sodium hydroxide

This should be made fresh from a 1M stock and degassed

6. Sample overlay buffer:

8M urea

1% Ampholines (0.8% v/v, pH range 5–7, 0.2% v/v, pH range 3–10)

This solution may be stored in frozen aliquots.

7. Sodium dodecyl sulfate (SDS) sample buffer.

0.06M Tris-HCl, pH 6.8 at 20°C

2% w/v SDS

5% v/v β-mercaptoethanol

10% v/v glycerol

8. Ampholines are used as supplied in 40% solutions. They should be kept sterile and stored at 4°C.

9. Ammonium persulfate:

10% w/v solution made up fresh every week and stored at 4°C

10. Pancreatic ribonuclease (50 mg/mL) and deoxyribonuclease (1 mg/mL) in sonication buffer. Store frozen in aliquots.
11. Sonication buffer

0.01M Tris-HCl, pH 7.4 at 20°C

0.005M MgCl₂

Two-Dimensional SDS-Slab Electrophoresis

12. 30% Acrylamide solution:

29.2% w/v acrylamide
0.8% *N,N'*-methylene-bisacrylamide

13 Running buffer:

0.025M Tris base
0.192M glycine
0.1% w/v SDS

For 3 L: 43.2 g glycine, 9.0 g Tris base, and 3.0 g SDS gives the correct pH, do not titrate this or add any salt.

14 Separating gel buffer:

1.5M Tris-HCl, pH 8.8, at 23°C
0.4% w/v SDS

15 Stacking gel buffer.

0.5M Tris-HCl, pH 6.8, at 23°C
0.4% w/v SDS

16. Bromophenol blue

0.1% w/v bromophenol blue in water

Solutions 2, 3, 7, 8, 9, 11, 12, 14, and 15 are stored at 4°C, and 4, 5, 13, and 16 at room temperature. All other solutions are stored as indicated.

Method

Sample Preparation

The principle to follow here is to avoid causing any chemical modification of proteins since any charge change will be detected on the gel and will result in an aberrant pattern. The following method should give satisfactory results

1. Pellets of cells should be resuspended in 100 μ L or less of sonication buffer at 4°C. They are sonicated with the microtip of an 150W MSE sonicator for 6 5-s bursts at 8 μ m peak to peak. Be very careful not to overheat the sample.
2. Add 2 μ L of RNase and DNase to 100 μ L and incubate for 5 min at 4°C.
3. Add solid urea to bring the sample to 9M (1 mg urea/ μ L of sample), which approximately doubles the volume. Add to this mixture one volume of lysis buffer, take it off the ice, and solubilize the urea with the palm of the hand. Do not overheat since this can result in modifications of the protein. Samples can now be used directly or stored at -70°C until needed.

One-Dimensional Gels (Isoelectric Focusing)

1. Standard glass tubes, 1–1.5 mm internal diameter and 10–15 cm long are thoroughly cleaned in chromic acid (5% Decon may be used, but beware of precipitation).
2. The tubes are rinsed in water, placed in fresh KOH in ethanol (0.4 g KOH to 20 mL ethanol) for 20 min, rinsed thoroughly first with distilled water, then ethanol, and finally allowed to air dry.
3. When dry, the tube bottoms are sealed with three layers of parafilm and lined up vertically around a 250 mL beaker with elastic bands. The tubes are marked to the same point with a felt-tipped pen to ensure that gel lengths are uniform (this is important to facilitate reproducibility between runs).
4. To make 10 mL of isoelectric focusing gel solution add to a 125 mL flask

Urea	5.5	g
Acrylamide stock (2)	1.33	mL
NP-40 (3)	2.0	mL
Water	1.7	mL
Ampholines (pH 5–7)	0.6	mL
Ampholines (pH 3–10)	0.115	mL

Gels may be made easier to remove by increasing the nonidet concentration by 10%.

- 5 To make non-equilibrium polyacrylamide gels (NEPHGE), weigh out the same quantity of urea, add the same volume of acrylamide stock and NP-40, but add 2.0 mL of water, 0.25 mL ampholine (pH 7–9), and 0.25 mL ampholine (pH 8–9.5).
6. Dissolve the urea by swirling in a water bath whose temperature is set not higher than 37°C and then briefly degas under vacuum.
7. Add 7 μL (IEF) or 14 μL (NEPHGE) of TEMED and 10 μL (IEF) or 20 μL (NEPHGE) of 10% ammonium persulfate solution. You now have about 15 min to pour the gels at normal room temperature (20°C), but if room temperatures are substantially higher, reduce the TEMED concentration to about two-thirds.
8. Using a long narrow-gage hypodermic needle (or Pasteur pipet) fill the tubes to about 1 cm from the top, being careful to avoid trapping air bubbles (these may be removed by carefully tapping the tubes).
9. Overlay the gel mix with 10 μL of water, allow it to polymerize for 1 h, and then carefully remove parafilm to avoid damage to the bottom of the gel. The parafilm may be replaced by dialysis tubing clipped on with a rubber O-ring to prevent the gel slipping out, but generally this precaution is unnecessary.
10. The gels are placed in the electrophoresis tank (we use an E-shaped tank rather than the conventional round one, since these are easy to manufacture and operate, and can be fitted with locking devices to prevent electric shocks to the operator).
11. The lower chamber is filled with 10 mM phosphoric acid (IEF) or 20 mM NaOH (NEPHGE), and any trapped air bubbles removed from the end of the gels by a gentle stream of fluid using a bent syringe.
12. For IEF, the water is removed from the top of the gel with a Pasteur pipet and replaced with 10 μL lysis buffer, 10 μL overlay buffer and 20 mM NaOH to fill the tubes.
13. The cathode solution is added carefully to the upper chamber and connected to the cathode and the lower chamber to the anode and the gels pre-run at 200 V for 15 min, 300 V for 30 min, and 400 V for 1 h or more. At the end of this procedure, there should not be more than 1.5 mA/5 tubes or 2 mA/12 tubes. Do not cool the

gels during this procedure since the urea will crystallize out

14. After the pre-run remove the NaOH from the upper chamber carefully so as to avoid contact with the gel surface and discard. Remove the liquid from above the gels and wash the tops with three washes of 20 μ L of water
15. The samples are loaded in a volume of 5–50 μ L with a syringe and overlaid with 10 μ L sample overlay buffer and 20 mM NaOH to fill the tubes
16. The upper electrophoresis chamber is filled with 20 mM NaOH, reconnected to the anode and electrophoresed for 18 h at 400 V. One hour before the termination of the run, turn the voltage up to 1000 V to increase the band sharpness. But do not exceed 10,000 V/h since bands will become distorted.
17. Turn off the power-pack and wait a few seconds, then remove the tubes and force the gels out onto parafilm troughs with a syringe full of water connected to the tubes via a flexible plastic tubing.
18. Put the gels (using the parafilm to handle them) into capped tubes containing 5 mL of sample buffer with 0.002% bromophenol blue. Leave for minimally 20 min without agitation and at this point the gels may be stored at -70°C indefinitely. After defrosting, it is advisable to exchange the sample buffer and leave for a further 30 min before loading onto the second dimension, by which time the second dimension slab gel will have been prepared.
19. For NEPHGE there is no pre-running since a stable pH gradient is not formed; thus samples are loaded directly onto the gels, overlaid with 10 μ L of sample overlay buffer and 10 mM phosphoric acid to fill the tubes. The upper reservoir is filled with 10 mM phosphoric acid and connected to the anode and the gels are run at 400 V for 4–5 h. However, since this is a non-equilibrium procedure, very basic proteins (for example, some ribosomal proteins or histones) may migrate off the end. Thus shorter total voltage-hours need to be used.
20. Investigators may wish to test their procedures at this point and gels may be stained as described in Chapter 6. The bands should be razor sharp. pH gradients

may also be tested by slicing the gel into 0.5 cm pieces, then equilibrated for 1 h in degassed water, and the pH is read using a microelectrode on pH meter. Similarly, a pH contact electrode may be used or alternatively visible pH markers may be bought from a range of suppliers. pH ranges should be from about 3.5 to 7.5 for the IEF gels.

Two-Dimensional SDS–Slab Gel Electrophoresis

- 1 Assemble the slab gel apparatus (we use the Bio-Rad/Hoeffler design, which gives consistently good results, but any homemade slab gel system is suitable), ensuring that there is a good seal. Make it level and vertical. The plates should have been thoroughly washed in 5% Decon, rinsed with water and ethanol, and air dried. Do not wipe with tissues that leave lint since this will interfere with gel polymerization.
2. The SDS separating gel is made as follows for 100 mL of a 10% gel.

Acrylamide stock (12)	33.3 mL
Water	41.7 mL
Separating gel buffer	25.0 mL

The solutions are mixed and degassed under vacuum.

- 3 Fifty μL of TEMED and 333 μL of 10% ammonium persulfate are added and the gel is poured by pipeting the solution down the side of the gel plates to about 2.5 cm from the top. It is overlaid with deionized water applied at one end and left to polymerize for 1 h until the gel interface can be seen as a sharp straight boundary. The gel may be left to stand overnight.
- 4 The water overlay is removed, the gel surface is washed with water, and the stacking gel poured. This is prepared fresh and to make 25 mL of solution add

Acrylamide stock (12)	4.0 mL
Water	14.75 mL
Stacking gel buffer	6.25 mL

- 5 Briefly degas, add 25 μL TEMED and 87.5 μL ammonium persulfate, and pour (taking care to avoid

trapping any air bubbles) up to a level 1 mm below the gel plate edge before inserting a Teflon edge. Overlay with water and allow this solution to polymerize for 30 min.

6. Remove the Teflon strip, rinse the surface of the gel with water, and load the first dimension on top of the second
7. Take the defrosted, re-equilibrated gel and straighten it in a trough of parafilm. Remove the liquid and apply it directly to the top of the stacking gel. Press down gently with a curved spatula tip from one end to ensure the removal of air bubbles, but be careful not to stretch the gel since this is one stage where variability can be introduced into the procedure.
8. If the gel adheres; however, the more conventional but slower method is to overlay the gel with 1 mL 1% agarose in SDS sample buffer (not too hot) and then to allow it to set for 5 min. If bromophenol blue had not been included in the sample buffer, add a few drops of 0.1% bromophenol blue over the total length of the gel.
9. The slab gel system is now assembled, the reservoirs filled with running buffer, and any air bubbles trapped under the gel removed with a bent syringe needle or Pasteur pipet. The lower gel reservoir is connected to the anode and the upper reservoir to the cathode, and the gels are then electrophoresed at 9 mA/gel overnight or at 20 mA/gel for 4–5 h until the bromophenol blue reaches the bottom of the gel. It is essential to avoid overheating and thus if the gels are run fast, the whole lot can be run in the cold room or, alternatively, an efficient cooling system may be incorporated into the gel apparatus. After 1–2 h, change the upper reservoir buffer (or alternatively, continuously mix the upper and lower reservoirs with a pump-siphon system) to prevent vertical tailing of the protein spots.
10. At the end of the run, turn off the power, remove the plates, separate them with gentle leverage, and process the gel for staining (Chapters 6 and 13) or autoradiography (Chapters 16 and 17).
11. Briefly, gels may be fixed in 45% (v/v) methanol and 7.5% (v/v) glacial acetic acid (or 15% TCA) and stained

with 0.2% (w/v) Coomassie blue (prepared by dissolving it in a small volume of methanol) in 45% (v/v) methanol and 10% (v/v) glacial acetic acid for 30 min to 1 h. Gels may be destained using several changes in the fixer, the latter may be regenerated by absorbing out the Coomassie blue by passing it through an activated charcoal filter.

12. Gels are then dried down using a gel drier (*see* Chapter 16) and autoradiographed using Kodak-XO-mat-AR film for several different lengths of time. Owing to the saturation of the film, this allows visualization and qualification of a range of proteins with different abundances.
13. Autoradiographs may be quantified using scanning gel densitometry (Chapter 15) or more accurately by the slower method of quantitative rooster scanning. However, a better, rapid, and accurate method for handling a large number of proteins is to prepare transparent plastic sheets (the sort used for overhead projector transparencies) as templates. The position of the protein spots on the autoradiogram are marked on the sheet and they are then cut out. This template is used to punch out areas of the gel and these areas are then counted in a scintillation counter. With appropriate labeling the plastic template becomes a permanent record of the protein spot position and may be filed with the resultant counts and autoradiogram. This enables long-term handling of data from a larger number of gels for metabolic experiments. A typical gel pattern is shown in Fig. 1.

Notes

1. The major problems associated with two-dimensional gels are spot streaking and artifactual charge heterogeneity. The principal charge artifacts are produced by post-translational modifications of proteins, for example, the spontaneous deamination of asparagine and glutamine or the oxidation of cysteine to cysteic acid. Care should be taken when preparing extracts and they should always be maintained in the cold and stored in lysis buffer. Proteins may also be carbamylated by

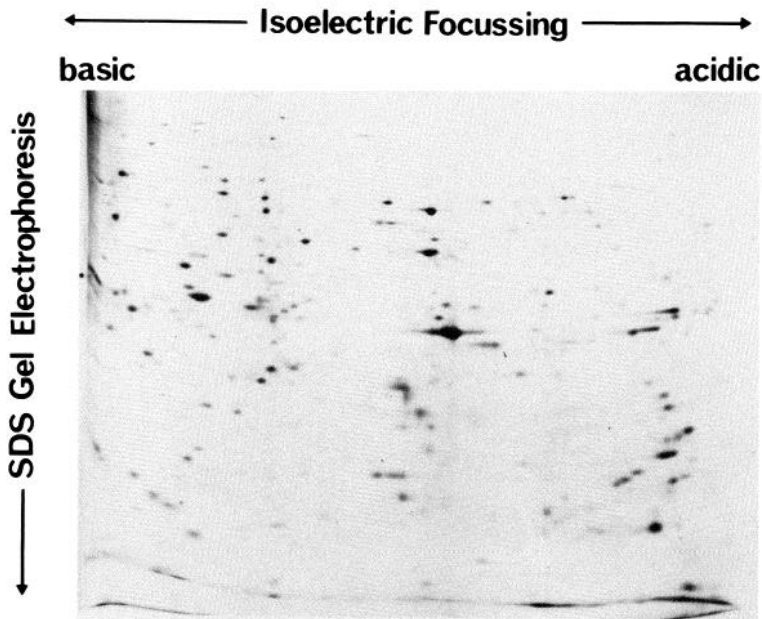


Fig. 1. An autoradiograph of a 10% two-dimensional polyacrylamide gel of ^{35}S -methionine labeled cellular proteins from a temperature-sensitive leucyl-tRNA synthetase mutant of Chinese hamster ovary cells.

About 5×10^5 cpm of ^{35}S -labeled proteins were loaded on the gels that were exposed to the X-ray film for 5 d. Note the characteristic protein spot shape and the distortion at the basic end of the gel caused by edge effects.

isocyanate impurities in the urea solution, and therefore it is worth investing in the highest quality urea, preparing it fresh, and storing it at -20°C . It is also advisable to have ampholines present wherever proteins are in contact with urea and to prerun the isoelectric focusing gels to remove isocyanate contamination. Basic ampholines may precipitate nucleic acids, with the resultant binding of protein producing streaking, but this may be easily solved by treating the sample with ribonuclease and deoxyribonuclease. Nevertheless, sample preparation by the methods described above should result in trouble-free gels.

2. Vertical streaking can also occur and this is caused either by not changing or recycling the running buffer or, more commonly, by poor equilibration of the first-

dimension gels. Occasionally, horizontal streaking can occur owing to poor solubility of proteins, but high concentrations of NP-40 and urea usually increase solubility sufficiently

- 3 Spot size increases nonlinearly with high protein concentrations and overloading may cause precipitation at the top of the isoelectric focusing gel, which can streak across the pH gradient. Overloading may also cause pH inversions in the isoelectric focusing dimension. The best separation and resolution is therefore obtained when the lowest amounts of protein are applied to the gel, but concentrations up to 100 $\mu\text{g/gel}$ are tolerated. Obviously, the best results are obtained with small amounts of protein having high specific activities, although the evolution of highly sensitive silver stains may diminish this requirement. If samples are too dilute or at too low a specific activity, they may be precipitated with ammonium sulfate or better, lyophilized, followed by solubilization of the pellet directly in lysis buffer. But remember that any salt will also be concentrated by these techniques and the salt may cause artifacts on the gel. Cells, if they are grown in a small area, for example in a microwell (*see* Chapter 9) may also be taken up directly in lysis buffer and applied directly to the gel, consequently avoiding dilution during sample preparation. Overloading can be particularly a problem with serum because of the enormous concentration of albumin. Methods such as immunoprecipitation of the major protein might improve resolution of minor components. Interestingly, it is possible to polymerize proteins within the isoelectric focusing gel, and this increases capacity, ameliorates the problem of precipitation, and gives acceptable gels.
- 4 We have found that, when cells in culture are trypsinized, if they are not thoroughly washed, the trypsin may carry over and cause streaking and artifactual gel patterns. Protease degradation, particularly of nuclear proteins, has also proved a problem and this has been remedied by adding protease inhibitors such as sodium bisulfite and diisopropylfluorophosphate. These should be omitted from the 2-D sample gel preparation since they may cause modification of proteins.

5. Ampholines also run as small proteins, are acid precipitable, and will stain. They can be eluted from the gel using the fixative described above and so avoid masking the detection of small proteins migrating near the gel front
- 6 Workers often, when they first start running gels, find poor polymerization. This is usually caused by dirty plates or poor quality bisacrylamide or acrylamide. Commercial electrophoresis-grade chemicals are usually of high enough quality, but acrylamide may be purified by heating 1 L of chloroform in a water bath at 50°C and adding to it 70 g acrylamide. Filter through Whatman No. 1 grade filter paper, while it is hot, and leave the resultant liquor on ice for several hours to recrystallize. Collect crystals in a Buchner funnel and wash with cold chloroform. Leave the crystals to dry. Bis-acrylamide may be recrystallized by bringing 1 L of acetone to a boil in a water bath, adding 10 g bis-acrylamide and re-boiling it. Thereafter, continue as described for acrylamide, except that crystallization is performed at -20°C. But remember these chemicals are highly toxic. Finally, to improve polymerization, we make up all solutions in double glass-distilled or deionized water (but be careful of flaking of ion-exchange resin from the column) and we filter all gel solutions through 0.45 μm filters before use.
- 7 Reproducibility can also be a problem and this is usually caused by batch variation in ampholines. If it can be afforded, batch testing is advisable. Manufacturers have also promised to improve the reproducibility of their ampholines. To date, LKB manufactures the most consistent product. It is also important to keep gel lengths and run times the same from day-to-day and particularly to exercise care in handling the isoelectric focusing gel to avoid stretching it. During equilibration between 5 and 25% of protein may elute from the gel, and variable amounts of protein may also precipitate at the top of the gel during the run. This can be a problem when comparing, for example, time courses of protein synthesis when it is desirable to add constant amounts of protein to the gels. Low concentrations of protein in the sample reduce the precipitation problem and O'Farrell (1) has described a technique for rapid equili-

bration involving a high SDS buffer added directly to the isoelectric focusing gel already *in situ* on the slab gel

8. SDS slab-gel electrophoresis separates according to molecular weight since SDS binds to most proteins on a molar basis (14 to 1) giving a uniform charge-to-mass ratio. Thus, dependent on the size range of the proteins to be analyzed, a suitable percentage acrylamide gel may be selected. Gradient gels may also be run and these give greater accuracy to molecular weight determinations of glycoproteins. Nevertheless, anomalous results may be obtained with unusual proteins, for example those containing a large percentage of basic residues, or those with large amounts of carbohydrate. Care should be exercised by the investigator in the interpretation of the experimental results.
9. Given that all these procedures are correctly followed, however, the gels are remarkably reproducible and provide one of the major analytical tools available to protein biochemists.

Acknowledgments

These techniques were derived from the original O'Farrell papers, simplified with the help of J. Parker, J. Friesen, R. Bravo, and J. Celis. This paper was prepared while my work was funded by the MRC (UK), the Cancer Research Campaign and the Central Research Fund of the University of London

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Chapter 11

Starch Gel Electrophoresis of Proteins

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Introduction

Starch gel is one of a wide variety of supporting media that can be used for horizontal zone electrophoresis. Such gels are prepared by heating and cooling a quantity of partially hydrolyzed starch in an appropriate buffer solution. The choice of buffer is somewhat empirical and a wide variety of compositions have been used successfully.

A characteristic feature of starch gels is that they exhibit molecular sieving effects. Separation of proteins is achieved, therefore, not only on the basis of differences in charge, but also of differences in molecular size and shape. The contribution of these factors, however, is difficult to control because a starch gel of any particular concentration will contain a range of pore sizes and there is no way of knowing what these are.

The gels may be prepared with total concentrations between 2 and 15% w/v, but 10% is a good starting point for most separations. They are usually cast in rectangular slabs between 1 and 6 mm thick. The 1-mm thick gels (thin layer) are most suitable for the electrophoretic separation of small quantities of protein mixtures, as encountered in forensic analysis. The 6-mm thick gels (thick layer) are used where the availability of sample is not a critical factor. They have been widely used in human genetics to study a variety of protein and enzyme polymorphisms. The preparation of both types of gel is described here.

After electrophoresis, the gels are stained to reveal the separated components. General protein stains can be used, but starch gels are commonly used in conjunction with specific staining procedures in which only one or a group of closely related proteins is located.

Materials

1. Gel and tank buffers are selected empirically to give the best separation for the protein system under investigation. Some compositions that have been used successfully are as follows:
 - a. Tank buffer, pH 7.4: 12.11 g Tris, 11.62 g maleic acid dissolved in 1 L water and adjusted to pH 7.4 with 10N sodium hydroxide. Gel buffer, tank buffer diluted 1 in 15 with distilled water.
 - b. Tank buffer, pH 8.5: 4.41 g sodium barbiturate, 2.34 g diethylbarbituric acid dissolved in 1 L distilled water and adjusted to pH 8.5 with 0.1N sodium hydroxide.
 - c. Tank buffer, pH 4.9: 28.72 g citric acid; 10.69 g sodium hydroxide dissolved in 1 L distilled water. Gel buffer, pH 5.0: 0.945 g succinic acid, 1.113 g Tris dissolved in 1 L distilled water.
 - d. Tank buffer, pH 7.2: 27.21 g boric acid; 1.68 g lithium hydroxide dissolved in 1 L distilled water. Gel buffer, pH 7.4: 0.818 g Tris; 0.378 g citric acid; 0.136 g boric acid dissolved in 1 L distilled water.
 - e. Tank buffer, pH 7.9: 18.6 g boric acid; 2 g sodium hydroxide dissolved in 1 L distilled water. Gel

buffer, pH 8.6: 9.2 g Tris, 1.05 g citric acid dissolved in 1 L distilled water.

2. Hydrolyzed starch for electrophoresis.
3. Glass plates, 22 cm long, 12 cm wide, and approximately 4 mm thick.
4. Glass edge strips, 22 and 14 cm long, 5 mm wide, and approximately 1 and 3 mm thick.

Methods

Preparation of Starch Gels

Thin Layer Starch Gels

1. Prepare the gel mold by sticking the 1-mm thick glass edge strips to one of the glass plates to give a shallow tray 21 cm long, 14 cm wide, and 1 mm deep. The UV light-sensitive glass adhesives are most suitable for this purpose.
2. For a 10% gel, mix 4 g of the dry starch with 40 mL gel buffer in a 250 mL conical flask to give a lump-free suspension.
3. Heat the suspension over a Bunsen flame while gently swirling the flask. As the temperature rises, the starch begins to dissolve and the solution becomes viscous. It is important to keep the flask moving at this stage since localized heating will cause the starch to burn. Continue the heating and at a certain point the viscosity of the solution will fall very rapidly. Continue to swirl the flask and bring the starch solution to a boil. Continue to boil for a few seconds. The bubbles will be seen to change from a frothy opalescent appearance to a clear transparent form.
4. Degas the solution by applying a vacuum to the flask for approximately 5 s. The solution will boil again under the reduced pressure. Release the vacuum slowly or else the hot starch solution will be violently mixed and can spit out of the flask.
5. Place the mold on a horizontal surface and pour the hot starch solution across the width of the plate at one end. Spread the solution across the plate to form an even layer. This is best achieved by using a bevelled Perspex

scraper placed across the width of the plate and drawn down its length, allowing the excess starch solution to flow off the end.

6. Leave the plate for 5 min to allow the starch solution to cool and gel. Transfer the gel plate to a moisture cabinet at 4°C.

Thick Layer Starch Gels

1. Prepare the gel mold by sticking the 3-mm thick glass edge strips to one of the glass plates. Place a second set of edge strips on top of the first. These are held in place by smearing the joining surfaces with a non-silicone grease. This will give a tray 21 cm long, 14 cm wide, and 6 mm deep.
2. For a 10% gel, mix 20 g of the dry starch with 200 mL gel buffer in a 2 L conical flask to give a lump-free suspension.
3. Heat the suspension with continuous vigorous swirling and bring the solution to a boil. Continue boiling for a few seconds and then degas the solution by applying a vacuum to the flask.
4. Place the mold on a horizontal surface, pour the hot starch solution into the center, and allow it to spread over the surface of the mold. If any bubbles form or if the solution does not reach the edge, touch the solution at the point with a rubber-gloved finger.
5. Leave the plate for 30 min to allow the starch solution to cool and gel. It is important that the plate not be disturbed during this period or the gelling solution will ripple. Transfer the gel plate to a moisture cabinet at 4°C.

Sample Application

Thin Layer Gels

1. Make sample application slots in the gel across the width of the plate approximately 6 cm from the cathode end. This is most easily achieved by using a comb made of thin (0.5 mm) plastic, with teeth 1 cm wide and 0.5 cm apart. Press the comb into the gel and remove it cleanly.

2. Soak pieces of 1-cm long cotton thread (pulled from a plain cotton bed sheet) in the sample solution and insert into the sample slots. Each thread absorbs approximately 2–3 μL of sample solution. Ensure the threads are pressed well down to touch the base glass plate. The threads are best held and manipulated using a pair of fine forceps

Thick Layer Gels

1. Make sample application slots in the gel across the width of the plate approximately 6 cm from the cathode end. These can be made individually by inserting a piece of razor blade, approximately 1 cm wide, vertically into the gel or using a comb as described for the thin layer gels
2. Soak a piece of Whatman 3 mm filter paper, approximately 1×0.5 cm, in the sample solution and insert into a sample slot. Ensure the inserts reach the base glass plate. The inserts absorb approximately 15–20 μL of sample solution

Electrophoresis

1. Place the gel plate in an electrophoresis tank and apply wicks to both ends of the gel. The wicks, consisting of several thicknesses of filter paper or pieces of sheet sponge and cut to the width of the gel, should be soaked in tank buffer, and overlap the gel surface by 1–2 cm. They can be held in place by a plain glass plate, 22×15 cm, which will also cover the gel and reduce evaporation
2. The gels must be cooled during electrophoresis by placing the whole tank in a cold room or refrigerator at 4°C , or by using a cooling plate through which water is circulated at 4 – 8°C .
3. It is difficult to generalize about the conditions of electrophoresis: time, voltage, and current are selected to give the best separation for the protein system being studied. As a starting point, however, potentials between 2 and 20 V/cm are used with running times between 2 and 16 h.

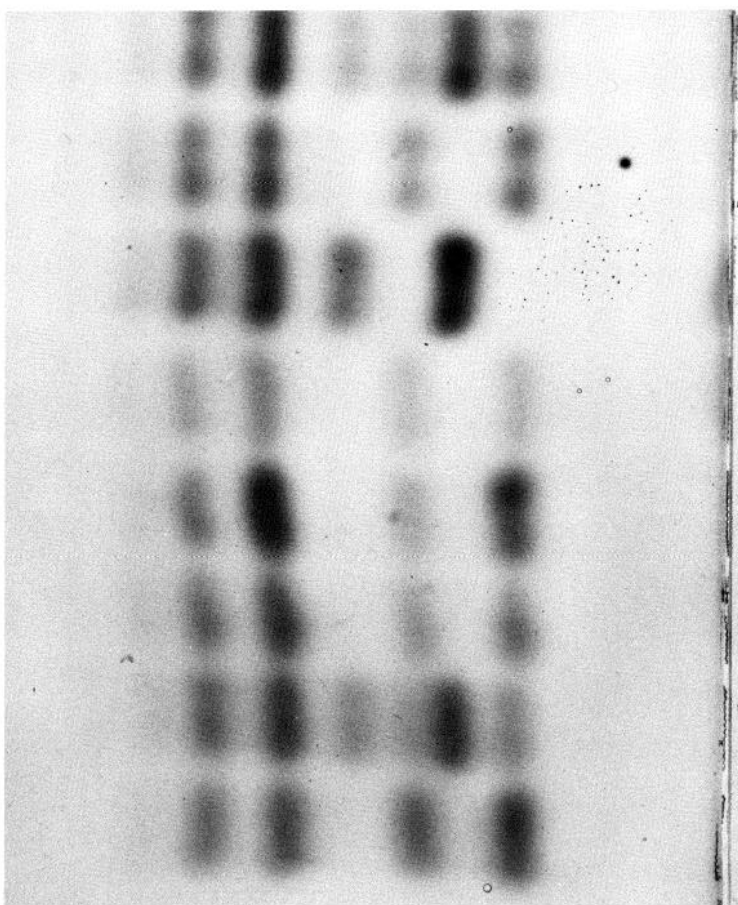


Fig. 1. Part of a thin-layer starch gel electrophoresis plate used to detect phenotypic variation in human red cell phosphoglucomutase (PGM). The samples were approximately 2–3 μ L of human red cell lysate applied to the gel on 1-cm cotton thread inserts. Electrophoresis was performed at 6 V/cm for 16 h in a cold room at 4°C. The plate was then developed for PGM activity using a zymogram agar overlay (*see also* Chapter 12 for the same analysis by isoelectric focusing).

4. After electrophoresis, remove the top plate and wicks, lift the gel plate from the electrophoresis tank, and apply a suitable stain. A typical electrophoretic run is shown in Fig. 1. The stain may be applied to the top surface of the thin layer gel, but the thicker gels must

be sliced. To do this, remove the filter paper inserts and the second set of edge strips. Place a glass plate on the surface of the gel and draw a thin wire or cutting blade through the gel. During the cutting process, ensure that the wire or blade is kept firmly pressed to the surfaces of the bottom edge strips. Remove the top glass plate and roll a piece of damp filter paper onto the gel surface. The upper half of the gel will adhere to the paper and can be peeled away from the lower half. The cut surfaces of each half can now be stained.

Notes

1. A wide variety of buffers have been successfully used. The pH, ionic strength, and buffer constituents are all important considerations when trying to achieve a particular separation. When studying enzymes, start with a buffer whose pH is near the optimum and use short running periods with low voltages, e.g., 2–4 h at 2–4 V/cm. It is sometimes necessary to incorporate known activators or cofactors into the buffers, e.g., Mg^{2+} ions or EDTA, since this can improve the resolution into distinct zones and increase the staining intensities of enzymes.
2. Hydrolyzed starch suitable for electrophoresis can be prepared in the laboratory by warming granular potato starch in acidified acetone. Most laboratories, however, use hydrolyzed starch powder that is commercially available. The methods described here have assumed 10 g/100 mL to give a 10% gel, but the exact amount of powdered starch required varies from batch to batch and is always given with the manufacturers instructions.
3. The starch solutions take between 5 and 30 min to gel, but it has been noticed on numerous occasions that superior resolution of protein bands is achieved if the gels are stored in a moisture cabinet overnight.
4. After preparation, the gels should appear clear or slightly opalescent and homogenous. The presence of undissolved starch, which appears as white speckles, bubbles, bacterial growth, or any other artefact renders the gel unsuitable.

5. The sensitivity of the method is ultimately dependent on the staining procedure used. Samples that have a total protein concentration of between 1 and 100 mg/mL are generally used

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Chapter 12

Isoelectric Focusing in Ultrathin Polyacrylamide Gels

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Introduction

Isoelectric focusing is an electrophoretic method for the separation of amphoteric macromolecules according to their isoelectric points (pI) in a stabilized natural pH gradient. It is a sensitive and reproducible technique, particularly valuable in the separation of closely related proteins that may not be easily separated by other techniques.

The method consists of casting a thin layer of polyacrylamide gel containing a large series of carrier ampholytes. A potential difference is applied across the gel and the carrier ampholytes arrange themselves in order of increasing pI from the anode to the cathode. Each carrier ampholyte maintains a local pH corresponding to its pI , thus creating a uniform pH gradient across the plate. Samples are applied to the gel surface and under the

influence of the electric field each individual protein component migrates to the region in the gradient where the pH corresponds to its isoelectric point. At its pI , a protein is electrically neutral and it becomes stationary or focused at that point in the gel

When focusing is complete, the separated components are detected by applying a general or specific stain to the gel surface.

Materials

1. Acrylamide, N,N' -methylene bisacrylamide and analytical grade sucrose.
2. Stock riboflavin solution, 1 mg in 10 mL distilled water. Store the solution in a brown glass bottle at 4°C
3. Carrier ampholytes of the desired pH range. The author has experience only with those available from LKB, Sweden (Ampholine^R). They are supplied as 40% w/v solutions. Store at 4°C.
4. Anolyte and catholyte wick solutions consisting of 1% v/v acetic acid and ethanolamine, respectively.
5. Plain glass plates, 22 cm long, 15 cm wide, and approximately 4 mm thick.
6. Fixing solution: 5 g sulfosalicylic acid and 10 mL trichloroacetic acid in 90 mL distilled water
7. Destaining solution. 300 mL methanol, 100 mL glacial acetic acid, and 600 mL distilled water.
8. Staining solution: 0.2 g Coomassie Brilliant Blue R.250 in 100 mL destaining solution. The solution should be filtered before use.

Method

1. Prepare the ultrathin mold by sticking strips of PVC electric insulating tape, 1 cm wide and approximately 0.15 mm thick, around the edge of a clean glass plate. This will give a very shallow tray, 18 cm long, 13 cm wide and 0.15 mm deep. Avoid overstretching the tape since it will later shrink and gaps will occur at the corners. Small gaps are tolerable, but larger ones (1

mm or more) will hinder polymerization of the acrylamide and cause an unevenness in the gel at these points.

2. Prepare the following solution, which is sufficient for making 12 plates. 3.88 g acrylamide, 0.12 g *N,N'*-methylene bisacrylamide, 10 g sucrose, and 80 mL distilled water. Ensure that all of the components are completely dissolved. This is best done with a magnetic stirrer, but vigorous mixing of the solution should be avoided. The solution should not be stored.
3. Add 0.6 mL riboflavin solution. The riboflavin does not completely dissolve in the stock solution, so that this solid material should not be picked up
4. Add 4 mL of carrier ampholyte solution of the appropriate pH range. The final concentration of carrier ampholytes is approximately 2% w/v. Ensure all the components are uniformly mixed by gently swirling the flask. Degassing has been found to be unnecessary so long as violent mixing has been avoided.
5. Place the glass mold in a spillage tray (photographers' developing trays are ideal for this purpose) with the taped surface uppermost. Wipe the surface with an alcohol moistened tissue to remove any traces of grease.
6. Using a disposable plastic syringe, transfer approximately 7 mL of the acrylamide–ampholyte solution to along one of the short edges of the glass mold.
7. Take a plain glass top plate (20 × 15 cm), wipe the surfaces with alcohol, and place one of the short edges along the taped edge of the mold adjacent to the acrylamide–ampholyte solution. Gradually lower the top plate and allow the solution to spread over the mold. Care must be taken to avoid trapping air bubbles between the two glass plates. If this happens, raise the top plate to remove the bubble, then lower it again. Excess solution will flow into the spillage tray. Press the top plate into firm contact with the taped edge of the bottom plate.
8. Lift the complete mold and top plate out of the spillage tray. Gently scrape the bottom of the mold on the edge of the tray to remove excess solution.
9. Place the whole plate in UV light for at least 3 h to allow photopolymerization to occur. A viewing table il-

luminated by white strip lights or bright direct sunlight provides sufficient UV light. To save space plates may be stacked, but no more than three in a pile, and each plate must be separated from the next by glass strips to prevent the plates sticking to each other.

10. Pour the excess acrylamide–ampholyte solution from the spillage tray into a clean flask. The dregs from four plates will be sufficient to pour one further plate.
11. Remove the plates from the UV light source and wipe the outside surfaces with a wet tissue to remove any solid material. This is important since the whole surface of the bottom plate must be in intimate contact with the cooling plate during the focusing run.
12. Store the plates in the dark at 4°C. They can be kept for several months, but should not be used if the gel has discolored or dried. If the plates are used immediately, it is still advisable to cool them to 4°C since this facilitates removal of the top plate.
13. Remove the top plate immediately prior to sample application by placing the whole plate on a horizontal surface and inserting a scapel blade between the top and bottom plate in one of the corners. Carefully twist the blade and pry the top plate away. Sometimes the gel sticks to the top plate, but as long as it remains in contact, the top plate plus gel may be used for focusing. Occasionally, part of the gel will stick to the top plate and part to the bottom plate. The whole lot will then have to be discarded.
14. Absorb sample solutions on 5 × 5 mm pieces of Whatman No. 1 filter paper and lay these on the gel surface down the length of the plate at approximately 5 cm from the anodal edge. Each piece of filter paper absorbs approximately 5–8 µL of sample solution. Samples that have a total protein concentration of between 0.05 and 10 mg/L are generally used.
15. Place the gel plate on a cooling plate through which water is circulating at approximately 4–8°C.
16. Apply electrode wicks to the anode and cathode edges of the gel. These consist of 18 × 1 cm strips of Whatman No. 17 filter paper saturated with 1% acetic acid (anolyte) and 1% ethanolamine (catholyte), respectively. Uniform soaking of the wicks is important in preventing wavy iso-pH bands.

- 17 Place a platinum electrode on each wick. These can consist of an 18 cm-long strand of platinum wire stretched and held taught across an 18×1 cm piece of Perspex. Hold the electrodes in place with a glass top plate.
18. Apply a potential difference of between 130 and 160 V/cm across the plate. This represents (for a 12 cm electrode gap) a working potential of approximately 1600–2000 V. A special high voltage power pack will, therefore, be required. Starting currents are usually between 5 and 8 mA.
19. Focusing takes between 3 and 6 h. This can be checked by placing identical samples on the cathode and anode sides of the gel. Equilibrium focusing conditions have been reached when the same components in each sample are seen to coincide.
- 20 When focusing is complete, disconnect the power supply, remove the top plate, electrodes, and wicks, lift the gel plate from the tank, and apply a general or specific stain to the gel surface. A typical example is shown in Fig. 1.
- 21 Many general protein stains form insoluble complexes with the carrier ampholytes and these must be removed from the gel before the stain is applied. A suggested procedure is as follows. Place the gel plate in the fixing solution for 20 min; transfer the plate into destaining solution for 15 min, transfer into staining solution for 30 min, then finally into destaining solution for 20 min. All stages are performed at room temperature.

Notes

1. When making ultrathin gels, some workers recommend employing the process of silanization (in which the polyacrylamide gel is chemically bonded to the glass plate, thus preventing the gel from becoming damaged or torn during use). In practice however, we have found this an unnecessary step.
2. Prefocusing of the plates prior to sample application, although carried out by some workers, is an unnecessary step in most instances.

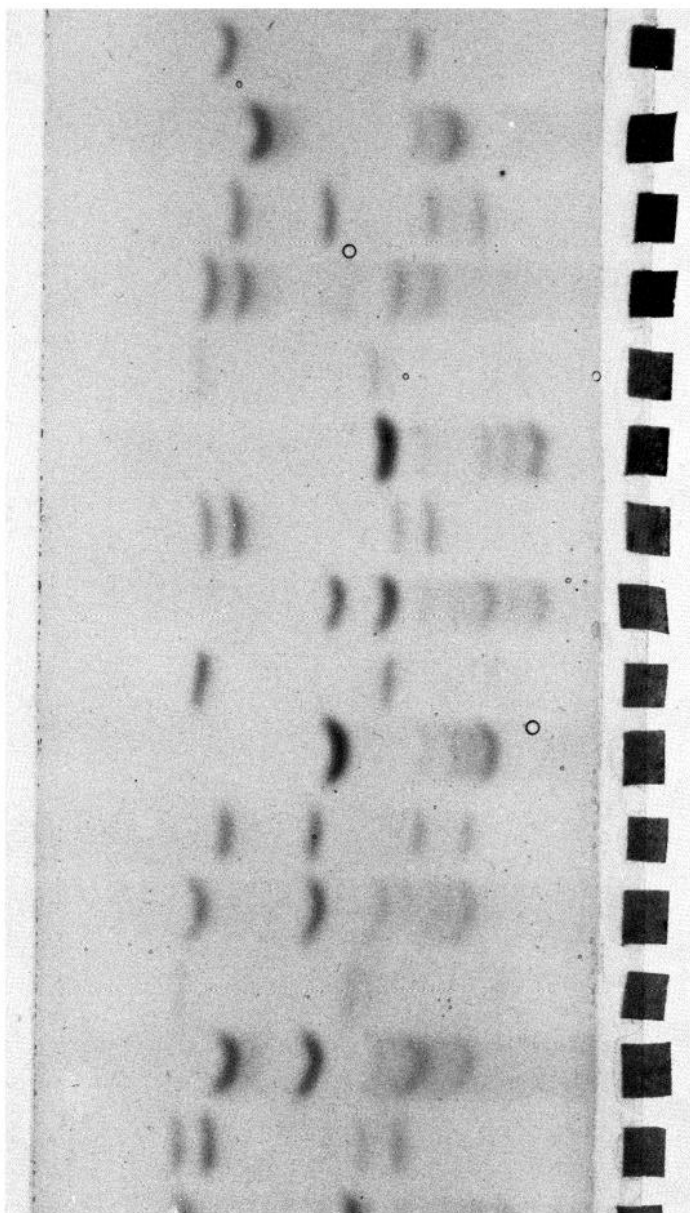


Fig. 1. Part of an ultrathin IEF plate used to detect phenotypic variation in human red cell phosphoglucosmutase. The samples were approximately 5–8 μ L of human red cell lysate (1:5 with distilled water) applied to the gel surface on filter paper squares at 5 cm from the anode edge. They were subjected to electrophoresis at 140 V/cm for 3 h in a pH 5–7 gradient. The run was stopped before focusing was complete and the plate was developed for PGM activity using a zymogram agar overlay.

3. Theoretically, it should make no difference where the samples are applied since all components will move in the pH gradient to positions corresponding to their isoelectric points. In reality, this is not always the case. Cathodal application may be more appropriate, for example, if a particular protein is easily denatured at acid pH values.
4. The electrode wick solutions (acetic acid and ethanolamine) have been used successfully for separations in the following gradients: pH 3–10, pH 5–7, pH 4–8, and pH 4–6
5. Adequate cooling is essential, but remember that at low temperatures proteins take longer to focus. For some systems, overcooling produces an inferior resolution of the components.
6. Considerable advantages can be obtained by not allowing the proteins to reach their isoelectric points. The runs are shorter, resolution can be improved, and an overall increase in clarity is sometimes observed. Under such conditions, the method is really one of electrophoresis in a pH gradient rather than true isoelectric focusing.

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Chapter 13

High-Sensitivity Silver Staining of Proteins Following Polyacrylamide Gel Electrophoresis

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Introduction

Polyacrylamide gel electrophoresis is a simple, inexpensive, yet highly versatile and powerful method for the analysis of complex mixtures of proteins. In part, the success of this method has resulted from the ease with which the fractionated proteins can be detected with the blue dye, Coomassie brilliant blue R 250. However, though this stain has proved to be ideal for many of the more traditional applications of this method, it is of limited sensitivity. In particular, the recent development of two-

dimensional gel electrophoresis (Chapter 10) and *in situ* peptide mapping techniques (Chapter 22) have demanded increasingly more sensitive detection methods. In part, this requirement has been met by the use of radioactively labeled proteins, followed by either autoradiography or fluorography (Chapter 9). The principal drawback of this approach is that it involves additional sample preparation and it is often difficult to label the proteins to a sufficiently high specific activity, particularly where proteins are obtained from dilute physiological samples, and so on.

The recent development of a number of very sensitive silver stains, which are between 20 and 200 times more sensitive than Coomassie brilliant blue (*see*, for example, Fig. 1 of Chapter 22), has been a very valuable addition to the range of protein detection methods that can be used with polyacrylamide gel electrophoresis. These silver staining procedures have been developed from a variety of histochemical and photochemical techniques, where silver has long been recognized as a valuable and sensitive reagent. Silver staining has the considerable advantage of not requiring any special sample preparation and it can also be used in conjunction with standard Coomassie stains. A number of problems arise from the novelty of this method and our limited understanding of the precise chemistry of the reactions involved. The most significant of these are: lack of reproducibility, nonlinear staining, preferential staining of different proteins, and nonspecific background staining. However, silver staining remains a valuable method for the detection of proteins in polyacrylamide gels.

The procedure described here is based on a method originally published by Oakley et al. (1). Although this method is neither the most sensitive or rapid of the methods that are currently available, we have found that it is both reliable and easy to use.

Materials

With the exception of Solution C, all of the following solutions are prepared from analytical grade reagents. De-ionized, glass-distilled water is used for the preparation of

all solutions. The volumes of these solutions are sufficient to stain a gel measuring approximately 15 by 15 cm; where either larger or smaller gels are to be stained, these volumes should be adjusted accordingly

1. *Solution A* 500 mL of 50% (v/v) methanol, 10% (v/v) acetic acid
2. *Solution B*. 2000 mL of 20% (v/v) methanol, 7% (v/v) acetic acid
3. *Solution C* 500 mL of 10% (v/v) glutaraldehyde is prepared by diluting 100 mL of 50% glutaraldehyde solution (laboratory reagent grade) to 500 mL with water.
4. *Solution D*. 200 mL of ammoniacal silver nitrate solution is prepared as follows to approximately 170 mL of water, add 2.8 mL of concentrated ammonia solution (0.88 sp.gr) and 3.75 mL of 1N NaOH solution. A volume of 8 mL of 20% (w/v) silver nitrate solution is then slowly added to this solution while vigorously mixing the solution. The solution is then made up to 200 mL with water
5. *Solution E* 1000 mL of 0.005% (w/v) citric acid, 0.019% (v/v) formaldehyde [0.5 mL of 38% (v/v) formaldehyde per liter]

Solutions A and B can be kept for up to 2 wk before being used. All other solutions are prepared as required and used immediately. Both the concentrated ammonia and glutaraldehyde solutions should be stored at 4°C when not in use

6. The only items of specialized equipment required for the following protocol are an orbital shaker table and a number of *clean* glass "gel dishes", shallow baking dishes are ideal for this purpose.

Method

In order to prevent the surface of the gel becoming contaminated by finger marks and so on (which are readily detected by this method), it is essential that talc-free plastic gloves are worn whenever the gel is handled. However, in the majority of the following steps, the various so-

lutions can be removed with the aid of a water pump, thus avoiding any direct contact with the gel

With the exception of Solution D (Step 5) where only 200 mL of this reagent is used, approximately 500 mL of each solution is used for each wash and the gel is continually agitated at between 20 and 50 rpm on an orbital shaker table.

1. Proteins are separated by polyacrylamide gel electrophoresis (*see* Chapters 6–10) and the gel is then placed in a glass dish and washed for 2–16 h in Solution A
2. The gel is then washed for a further 2–16 h in Solution B.
3. This solution is then replaced with Solution C and the gel is washed for a further 30 min, after which time it is washed with at least four changes of distilled water for a minimum of 2 h and preferably overnight.
4. Upon completing the wash step, the ammoniacal silver nitrate solution (Solution D) is poured into the dish and the gel is vigorously agitated (~200 rpm) for 15 min. The gel is then briefly rinsed with two changes of distilled water and transferred to a clean gel dish between the first and second rinses.
5. The proteins are then visualized by the addition of Solution D and gently rocking the gel dish. When the image has developed to the required intensity (usually within 5 min), the gel is again washed in Solution B for about 10 min before finally being stored in this solution in the dark at 4°C. Although the silver-stained proteins remain clearly visible for at least 18 h, there may be some slight change in color as well as an increase in the nonspecific background staining during this time and the gel should therefore be photographed as soon as possible after staining.
6. During the above staining procedure the gel will undergo a considerable increase in size and the proteins frequently appear as diffuse bands. This effect can be reversed by leaving the gel for 2–16 h in Solution A (in the dark). The high concentration of methanol in this solution will dehydrate and shrink the gel and the proteins will then appear as intense, sharp bands

Notes

1. The importance of wearing talc-free plastic gloves and of touching the gel as little as possible has already been stressed. In addition it is important that all of the glassware used for the preparation of the gel, as well as the silver staining solutions, is scrupulously clean
2. Ammoniacal silver nitrate solution (Solution D) is potentially explosive when dry, and therefore should be collected after it has been used for staining the gel and precipitated by the addition of an equal volume of 1N hydrochloric acid. When large numbers of gels are being silver stained, sufficient silver chloride may be collected to make the recovery of the silver worthwhile. However, if this method is used only occasionally, the silver chloride can be washed down a drain with a large volume of cold water
3. In gels where either the proteins have been stained too heavily or there is too much background staining, the staining can be reversed with Kodak rapid fixer, as described by Oakley et al. (1)
4. If gels are to be pre-stained with Coomassie brilliant blue, gels are treated with a 0.5% (w/v) solution of Coomassie brilliant blue R 250 in Solution A for 2–18 h, followed by destaining against several changes of Solution B. The gel can then be silver stained by starting either at step 3 (*see Method*) or the gel can be “totally” destained by washing for 1–2 d in Solution A and then continuing from step 2 (*see Method*). This “total” destaining step will remove the Coomassie stain from all but the most intensely stained proteins and will ensure a more uniform deposition of silver.
5. The above procedure is designed for slab gels that are up to 1 mm thick. Where thicker gels are to be stained this method may fail to produce satisfactory results, with only the proteins at the surface of the gel being stained. Marshall and Latner (2) have recently published a technique designed for staining thick gels, and this should be used when the method described here proves unsatisfactory
6. Most radioactively labeled proteins (^{125}I , ^{32}P , ^{35}S , and ^{14}C) can be detected by either autoradiography or

fluorography, as appropriate, following silver staining. However, gels containing proteins that have been labeled with tritium are unsuitable for fluorography following silver staining because the silver grains absorb most of the weak beta emissions from this isotope.

References

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Chapter 14

Quantification of Proteins on Polyacrylamide Gels (Nonradioactive)

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Introduction

It is frequently necessary in biochemical experiments to quantify proteins. There are various methods for estimation of the concentration of total protein in a sample, such as total amino acid analysis, the Biuret reaction, and the Lowry method [see ref. (1)], but these do not allow quantification of one protein in a mixture of several. This may be done by chromatography and estimation of the content of the appropriate peak in the elution profile by virtue of its absorption of light at, say, 220 nm. However, it is usually quicker, easier, and more economical to quantify proteins that have been separated from each other on

a polyacrylamide gel. This is done by scanning the gel and by densitometry of the stained bands on it. Microgram quantities of protein may be quantified in this way. The method described below for quantitative staining uses Procion Navy MXRB and is suitable for proteins on acid/urea or SDS polyacrylamide gels (see Chapters 6 to 8)

Materials

1. A suitable densitometer, e.g., the Gilford 250 spectrophotometer with scanning capability and chart recorder, or the Joyce-Loebl "Chromoscan"
2. Protein stain.

Procion Navy MXRB 0.4 g

Dissolve in 100 mL methanol, then add 20 mL glacial acetic acid and 80 mL distilled water. Make fresh each time.

3. Destaining solution

Methanol	100 mL
Glacial acetic acid	100 mL
Distilled water	800 mL

Method

1. At the end of electrophoresis, immerse the gel in Procion Navy stain, and gently agitate until the dye has fully penetrated the gel. This time varies with the gel type [e.g., 1.5 h for a 0.5 mm-thick SDS polyacrylamide (15%T) gel slab], but cannot really be overdone.
2. At the end of the staining period, decolorize the background by immersion in destain, with agitation and a change of destain whenever the destain becomes deeply colored. This passive destaining may take 24–48 h even for a 0.5 mm-thick gel. Thicker gels will take longer.
3. Measure the degree of dye binding by each band of protein by scanning densitometry of the gel at a wavelength of 580 nm. The total absorption by the dye

in each band (proportional to the area of the peak in the scan profile) may be automatically calculated by integration of peaks in the scan profile, but if not then the peaks in the chart recording may be cut out and weighed. For standard curves, the absorption at 580 nm (i.e., peak size) is plotted versus the weight of protein in the various samples.

Notes

1. For quantification of proteins on gels, three simple conditions need to be fulfilled.
 - (i) Protein bands on the gel should be satisfactorily resolved
 - (ii) The dye used should bind to the protein of interest
 - (iii) Since one sample will be compared to others, to overcome errors caused by differences in sample sizes, the binding of the dye should be constantly proportional to the amount of protein present over a suitably wide range.
2. Errors that are difficult to eradicate arise when the above condition (i) is not fulfilled. Ideally, another electrophoresis system that *does* give sufficient resolution should be used, but otherwise the operator has to decide where the division between two overlapping peaks should be and where the baseline is. Use of a narrower beam of light for densitometry will improve resolution, but may worsen the baseline because of the increased effects of bubbles, dust, and so on.
3. Conditions (ii) and (iii) should be checked by construction of a standard curve for the protein(s) of interest. The range of protein quantities used should cover that to be found in experiments, and for accurate results the plot of dye-binding (measured by densitometry at a particular wavelength) versus protein quantity should be linear (or approach close to it) over that range.
4. Any stain may be used for quantification provided that it meets the above requirements (ii) and (iii). However, various factors may influence the choice of dye to be used. This can be illustrated by comparing Coomassie

Brilliant Blue R250 (BBR 250) with Procion Navy MXRB. Firstly, given time, Procion Navy can penetrate dense bands of protein and stain them, but Coomassie BBR 250 seems unable always to do this, so that the size of band that the protein forms may affect its staining. Thus, since the size of a protein band will generally increase with migration during electrophoresis, the extent of electrophoresis may have a small effect on staining by this dye. Secondly, when Coomassie BBR 250 binds to a protein, the resulting complex may be colored blue or red or anything in between, depending on the chemical structure of the protein. Such production of a variety of colors upon complexing of dye with proteins is called the "metachromatic effect." Maximum sensitivity is achieved in this technique by densitometry at the wavelength at which the dye-protein complex absorbs maximally. The metachromatic effect, therefore, dictates multiple estimations at wavelengths optimal for each complex, or gives suboptimal sensitivity. Procion Navy gives the same color with every protein. Thirdly, Coomassie BBR 250 binds proteins by electrostatic forces and can be completely removed from protein by extensive destaining. Thus, destaining may introduce an element of variability to the experiment. Results are more consistent with Procion Navy, which binds covalently to proteins so that prolonged destaining does not remove it. Fourthly, the efficacy of Coomassie BBR 250 varies from batch to batch of the solid dye and also with the method of its dissolution in making the stain (*see* Chapter 6). We have not seen this with Procion Navy MXRB. From the above it may be seen that Procion Navy MXRB (and probably other members of the Procion dye family) is potentially the better choice for quantification purposes. For the proteins studied in our own laboratory, at least, it also gives stoichiometric binding to proteins over a wider range than does Coomassie BBR 250 and so it is the dye of choice for quantification, even though it is severalfold less sensitive than Coomassie BBR 250.

5. Standard curves of Procion Navy MXRB binding to various proteins are straight lines passing through the

origin, although they do plateau at higher protein quantities. The upper limit is dependent on the size of the gel, but in a 0.5 mm-thick gel the standard curve was found to be linear beyond a loading of 6 μg of histone H1, and up to about 30 μg for a mixture of four proteins (histones) together (2). On such gels, a loading of less than 0.5 μg is detectable and 1 μg is sufficient for quantification.

6. The whole process of quantification should be standardized to reduce variation. Sources of error should be recognized and countered if results are to be accurate. Thus, the densitometer should give a linear response over the range of dye densities measured. Sampling errors are inevitable, but their effect can be reduced by repetition and averaging the results. The process of electrophoresis should be standardized, using samples of the same volume and electrophoresing for the same time at the same voltage in each experiment. Multisample slab gels are ideal for this purpose. The staining time should also be the same each time. Since variation in band widths will still occur, in a slab gel at least, and because irregularities in band shape will occur occasionally in any case, the whole of every stained band should be quantified. If the band width is greater than the width of the densitometer's light beam, and the raster type of scan is not available, the whole gel may be sufficiently reduced in size by equilibration in aqueous ethanol. Too much ethanol causes the whole polyacrylamide gel to become opaque, but if this happens the gel may be rehydrated in a lower-percent ethanol solution. For shrinking 15% polyacrylamide gels, 40% v/v ethanol in water for 1 h or so is suitable, reducing their size by about a third. If the sample width is still too great, it can be divided up and each section estimated separately. These estimates are then summed for an estimate of the whole sample.

Despite efforts to reduce variations between experiments, they may still occur. In this case it may be necessary to construct a standard curve for each experiment. The type of standard curve required differs according to the aim of the experiment, which may be of the following types:

- (a) Simple comparison: determination of the relative concentrations of the same protein in two or more samples
- (b) Complex comparison: determination of the relative concentrations of two or more proteins in the same or different samples.
- (c) Absolute determination: determination of the amount of a protein in one or more samples (that is, on a weight or molar basis).

In experiments of type (a) the dye-binding capabilities of the (identical) protein in the different samples will be the same and all that is needed is to know that dye binding is stoichiometric over the appropriate range of quantities. Thus, the standard curve can be set up with a serially diluted (or concentrated) sample, without any accurate protein quantification involved. To compensate for sampling errors, it is often useful to relate the protein of interest to an "internal standard" of another protein that is known to be at the same concentration in all samples. This internal standard will indicate the magnitude of difference in size between samples on the gel, and this factor can then be used to adjust the results of the experimental protein. It must be remembered that the internal standard protein should also have stoichiometric dye binding capabilities over the appropriate range.

Two different proteins will not necessarily have the same dye-binding capabilities, so if they are to be compared on a weight or molar basis [experiment type (b)] it is necessary to determine the characteristics of each. Thus, the pure proteins are isolated, and each is accurately quantified (say, by amino acid analysis) and used to construct a standard curve. Experimental samples can be compared with these standard curves to give accurate weight measurements or it may be sufficient to determine the ratio between dye-binding properties of the different proteins for subsequent use with only one of the standard curves.

Experiments of type (c) are similar to those of type (b) in that accurately constructed standard curves are required using well-quantified protein standards. Note

that simple weighing is usually not accurate enough for such work as this, for it cannot distinguish between protein and nonproteinaceous constituents (such as contaminating dust)

For accurate results with experiment types (b) and (c), standard curves need to be run in each experiment (even if they are relatively crude, with only 4 or 5 points)

7. Destaining of Procion Navy-stained gels may be speeded up by driving free dye out of the gels by electrophoresis. The protein-dye complex remains in the gel. To do this, immerse the gel in destaining solution. Place a platinum electrode on either side of the gel and apply a dc electric field across the gel. Change the buffer frequently (as it becomes colored) or cycle the buffer through a decolorizing agent (i.e., an agent that binds the dye). Disposable paper tissues have proven satisfactory as a cheap, disposable decolorizing agent.

References

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Chapter 15

Computer Analysis of Gel Scans

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Introduction

Gels are frequently scanned either to obtain quantitative data from the patterns of stain or radioactivity or to facilitate comparisons between samples. Both these processes can be greatly enhanced by using a computer and I will describe the use of a desktop or microcomputer in this area. There are several programs available for scanning and analyzing two-dimensional gels and I will mention these briefly, in passing. These programs need extensive computer facilities and a full discussion of these systems is beyond the scope of this chapter (*see ref. 1 and references therein*). One-dimensional gel scans, usually individual tracks from a slab gel, are routinely used for estimating the quantities of specific proteins, or the amounts of radioactivity in specific proteins. I will describe an interactive in-

tegration program that provides very flexible procedures for determining the areas under peaks in the scan. A more automatic integration system can also be used and a number of such systems, designed for chromatography, are available. I will describe a peak picking routine that could be used to develop an automatic integration program if commercial software is not available. One characteristic of gel electrophoresis is that the conditions of electrophoresis, unlike those in most chromatographic procedures, are close to equilibrium, so that the resultant band shapes are close to Gaussian. This means that a complex of overlapping bands may be resolved into its underlying Gaussian components. I discuss approaches to this problem and describe a flexible, robust program suitable for a good microcomputer. Finally, the comparison of gel scans involves transformations to correct for variations in run conditions and scan parameters. I describe a program that aligns and "stretches" gel scans and allows mathematical operations, such as generating the difference between two scans as a "difference scan."

Gel scans must be stored in a computer before they can be used. In principle, a chart record can be digitized, but this is slow and inaccurate. It is much better to record the scan directly into the computer, or onto a data-logger that gives a machine-readable output. Similarly, the programs have to be fed into the computer. The programs I will describe have been implemented in BASIC on Hewlett-Packard desktop computers. The BASIC listings are available from me or from the Hewlett-Packard Desktop Software Catalog.

Main Menu

The following options are available:

1. Scan a gel and store the scan.
2. Plot a stored scan.
3. Determine areas under peaks in a scan.
4. Resolve overlapping peaks into Gaussian components.
5. Compare two scans.

Options

Scan and Store

A digitized gel scan is a list of absorbance readings taken at equispaced positions, Dx , along the gel starting at a particular point, $Xmin$. The list of absorbance readings is stored together with the parameters Dx and $Xmin$ and the number of readings in the list, $Npts$

In some cases it is very useful to be able to digitally reduce the "noise" in a gel scan. This is particularly appropriate for very weak bands or where there is a high background, as in fluorography. Many methods to smooth curves are available, but the simpler ones, like a moving average, are not very effective and also distort the data. The best choice is a low pass digital filter that removes all the components that appear to be narrower than a set value, such as the grains on a photographic emulsion. Figure 1 shows an example of the use of a 0.1 dB Chebychev

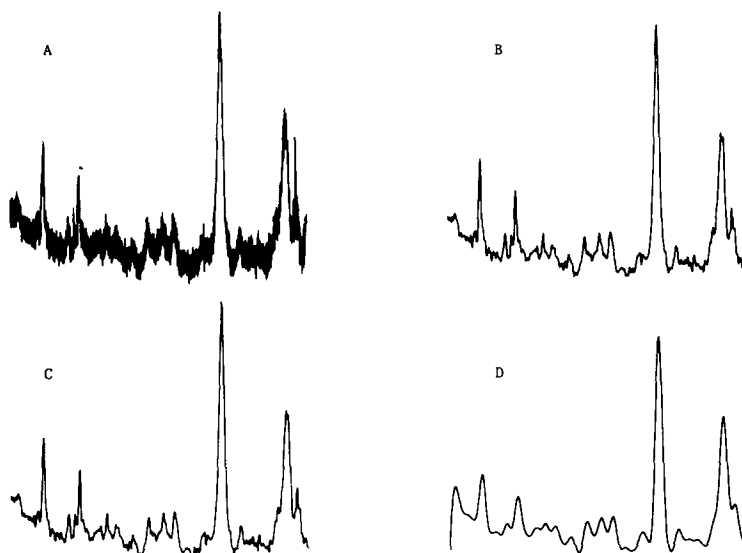


Fig. 1 Digital filtering. Scan A is the original data from a scan of a fluorograph of soluble histones (7). Scan B shows the result of passing the data through the digital filter with a relative cut-off frequency of 0.1. Scan C shows the result of reducing the number of data points in the filtered scan from 2840 to 568. Scan D shows the results of filtering the data of Scan A with a relative cut-off frequency of 0.02.

filter on a noisy gel scan. The program uses a second-order recursive filter, which is very stable, and passes the data through it four times to give the effect of an 8th-order filter. This technique is required for desktop computers because of their modest number precision. Phase shift is eliminated by passing the data alternately backwards and forwards through the filter

After filtering, it may be possible to reduce the number of data points without affecting the quality of the scan. At this stage (or before filtering), it may be necessary to invert the scan on either axis. For example, a photographic negative of a stained gel shows the bands as peaks below a high background. This may be inverted by the statement $\text{MAT Scan} = (\text{Ymax}) - \text{Scan}$ in Hewlett-Packard BASIC, where Scan is an array containing the absorbance values and Ymax is the maximum value in Scan. Inverting the X-axis may be necessary if some gels were scanned in different directions. Usually, gel scans are plotted with the top of the gel on the left. Inverting the X-axis requires a loop to move all the data points in the array as:

```

FOR I = 1 TO INT(Npts/2)      ' Lower subscript of
    J = Npts - I + 1          ' Scan is 1
    Temp = Scan(I)
    Scan(I) = Scan(J)
    Scan(J) = Temp
NEXT I

```

where I, J, and Temp are temporary variables and Npts and Scan have been defined above. Also at this stage it may be necessary to discard data points that are not needed at either end of the scan. This can be carried out by entering the X-axis values at the ends of the required part of the scan from the keyboard, or by digitizing on a plot of the scan. Digitizing in this application is achieved by placing a "+" or cursor on the scan as it appears on the graphics display. Keyboard controls are used to move the cursor, which moves along the scan, and hence to select the required section of the scan. The digitization subprogram is widely used in this set of programs to enter instructions. A modified version allows the cursor to be moved anywhere on the screen, in two dimensions, rather than being constrained to move along the scan.

Finally, changes to the baseline may be made, if desired.

The processed scan is then stored on a disk file and is available for analysis by subsequent programs. An integer format is used in order to conserve storage space. Sufficient descriptive information should be stored with the scan to allow it to be readily identified later.

Plot

A versatile plotting program is needed to prepare plots of the scans for discussion and for publication (e.g., 2). The program should allow for both rapid plotting, with minimal operator input, and for highly formatted plotting, where the user exercises extensive control over the plot. This is achieved by offering default parameters and making some sections of the program optional. Ideally, the plot should be developed on the CRT screen and then dumped, when perfect, to a graphics printer or a pen plotter. Figure 2 shows an example of a highly formatted plot, where the user controlled almost all of the parameters. Figure 1 is an example of plots produced automatically, with no user input. Axes may also be generated automat-

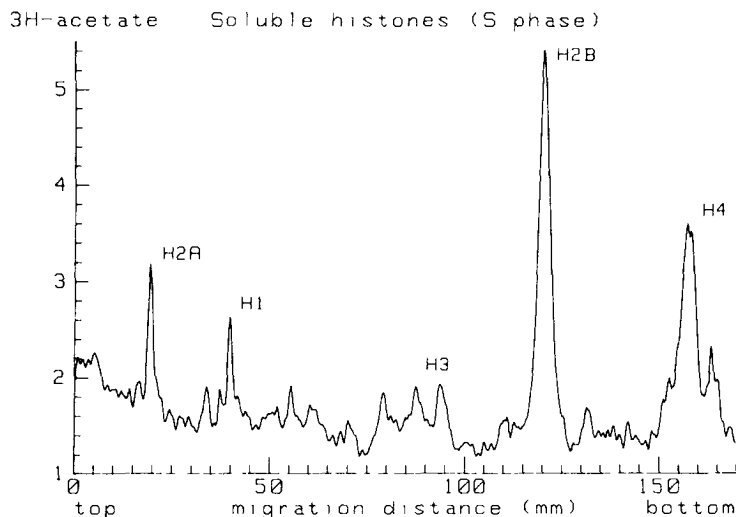
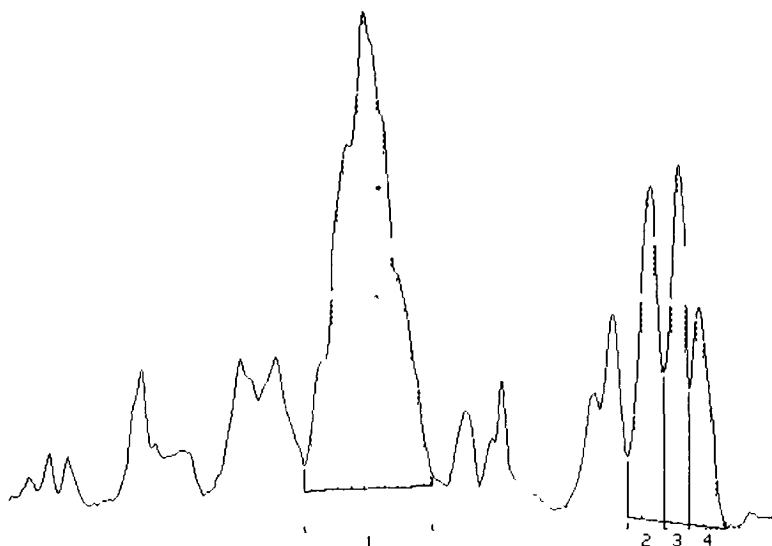


Fig 2 Figure 1B replotted with axes and labels



The range of this plot is 24mm to 172.44mm

Demonstration of results of integration program

Peak #	Area mm abs	Start mm	End mm	% of integrated area	normalized	
					to band 1	to total
1	25 4200	66 84	96 64	62.1	100	62063
2	6 3751	140 04	146 14	15.6	25.00	15565
3	5 4877	148 14	155 34	13.4	21.588	13398
4	3 6755	153 84	161 34	9.0	14.459	889738

Notes:

The last two columns show the areas normalized to either the area of the first integrated band (set to 100 from the keyboard) or to a total area of 1 (also entered from the keyboard).

Fig. 3 Integration of selected peaks on a gel scan. The integrated areas are shown by shading and the results are given in the table below the plot.

ically, if required. Figure 3 shows an example of a plot dumped to a graphics printer.

Areas

The main problem with determining areas on a gel scan is usually fixing the baseline. We use one of four options. Option 1 is to use the X-axis, which is arbitrarily

drawn through the lowest point on the scan. Option 2 is to set a baseline by digitizing two points on the baseline and using them to define the baseline for the whole scan or part of the scan. Option 3 is to set the baseline as in option 2, but to set it separately for each area determined. Option 4 sets the baseline automatically for each area by drawing a straight line through the initial and final points of the area. Other programs for automatic integration may set the baseline by sensing parts of the scan where the first derivative is essentially zero.

A second problem with identifying the peaks interactively, i.e., with the digitizer is the lack of resolution of most microcomputer graphics displays. This is overcome by allowing the operator to choose part of the X-axis for display and automatically scaling the display so that the largest peak visible goes to the top of the screen. If only a small number of data points (< 100) is on the screen, they are joined by a cubic spline curve. The actual points are shown if there are < 50 points.

Each area determined is stored in memory with its associated parameters: starting X-value, finishing X-value, and baseline. The results are tabulated and related to standards by three methods: (i) each area is given as a percent of the total area determined, (ii) these percent values are multiplied by a constant; (iii) each area is given as a fraction of the area of a specified standard peak and these fractions may be multiplied by a constant. The table may be stored on disk and recalled for later study or additions or deletions.

The integration routine uses a cubic spline approximation to the peak if there are < 100 data points in the area chosen. For more than 100 points, trapezoidal integration is used in order to reduce memory requirements and increase speed. If the starting and finishing points lie between data points, then the starting and finishing points are determined by interpolation (usually cubic spline interpolation).

Overlapping Bands

There are general procedures available for resolving overlapping bands (e.g., 3) and these must be used if non-Gaussian band shapes are encountered. However, in most

cases the bands on electrophoresis gels that have not been overloaded or run too fast are Gaussian. Overlapping bands can be resolved by trial and error using a DuPont curve analyzer, which is an analog device that generates a composite of Gaussian curves that the operator compares with the gel band and then modifies the parameters of the Gaussian curves to get a good visual fit. A similar approach can be used on a digital computer (e.g., 4). A digital computer may also be used to determine the parameters that automatically give a statistical best fit. The general approach is to minimize the sum of squares of differences between the actual absorbance values and the predicted absorbance values

$$\sum_{I=1}^{Npts} (\text{Scan}(I) - \text{Sum}(I))^2$$

where Sum is an array of predicted absorbance values for each of the Gaussians involved. A sum of Gaussians is mathematically nonlinear, so that an iterative process is used to minimize this function. A general nonlinear least squares fitting program can be used on a large computer (5) but, in my experience, the lower numerical precision found on microcomputers leads to problems in matrix inversion, so that the program frequently stops prematurely because of indeterminate equations. A simple intuitive procedure that works in simple cases is to fit one Gaussian, by linear least squares, in a region where it is not much affected by the overlapping Gaussians. Then extrapolate this fit into the overlap area and subtract it from the experimental scan. Then fit a second Gaussian to the remainder. This type of approach has been used successfully in simple cases (1), but I have found it to be unstable when extended to several Gaussians and substantial amounts of overlap. I have used a general pattern-searching minimization program, which avoids the problem of matrix inversion instability, but it converged very slowly when run on the microcomputer used for this application. These difficulties led to the development of another minimization routine that has been found extremely stable and converges at least as rapidly as the less stable routines mentioned above. The routine minimizes the

function, one parameter at a time, by calculating the function at the current value of the parameter and at two values a specified distance on either side of the current value. A parabola is drawn through the three points and the minimum value determined. This is repeated until the function is below a pre-set threshold or a given number of iterations is exceeded. The pattern of iteration is shown in the flow diagram (Fig. 4). This program makes it easy to choose some parameters from the keyboard and to introduce constraints such as making the bands equispaced and/or all the same width. It can also be advantageous to mix the automatic minimization with an interactive approach. For example, an operator can make changes in two parameters simultaneously and bypass quite a number of iterations of the automatic routine.

The initial input to this program is an estimate of the parameters of the Gaussians and the background. The minimization routine is not very sensitive to these and they can usually be determined by a peak-picking routine that runs through the first and second derivatives of the experimental scan to locate the peaks (first derivative zero, second derivative negative) and their widths (distance between points of inflexion, i.e., second derivative zero). The program allows the user to add, delete, or change Gaussians found by this peak-picking routine before entering the least squares minimization routine.

Output from the program is a table of the parameters of the Gaussians and their areas, and the parameters of the baseline plus plots of the experimental scan, the sum of Gaussians, the individual Gaussians, and the baseline. Figure 5 shows an example.

Compare

This program takes one scan as a standard scan, compares a second scan with it and, if required, generates a difference scan. First, the digitizer routine is used to identify a peak present on each scan that should be aligned. The computer aligns them. Second, the digitizer is used to indicate a second peak present on each scan that should be aligned. The computer scales (stretches or compresses the X-axis) one scan so that both pairs of peaks are aligned.

1. Check input parameters (avoid negative or zero width)
2. Initialize number of iterations
3. Increment number of iterations
4. Is the number of iterations >2? --- yes --> 12
5. Initialize number of new starts
this iteration
6. Calculate mean square error for
 - (i) parameter value passed in
 - (ii) parameter value passed in + increment
 - (iii) parameter value passed in - increment
7. Print the parameter values and errors on the CRT monitor
8. Considering an imaginary parabola through the 3 points
(error, parameter value)
 - (i) is there a maximum within the range? --- yes --> 13
 - (ii) are the 3 calculated errors equal? --- yes --> 13.
 - (iii) is the minimum of the parabola --- yes --> 13
outside the range?
9. There is a minimum in range so
Halve the increment for determining range & new parameter values &
Up-date the current best parameters
10. Is the current best error below threshold? --- yes --> 12
11. Was there a worthwhile reduction in error? --- yes --> 3
12. Print the best estimate
Pass out the new parameters
13. Increment the # of new starts
14. Make the best end of the old range into the center of the new range.
15. Calculate parameter for new range
16. Is # of new starts >2? --- yes --> 12.
17. Calculate new mean square error
18. GOTO 7

Fig 4 Flow diagram for minimize routine

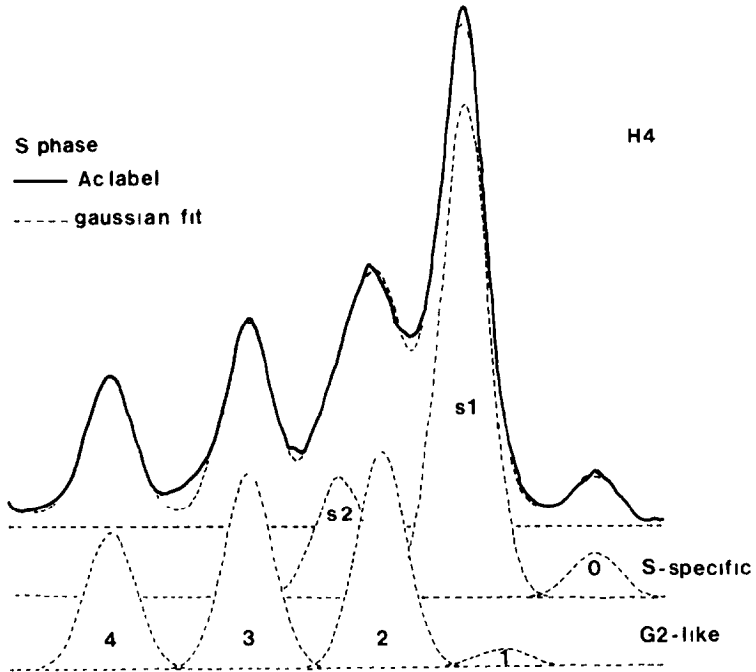


Fig. 5 Example of resolving a complex band into Gaussian components. The solid line shows the original scan and the broken line superimposed upon it shows the sum of the seven Gaussian bands shown below.

This program loads an upper and a lower scan from disc. The scans are plotted and you can then align the lower scan with the upper scan. The lower scan can be subtracted from the upper scan and the subtracted curve becomes the upper scan.

Program codes (return to main program is 15)

Upper scan load from disc (1) plot (2) store on disc (7)
Lower scan load from disc (4) plot (5) store on disc (6)

Select a part of the scans (7)

Change plotting offset (8) scale (9) line types (10)

X-axis align (11) scale (12)
Y-axis scale (13)

Subtract lower from upper - difference curve replaces upper scan (14)

Enter program code required, then press ENTER

Fig. 6 Menu for the program that aligns scans for comparison and calculates a difference scan

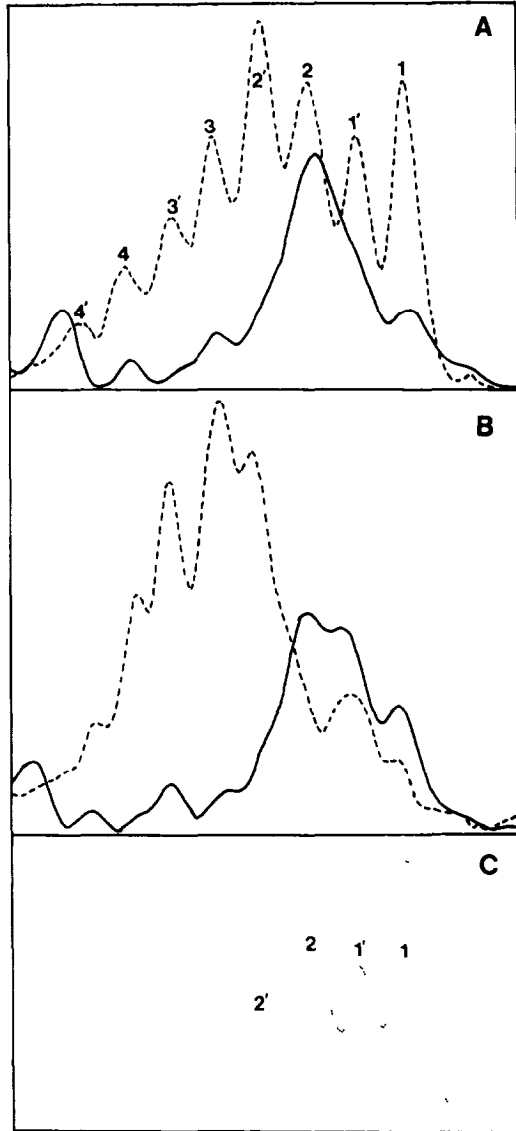


Fig 7 Example of a difference scan. A Part of a gel scan showing histone H3 pulse-labeled with ^3H in S phase of the cell cycle and separated on a Triton-acid-urea gel. Solid line Coomassie blue stain. Broken line fluorograph. B As A except that the pulse of ^3H -acetate was given in G_2 phase of the cell cycle. C Difference between the fluorograph scans in A and B. The numbers, 1, 1', etc., refer to the numbers of acetate groups per H3 molecule in the band indicated.

Third, the digitizer is used to indicate a peak that should be the same height on each scan. The computer scales the Y-axis of one scan. Each of these processes is optional. To generate a difference scan, the program takes each data point on the lower scan, then interpolates on the upper scan to find the absorbance value at the same X-axis value and then subtracts the absorbance values. The resultant array is the difference scan.

The mathematical manipulation of gel scans can be extended. For example, Cooney et al. (6) corrected a pair of gel scans of DNA restriction fragments for impurities in the samples by solving a pair of simultaneous equations at each point along the gel.

Figure 6 shows the main menu for this program and Fig. 7 shows an example of a difference scan.

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Chapter 16

Drying Gels

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Introduction

There are several reasons why it may be desirable to dry a gel after its use for electrophoresis. Firstly, it is a convenient way of storing the end result of the experiment. Secondly, drying a gel that is fragile may make it easier to handle (say, during optical scanning). Thirdly, and most importantly, it may be necessary to dry a gel to allow the most efficient detection of radioactive samples on it by autoradiography or fluorography. The method described below is rapid and generally applicable to all types of slab gel, although the description below applies specifically to a thin (0.5–1 mm thick) polyacrylamide (15%) gel being prepared for fluorography (*see* Chapter 17).

Materials

1. A high vacuum, oil pump. Protect the pump and its oil from acid and water by inclusion of a cold trap in the vacuum line.

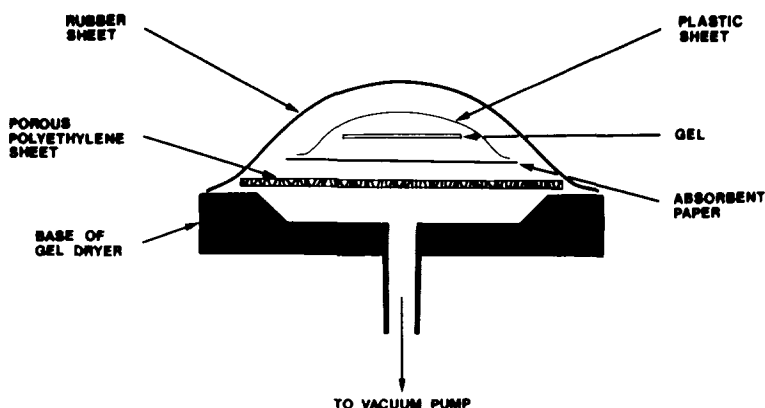


Fig. 1. Diagram showing the construction of a simple gel dryer

2. A heat source (such as a hot air fan, infrared lamp, electric hot plate, or a steam/hot water bath)
3. A gel dryer, available commercially from various sources or readily made in the laboratory, along the lines of the dryer shown diagrammatically in Fig. 1. Essentially, a gel dryer has the gel placed on top of absorbent paper, which in turn is supported by a firm sheet of porous polyethylene and/or a metal grille. As shown in Fig. 1, the gel is covered with "plastic sheet," such as Saran Wrap or Cling Film transparent domestic food wrapping. This construction is put under vacuum beneath a sheet of silicon rubber and is heated so as to help drive off moisture. Suitable polyethylene sheet, absorbent paper, cellophane, and other items may be obtained commercially. Preferably the absorbent paper, to which the dried gel adheres, should be about 1 mm thick. If it is thinner (as is, say, Whatman 3 MM paper), it has a tendency to curl up once removed from the gel dryer.

Method

1. The gel will already have been run and, if carrying out fluorography, suitably treated and then washed thoroughly with distilled water (*see* Chapter 17).

2. Place the gel on a piece of absorbent paper that is slightly larger than the gel itself. Do not trap air bubbles between them. Place them, gel uppermost, on top of the porous polyethylene sheet and cover the whole with some of the nonporous Cling Film plastic film. Place this construction on the gel dryer base and cover over with the sheet of silicon rubber, as shown in Fig. 1.
3. Apply the vacuum, which should draw all layers tightly together. Check that there is no air leak. After about 10–15 min, apply heat (say, 60°C) evenly over the face of the gel.
4. Continue until the gel is dry, at which point the silicon rubber sheet over the gel should assume a completely flat appearance. The time taken to dry a gel is dependent on various factors (see Notes), but a gel of 0.5–1 mm thickness, equilibrated with water, will take 1–2 h to dry.
5. When the gel is completely dry and bound onto the absorbent paper, it may be removed and the Cling Film over it peeled off and discarded.

Notes

1. The plastic Cling Film layered over the gel is used for two reasons. Firstly, it prevents sticking of the gel to the rubber sheet covering it. Secondly and more importantly, it reduces the likelihood of contamination of the dryer by radioactive substances from the gel. After drying, the plastic film may be discarded as radioactive waste.
2. If a suitable high vacuum pump is not available for use with the gel dryer, a good water pump may suffice instead, but in this case the gel will take longer to dry. As another alternative, the gel may be dried in atmospheric conditions, for instance as described in (1). In that case the gel is equilibrated with 2% (v/v) glycerol in water, covered with porous cellophane, and kept flat while it dries in the air at room temperature. This process may take a day or so.
3. Gels for storage can be dried, as for fluorography, onto paper. However, gels for transmission optical

scanning obviously need to be transparent. For this purpose the gel is sandwiched between two sheets of porous cellophane (with no trapped air bubbles), and then taken through the process described above. The dry gel will not adhere to the absorbent paper.

4. The main problem with this method is that gels may crack up and so spoil the end result. This may occur as a chronic process, during the drying. Gels of higher %T acrylamide and of greater thickness are particularly prone to suffer this fate. However, to alleviate this problem the gel may be treated before drying with one of several solutions, which are:

- (i) 70% (v/v) methanol in water (2). Equilibrate the gel by soaking in this solution (0.5–1 h for thin, 1-mm, gels or longer for thick gels) and then proceed with drying. This treatment will cause the gel to shrink. High %T gels may dehydrate and go opaque. This process may be reversed with water, but if it is too extreme the gel may crack. If this is a danger, use a weaker methanol solution (say, 40% v/v). This treatment speeds up the drying process somewhat since the methanol is driven off fairly quickly.
- (ii) Glycerol (1% v/v) and acetic acid (10% v/v) in water (3). Equilibrate the gel and proceed with drying.
- (iii) DMSO (2% v/v) and acetic acid (10% v/v) in water (3). Equilibrate the gel and proceed with drying.

Of these three, the DMSO solution is most likely to prevent cracking of difficult gels and the 70% methanol the least likely, but the former will take the longest time to dry down, and the latter the least time.

5. As a further precaution, and also to speed up drying, a second sheet of porous polyethylene may be used, so that the gel, *without* the overlaid, nonporous plastic Cling Film sheet, is sandwiched between the two polyethylene sheets. This arrangement provides a greater surface of gel for drying. However, ensure that the face of the polyethylene sheet that is in contact with the gel is very smooth, for otherwise the gel

will dry into it (as well as into the absorbent paper) and they will be difficult to separate. If this remains a problem, employ a sheet of porous cellophane between the gel and polyethylene. The cellophane may be removed after drying.

6. Another precaution is to use a lower temperature during the drying process, so that gradients of temperature and hydration through the system are less extreme.

Thus, probably the best approach to drying a difficult gel such as a 3 mm-thick 10% or 15%T acrylamide gel would be to use the DMSO (2%) soaking of the gel before drying, and two polyethylene sheets and less heat (say, 40°C) during the drying. Under these circumstances, a thick gel may take a whole working day to dry down.

7. Gel cracking may also occur as an acute phenomenon when the vacuum is released from a gel that is not completely dry. Thus, it is important to check for air leaks in the dryer and not to end the drying (i.e., release the vacuum) too soon. So that this does not happen, it is important to determine (by trial) the time required to dry down a gel in one's own gel-drying equipment. This time increases with increased %T of acrylamide, thickness and surface area of the gel, and with decreased temperature and vacuum during the drying.
8. A problem may arise if the gel is prepared for fluorography using DMSO as solvent. The gel must be thoroughly washed in water (or other solution, see Note 4) to remove the DMSO. This is because DMSO has a boiling point of 189–193°C and is difficult to remove under the conditions of drying. If remaining in the gel in significant amounts, the gel will remain sticky and the photographic film it contacts may become fogged.
9. Agarose gels may be dried in the manner described. Even 3-mm agarose gels dry quickly (about 0.5 h) and without cracking when using only one polyethylene sheet and room temperature for drying. Composite weak acrylamide–agarose gels are likewise readily dried down. When dealing with gels containing

agarose, beware the use of DMSO, which dissolves agarose.

10. If using EN³HANCE (New England Nuclear) in the gel, do not employ temperatures above 60°C for drying, for EN³HANCE is volatile above 65°C.

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Chapter 17

Fluorography of Polyacrylamide Gels Containing Tritium

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Introduction

Fluorography is the term used for the process of determining radioactivity in gels and other media by a combination of fluorescence and photography. Since most of the radiation of a low energy emitter will largely be absorbed by the gel, in the technique of fluorography a fluor (e.g., PPO) is infiltrated into the gel where it can absorb the radiation and re-emit light that will pass through the gel to the film. The resulting photographic image is analogous to an autoradiograph, but for a low energy β -emitting isotope like ^3H , the sensitivity of fluorography

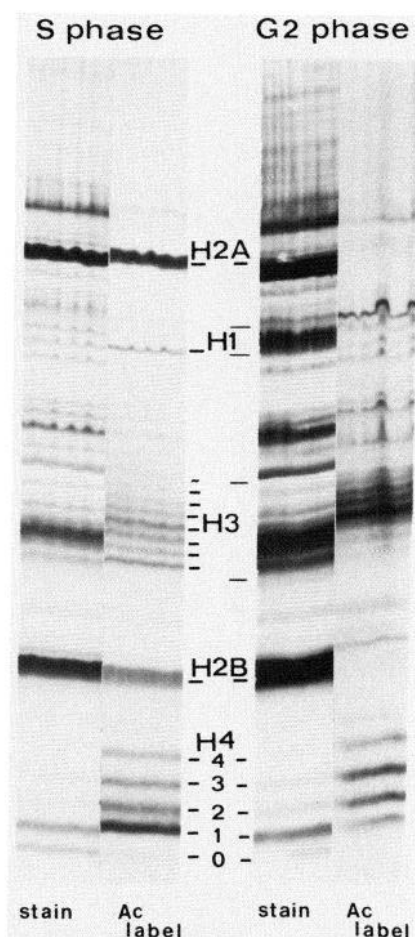


Fig. 1. This shows two examples of fluorography of protein bands labeled with ^3H . Basic nuclear proteins were isolated from the slime mold, *Physarum polycephalum*, pulse-labeled with ^3H -acetate in either S phase or G₂ phase of the naturally synchronous cell cycle. The proteins were analyzed by acrylamide gel electrophoresis in acetic acid, urea, and Triton X-100. After electrophoresis, the gel was stained with Coomassie blue, photographed, and then fluorographed. Individual lanes of the gel image were cut from the photograph (negative) and the fluorograph and then printed side-by-side to give the figure shown.

Notice that the stain patterns of the two lanes are practically identical, except for the loading, while the radioactivity patterns show major differences, for example, the absence of label in histones H2A and H2B in G₂ phase (4).

is many times the sensitivity of autoradiography. The fluorograph may be used directly, as a qualitative picture of the radioactivity on the gel. It may also be used to locate radioactive bands or spots that can then be cut from the original gel for further analysis, or be scanned to give quantitative information about the distribution of radioactivity. Figure 1 shows an example of a gel that was stained with Coomassie blue and then fluorographed. Notice that there is no loss of resolution in the fluorography of thin gels of normal size.

The procedures described here are based on those described by Laskey and Mills (1), Bonner and Laskey (2), and Randerath (3).

Materials

1. -70°C freezer
2. Film cassette, preferably with enhancing screen
3. Small photographic flash unit
4. Gel dryer
5. Film developer
6. Fixing solution: 7% acetic acid, 20% methanol in distilled water
7. Acetic acid, 25% (v/v)
8. Acetic acid, 50% (v/v)
9. Acetic acid (glacial), 100%
10. PPO solution: 20% (w/v) PPO (2,5-diphenyloxazole) in glacial acetic acid.
11. Film: Kodak XAR-5

Method

1. After electrophoresis, remove the gel from the apparatus and fix by soaking in Fixing solution for 1 h. If required, the gel may then be stained with Coomassie blue, destained, and photographed.
2. Dehydrate the gel by shaking it for 10 min each in 25% acetic acid, 50% acetic acid, and glacial acetic acid.

3. Completely cover the gel with not less than 4 gel volumes of PPO solution and shake the gel in the PPO solution for 2 h
4. Transfer the gel gently and evenly to a dish of distilled water and shake for 2 h. In water, PPO precipitates so that the gel turns opaque white
5. Dry the gel completely (*see* Chapter 16), using Saran wrap on one side, on Whatman 3MM paper
6. Open a film cassette and place the dry gel (after removal of the Saran wrap) in the cassette with the enhancing screen if used. Leave the top of the cassette beside the part with the gel in it with the white inner lining facing up.
7. In complete darkness, place a sheet of film on the white lining of the open cassette and pre-flash it (*see* Note 5)
8. Assemble the cassette with the film directly on the gel. Wrap the cassette with aluminum foil and place in the -70°C freezer.
9. Remove the cassette from the freezer after the appropriate exposure time. (*see* Note 7) and allow to warm up for about 2 h at room temperature
10. In complete darkness, open the cassette and develop the film.

Notes

1. Other fixing solutions may be used in step 1. For example, formalin can be used to fix peptides covalently in the gel
2. Coomassie blue staining gives minimal color quenching, but very heavily stained bands may show reduced efficiency for fluorography. Amido black gives more color quenching and is not recommended. Silver staining has not been tested. It is possible to re-swell the gel after fluorography and stain it then, but some loss of resolution occurs
3. The times given in steps 2 and 3 are for 0.5–1.5 mm thick gels containing 15% acrylamide. Thicker or more concentrated gels will require longer periods in all solutions.

4. Since PPO is expensive, it is normally recycled as follows: Set up four 4 L Erlenmeyer flasks with a large stirring bar and about 2.5 L of distilled water in each. Add about 0.25 L of used PPO solution, slowly, to each Erlenmeyer, stirring continuously. The PPO crystallizes out. Collect the PPO by filtering the solutions. Dry the crystals at 20°C for 2–3 d. Dissolve the crystals in a minimum volume of ethanol and precipitate, filter, and dry again. Finally, dry the PPO in a vacuum oven for about 1 wk, breaking up lumps at intervals. Glassware that was used for PPO should be rinsed in ethanol before washing.
5. *Pre-flashing* is used to improve the sensitivity and linearity of response of the film (1). Use a small, battery-operated, photographic flash unit. Tape a red filter and a diffusing screen of Whatman 3 MM paper over the flash window. Experiment with the number of layers of paper required to give an absorbance of about 0.1 when the film is pre-flashed. To pre-flash, hold the flash unit directly above the film, about 60 cm away, and press the manual flash button. Use a procedure that you can easily reproduce in the dark. Note that the first flash after the flash unit is switched on may be different from subsequent flashes, so avoid using the first flash. The white inner lining of the film cassette provides a uniform, reproducible background for pre-flashing.
6. Be careful not to place the cassette near penetrating radiation from sources such as a ^{32}P or ^{125}I autoradiograph or radioactive samples. This radiation will fog the fluorograph. Make sure the cassette is light-tight, too.
7. Recommended exposure times are of the order of 24 h for 1000–10,000 dpm ^3H and correspondingly longer for lower amounts of radioactivity. Exposure times of several months do not give a significant increase in background. Sensitivity for ^{14}C is reported to be about $10\times$ that for ^3H . Note that high acrylamide concentration such as the 50% gels used for peptide analysis severely quench the fluorescence and drastically reduce the efficiency of fluorography.
8. An automatic developer is most convenient. Manual development is described in Vol. 2, Chapter 8.

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Chapter 18

Recovery of Proteins from Dried Polyacrylamide Gels after Fluorography

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Introduction

Several methods are available for recovering proteins electrophoretically from polyacrylamide gels directly after electrophoresis or after staining for protein (*see*, for example, Chapter 19). Fluorography (*see* Chapter 17) is used to detect small amounts of proteins or specifically modified forms of proteins by labeling these with specific precursors such as ^3H -lysine or ^3H -acetate.

However, fluorography requires dried gels and thus prevents the use of existing methods for protein recovery from gels. We have found that dried gels containing PPO

(2,5-diphenyloxazole) can rapidly be reswollen in acetic acid. The gels may then be eluted electrophoretically so that the labeled (and also the unlabeled) proteins can be analyzed for amino acid composition, tryptic peptide analysis, and even Edman degradation to determine amino acid sequence (1,2).

Materials

1. Swelling solution: glacial acetic acid with 0.001 % (w/v) Coomassie Brilliant Blue.
2. Equilibration buffer (*see* Chapter 19). 1M acetic acid, 50 mM NaOH, 1% cysteamine.

Method

1. Determine the position of the protein band(s) of interest in the polyacrylamide gel, dried on Whatman 3MM paper, after fluorography (Chapter 17).
2. Use scissors or a razor blade to cut out the required piece of dried gel together with its paper backing.
3. Place up to 10 mL of gel pieces (*see* Note 4) in a 50 mL polypropylene tube with screw cap. Add 10 volumes of swelling solution to the tube, and gently rock, roll, or invert for 15 min.
4. Decant the solution from the swelling gel pieces and discard.
5. Repeat steps 3 and 4 once.
6. Equilibrate the swollen gel pieces for 30 min in 10 volumes of equilibration buffer, decant the equilibration buffer from the gel pieces, and discard.
7. Repeat step 6 once and use the reswollen and equilibrated gel pieces in the electrophoretic elution procedure described in Chapter 19.

Notes

1. This protocol for reswelling dried and fluorographed polyacrylamide gels is ineffective on dried gels that do not contain PPO.

2. Labeled protein bands can be precisely located in the dried gel by placing the fluorograph film on top of the dried gel. Generally, a marker pen injected with some ^{14}C -labeled compound is used to mark the filter paper at several positions next to the gel with sufficient cpm to give a small dark spot on the film after the fluorography. In gels previously stained with Coomassie Brilliant Blue, the residual stain can assist in precisely superimposing the film on the gel. Although gel staining prior to fluorography is not required when only labeled proteins are to be recovered, unlabeled proteins can only be isolated when they can be localized by their residual stain.
3. Cut the pieces of gel-on-paper small enough to assure homogeneous swelling, e.g., diameter less than 0.5 cm.
4. The "volume" of the pieces is measured as a loose layer. During the first wash the gel pieces will be seen to rapidly swell to up to twice the original gel thickness.
5. The gel pieces, in equilibration buffer, will generally display a white center of reprecipitated PPO. Gel pieces from thick gels may even be completely white. This PPO will not interfere with the subsequent electrophoretic elution, and reproducible and complete protein recovery is obtained when the gels are completely reswollen. For very thick gels (more than 2 mm thick) steps 4 and 5 may need to be repeated twice, and care should be taken to reduce the size of the gel pieces as much as possible. Loss of protein may occur when the number of washes in swelling solution is increased, especially when the trace of Coomassie is omitted from the swelling solution.

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Chapter 19

The Electrophoretic Elution of Proteins from Polyacrylamide Gels

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Introduction

The analytical power of acrylamide gel electrophoresis is one of the keys of modern protein chemistry. It is not surprising, therefore, that many methods have been described for converting that analytical power into a preparative tool. None of the available methods is entirely satisfactory for general use since loss of resolution or low recovery is often involved. The method described here has given both high resolution and good recovery but suffers from the disadvantage of being relatively laborious (1,2). In addition, although the recovered proteins are good for

peptide analysis or amino acid composition determination, we have found very low yields on Edman degradation of proteins eluted from gels (3)

The method described below works well for eluting proteins from acid-urea-Triton gels and should work equally well with acid-urea gels (Chapter 8). Wu et al. (1) describe an alternative set of buffers that can be used for SDS gels

The method uses the principle of isotachopheresis, as in the stacking gel portion of a discontinuous gel electrophoresis system (*see*, for example, Chapter 6). The gel pieces containing the protein of interest are embedded in agarose above an agarose gel column. A detergent, CTAB, is used to displace the Triton (or SDS) bound to the protein that is electrophoretically eluted into the agarose gel column. In the column, the protein is concentrated by stacking between the leading ion (Na^+) and the trailing ion (betaine). The protein dye (Coomassie blue or Amido black) stacks ahead of the protein. The concentrated protein band is cut from the agarose gel and recovered.

Materials

1. Lyophilizer
2. Vacuum pump
3. Electrophoresis apparatus capable of running tube gels, preferably in glass tubes with small funnels like the BioRad Econocolumn #737-0243 (which is 5 mm inner diameter and 20 cm long) with the lower end cut off.
4. Staining solution (Coomassie): 0.1% (w/v) Coomassie Brilliant blue R, 5% (v/v) acetic acid, 40% (v/v) ethanol, 0.1% (w/v) cysteamine in distilled water
5. Destaining solution (Coomassie): 5% (v/v) acetic acid, 40% (v/v) ethanol, 0.1% (w/v) cysteamine
6. Staining solution (Amido): 0.1% (w/v) Amido black in destaining solution (Amido) (note: add the cysteamine immediately before use)
7. Destaining solution (Amido): 7% (v/v) acetic acid, 20% (v/v) methanol, 0.1% (w/v) cysteamine.
8. Equilibration buffer: 1M acetic acid, 50 mM NaOH, 1% (w/v) cysteamine

- 9 Siliconizing solution: 1% (v/v) Prosil-28 in distilled water.
- 10 LMT agarose solution: 0.5% (w/v) low melting temperature agarose in 1M acetic acid, 50 mM NaOH.
11. HMT agarose solution: 1% (w/v) high melting temperature agarose, 1M acetic acid, 50 mM NaOH, 0.0005% (w/v) methyl green in distilled water (*see note 3*).
12. Upper reservoir buffer: 1M acetic acid, 0.1M betaine, 0.15% (w/v) Cetyl Trimethyl Ammonium Bromide (CTAB) in distilled water. Lower reservoir buffer: 1M acetic acid, 50 mM NaOH
- 13 Acidified acetone: add conc. HCl to acetone to give a concentration equivalent to 0.02N.
14. Elution buffer: 0.02N HCl with optionally 0.1% (w/v) cysteamine, 0.1M N-methyl-morpholine-acetate, pH 8.0
15. Agarose gels. (can be done 1 d before needed).
 - (a) Thoroughly clean and dry the tubes and then siliconize them, for example, by immersing in 1% Prosil-28 followed by thorough rinsing with water and drying.
 - (b) Soak small pieces (3 cm square) of dialysis membrane in distilled water and fix one piece over the end of each tube with an elastic band. This is intended to hold the agarose in the tube.
 - (c) Melt LMT agarose solution by heating to 65–100°C. Use 7 mL per tube. Fill each tube carefully, avoiding trapping air bubbles, just to the bottom of the funnel. Let the agarose gel at 4°C.

Method

1. Stain the gel just enough to visualize the band(s) of interest. Either Coomassie blue or Amido black staining may be used. If the gel is loaded heavily enough, then 15 min in either staining solution and no destaining is recommended.
2. Cut out the bands of interest and soak them for at least 1 h, or overnight, in equilibration buffer.
3. Chop the gel bands into small pieces with a razor blade and transfer them to the funnels on the tops of the agarose gels in the electrophoresis apparatus

4. Melt the HMT agarose solution in a 100°C bath and add between 2 and 5 mL to each funnel. Stir to remove air bubbles and achieve a uniform distribution of gel pieces in the agarose.
5. Let the agarose solution gel either at room temperature or at 4°C.
6. Place the bottom reservoir of the electrophoresis apparatus in a tray containing melting ice as coolant. Fill the upper and lower reservoir with upper reservoir buffer and lower reservoir buffer, respectively. Connect the negative terminal of the power supply to the lower electrode and the positive terminal to the upper electrode.
7. Start the electrophoresis, using 2.5–5 mA per tube (about 200 V).
8. During electrophoresis, the staining dye (Coomassie or Amido) migrates fastest, followed by the buffer discontinuity, the protein, and the methyl green marker dye. Continue electrophoresis at least until the staining dye front reaches the middle of the gel. If the buffer discontinuity is clearly separated from the staining dye, then proceed to step 9. Otherwise, continue electrophoresis until the staining dye is clearly separated.
9. Immediately remove the gel from the tube as follows.
 - (a) Use a spatula to remove the acrylamide gel pieces and their supporting HMT agarose gel.
 - (b) Push the gel column with a plunger from a disposable syringe out of the tube onto a clean glass plate.
10. Immediately cut the gel about 3 mm below the buffer discontinuity (above the staining dye) and about 5 mm above the buffer discontinuity. The resulting 8 mm section of gel contains the protein and the methyl green marker dye; the remainder is discarded. Place the gel section in a 1.5 mL polypropylene centrifuge tube.
11. Use the following procedure (steps 12–15) if the presence of agarose, and protein denaturation during drying, do not interfere with subsequent analysis. Otherwise proceed to step 16.

12. Add 1 mL of acidified acetone and leave overnight in the freezer, at -20°C .
13. Centrifuge (10,000g; 5 min) and slowly decant the acetone, which contains the dye, buffer salts, and detergent.
14. Cap the tube, pierce the cap, and dry the gel under vacuum using a vacuum pump until the vacuum falls below 100 mtorr.
15. Rehydrate the agarose with 20–50 μL of distilled water, or appropriate buffer, by incubation at 65°C for a few minutes. This solution will remain liquid at 37°C , allowing enzymatic digestion, or it may be diluted to give less than 0.1% agarose when it will no longer gel. This is the end of this procedure.
16. This step follows step 11 if steps 12–15 were unsuitable. If this procedure is used, then the methyl green marker dye should not be used (see note 3).
17. To the gel slice, add 0.5 mL elution buffer. Mix very gently without breaking up the agarose. Leave at room temperature for at least 1 h and preferably overnight.
18. Equilibrate a small ($\sim 1 \times 10$ cm) desalting column of Sephadex G-25 with 0.1M *N*-methylmorpholine acetate, pH 8.0.
19. Centrifuge (10,000g; 5 min). Carefully decant or pipet off the supernatant and set it aside.
20. Repeat steps 17 and 19, combining the supernatants. The extracted gel may be discarded.
21. Load the combined supernatants onto the Sephadex desalting column equilibrated with 0.1M *N*-methylmorpholine acetate, pH 8.0 (step 18). Elute with 0.1M *N*-methylmorpholine acetate and collect the material eluting in the excluded volume.
22. Lyophilize the protein to dryness. The resulting protein is salt-free since *N*-methylmorpholine acetate is volatile. This ends the alternative procedure.

Notes

1. Gels that have been normally stained and destained may be used but the time of agarose gel electrophore-

sis may have to be increased or the capacity of the system will be reduced (Note that cysteamine should not be included for Coomassie staining of histone H1)

2. For Amido black, but not for Coomassie, the equilibration step (step 2) also acts as a destaining step.
3. The presence of methyl green in the HMT agarose makes it easy to locate the buffer discontinuity during the subsequent electrophoresis and is mostly removed by acetone precipitation of the protein. Small residual amounts do not interfere with subsequent peptide mapping. However, if the protein is to be isolated and desalted by the Sephadex method (step 16 on) the methyl green should be omitted from the HMT agarose solution. (If methyl green *were* included it would elute partly with the protein and partly immediately after the protein). In this case the buffer discontinuity must be located directly from the refractive index change
4. At step 4, use the minimum volume of agarose required to suspend the gel fragments since the time of electrophoresis depends greatly on the volume of agarose plus gel fragments.
5. Betaine (rather than glycine) is used as the trailing ion because of its higher acetone solubility. It also increases the capacity of the system.
6. If ice cooling is not used during electrophoresis, reduce the current to 2.5 mA/tube
7. The distance between the lower edge of the staining dye and the buffer discontinuity depends on the amount of stain and how full the funnels are on top of the gel, varying from about 5 to about 40 mm. The distance between the upper edge of the staining dye and the buffer discontinuity is fairly constant at 5–10 mm (reduces to 2–3 mm if the betaine buffer is replaced by glycine).
8. The time of electrophoresis will be 2–4 h at 4 mA/tube for low amounts in the funnels, and up to overnight at 2.5 mA/gel for full funnels. Different tubes may require different running times. In this case, remove the tubes whose electrophoresis is complete as follows: (a) turn off the power, disconnect the power supply, and empty the upper reservoir buffer into a beaker

(b) Remove the required tube(s), block the resulting hole(s) with rubber bung(s), replace the upper reservoir buffer, and continue the electrophoresis, adjusting the power supply as necessary.

If large volumes of reservoir buffers (e.g., 2 L) are used, then reservoir buffer changes are not needed

- 9 It is important to remove the gel immediately after electrophoresis since the staining dye diffuses rapidly once the current is turned off. It may be possible to remove the gel in other ways (e.g., *see ref 1*), but we have found the method described in step 9 to be rapid and reliable
10. Slice the gel immediately after electrophoresis, to prevent loss of resolution by diffusion. In our experience, this 8 mm section of the gel contains at least 98% of the protein.
- 11 The simple acetone extraction and drying procedure (steps 12–15) is suitable if the protein is to be digested with enzymes and the products analyzed by gel electrophoresis. Note that the procedure described by Cleveland (*see Chapter 22*) provides another peptide mapping approach for some applications. The alternative procedure (step 16 on) is recommended if the protein is to be analyzed for amino acid composition or characterized by nuclear magnetic resonance or other techniques, for example, thin layer analysis of tryptic digests.
12. The protein will precipitate in the gel at step 12, possibly forming a white band, but precipitation may not be quantitative with extremely small amounts of protein
13. Drying of the extracted gel (step 14) takes several hours. This procedure may result in irreversible denaturation of the protein, but it will probably still be solubilized by enzymic digestion
14. At step 17, do not vortex or mix vigorously or you will get agarose in the final protein solution. Allow the protein to diffuse out. Do not include urea or other agents that would solubilize the agarose in the elution buffer
- 15 The second extraction (step 20) need only be 1 h. In our tests, less than 5% of the protein remained in the

gel after the two extractions, although the white appearance persisted. The protein concentration in the eluate cannot be determined by the Bradford method (4) since CTAB strongly interferes.

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Chapter 20

Transfer Techniques in Protein Blotting

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Introduction

Polyacrylamide gel electrophoresis is an extremely powerful tool for the analysis of complex protein mixtures. Although the value of this method cannot be questioned, it is restricted in that the separated proteins remain buried within the dense gel matrix and are not readily available for further investigation. A number of methods have been developed in order to try and overcome this problem, for example the elution of proteins from excised gel slices (*see* Chapter 19). Alternatively, proteins have been studied while they are still buried within the gel using a variety of *in situ* peptide mapping (*see* Chapter 22) and gel overlay techniques (for example, *see* ref 1). Unfortunately all of these methods have serious drawbacks. In the case of protein elution and *in situ* peptide mapping techniques, the resolution and number of bands that can be processed is

restricted, whereas the gel overlay techniques are generally time-consuming and insensitive

Recently a new technique—protein blotting—has been developed that promises to overcome many of these problems (*see* ref. 2 for a recent review). Using this method, proteins are transferred out of the gel and onto a filter or membrane, forming an exact replica of the original protein separation, but leaving the transferred proteins accessible to further study. A wide range of different probes, including antibodies, DNA, RNA, and lectins, have been used in protein blotting experiments and the potential applications of this method are only limited by the availability of suitable probes and assay systems.

As has already been emphasized, the great advantage of protein blotting is that the transferred proteins are no longer trapped within the gel, but are freely accessible to further analysis by being bound to the surface of a filter. It is clear therefore that the transfer step is the crucial stage in any protein blotting experiment and the resulting transfer should be a faithful replica of the original gel. The degree to which this is achieved is largely determined by the choice of transfer buffer, together with any pretransfer equilibration step, as well as the types of transfer media and transfer techniques that are used

In the following sections two widely used methods of protein blotting—electroblotting (3) and passive-diffusion blotting (4)—are described. In addition, a method for staining proteins after transfer to nitrocellulose filters is given.

Materials

Unless otherwise indicated, all stock solutions and buffers are prepared from analytical grade reagents and are made up in deionized, double-distilled water. Buffers used in electroblotting experiments are stable for at least 3 months when stored at 4°C whereas passive-diffusion blotting buffers should be freshly prepared as required

1. Equilibration buffer for electroblotting. The following chemicals are weighed out: 24.2 g Tris-base, 144.0 g glycine, and 10.0 g SDS (electrophoresis grade). They

are dissolved in approximately 7 L water, and to this solution 2 L methanol are added and the buffer is then made up to 10 L with water. The pH of this buffer is 8.3 and should not require any adjustment.

2. Transfer buffer for electroblotting. This buffer is identical to the equilibration buffer above, except that SDS should not be added (*see Note 4*)
3. Electroblotting equipment. A number of different types of transfer apparatus are now commercially available. Alternatively, equally satisfactory transfer assemblies can be made, without too much difficulty, in a laboratory workshop. One of the best designs for such an apparatus is described in a paper by Bittner et al (5). The basic design given in this paper can be improved by substituting stiff nylon netting for the dialysis membrane used on the inner faces of the cassette frame to support the gel and filter paper "sandwich" (*see Method*) Also the efficiency of the packing can be further improved by using two Scotch Brite pads as additional packing (*see Method*) and rubber bands, instead of the screws, can be used to hold the transfer assembly together
4. Equilibration buffer for passive-diffusion blotting. The following solutions are added to approximately 350 mL of water. 5mL 1.0M Tris-HCl, pH 7.0; 5 mL 5.0M NaCl; 10 mL 0.1M Na₂EDTA and 0.5 mL 0.1M dithiothreitol, followed by the addition of 120 g of urea ("Aristar" grade or similar). The solution is then mixed until all of the urea has dissolved and made up to a final volume of 500 mL.
5. Transfer buffer for passive-diffusion blotting. This buffer is identical to the equilibration buffer above, except that it contains no urea and a total of 4 L of buffer is required
6. Passive-diffusion blotting equipment. As yet, no commercial transfer apparatus is available for passive-diffusion blotting experiments. However, the limited amount of apparatus required for this type of transfer can be easily made in a laboratory workshop (*see Fig. 1*).
7. Transfer filters. Nitrocellulose filters are used for both electro- and passive-diffusion blotting experiments and

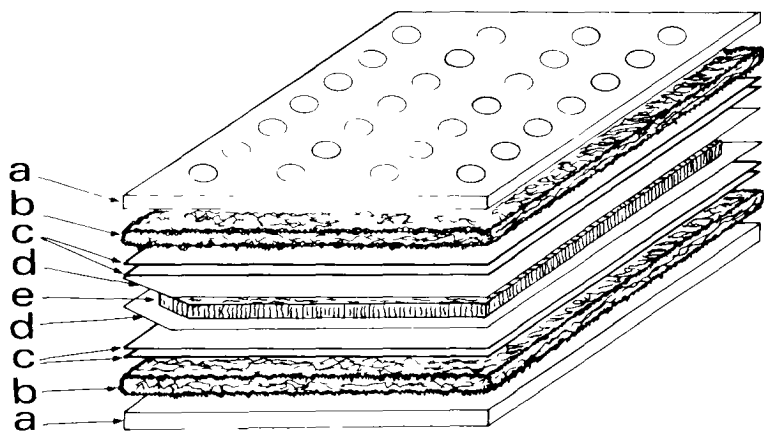


Fig. 1 Schematic representation of a transfer assembly for protein blotting by the passive-diffusion method. (a) rigid plastic sheets (20×20 cm, rows of 1-cm diameter holes at 2 cm intervals are drilled through these sheets in order to allow the free passage of transfer buffer), (b) Scotch Brite pads, (c) filter papers; (d) nitrocellulose filters and (e) the gel. N.B. The front left-hand corner has been cut off both the gel and the nitrocellulose filters in order to assist in the later orientation of the transferred proteins.

can be obtained from a number of suppliers. The majority of these suppliers produce filters with a $0.45 \mu\text{m}$ pore size and these are the most widely quoted in the literature. However, in common with a number of other workers, we prefer to use the smaller $0.2\text{-}\mu\text{m}$ pore size filters because of their greater binding capacity. Nitrocellulose filters should be stored in an airtight box at 4°C when not in use to prevent them from becoming contaminated by exposure to volatile chemicals, etc. Several other types of filters can be used in protein blotting experiments and these are briefly discussed in Note 3 below.

8. Amido black protein stain. About 30 min before the stain is required, a 1 L solution containing 20% (v/v) ethanol and 7% (v/v) acetic acid is prepared. Into 100 mL of this solution 0.1 g of Amido Black 10 B is dissolved by constant stirring for ~ 30 min, the remaining 900 mL of solution being held over for the destaining step.

Method

N.B. Disposable plastic gloves should be worn when handling either the gels or transfer filters in order to prevent them from being contaminated by grease and other substances

Electroblotting

1. Proteins are separated by SDS polyacrylamide slab gel electrophoresis (SDS PAGE; *see* Chapter 6). A single sample can be run on the gel where a variety of probes or binding conditions are to be studied. Alternatively, when only a limited number of probes and/or binding conditions are under investigation, a large number of different samples may be analyzed
2. At the end of the run, the position of the sample tracks is marked on both gel plates. The plates are then carefully separated using a large spatula as a lever, while a second smaller spatula is eased between the gel and one of the gel plates. (The separation of the gel plates can be further simplified by routinely siliconizing one of the plates.) The comb and stacking gel (if any) are then removed and in the case of single sample gels side strips are cut. Alternatively, where several sets of samples have been run on the gel, these are cut into separate strips. Polyacrylamide gels can be easily cut into strips with a large rounded scalpel blade. In order to prevent the gel from tearing, the scalpel blade is slowly brought across the gel by advancing with a rocking action of the blade.
3. Before removing the gel strips from the plate one corner should be cut off each strip so as to assist in their later orientation
4. The side strips or duplicate gels are then stained in Coomassie brilliant blue R 250 or investigated by any other suitable detection system (*see* Chapters 6 and 17)
5. The remaining gel strips are placed in a large glass dish containing approximately 750 mL of equilibration buffer. The gel is then gently agitated in this solution on a rotary shaker table for 60 min.

- 6 Open the transfer cassette, and place it in a large glass dish. The cassette is then assembled by layering the following, starting from the anode side: one Scotch Brite pad, two sheets of thick chromatography paper (e.g., Whatman 3MM grade), nitrocellulose filter, the gel to be blotted, followed by two further sheets of filter paper and a final Scotch Brite pad. The filter papers and the nitrocellulose filter should be thoroughly saturated with transfer buffer before being placed in the cassette. Uneven wetting of the nitrocellulose filters frequently presents a problem at this stage. However, this problem can be easily overcome by floating the filter across the surface of a dish containing transfer buffer. After a few seconds, the filter rapidly loses its pure white color, turning a translucent gray-white color. Once the filter is uniformly saturated with buffer, the dish is gently rocked, submerging the filter and ensuring its complete saturation. The Scotch Brite pads are very porous, quickly taking up the buffer when submerged in the transfer chamber, and consequently do not require any pre-wetting. As the transfer cassette is assembled, it is vital that no air bubbles are trapped between either the filters or the gel, otherwise an uneven transfer will result. The transfer cassette is then closed and secured with rubber bands (if required).
7. The assembled cassette is placed in the transfer chamber containing transfer buffer and lifted in and out of the buffer a few times to ensure that no air bubbles are trapped in the Scotch Brite pads. The lid of the transfer chamber is then put in place and the electrodes connected (the right way round!)
- 8 The transfer is performed at a current of ~ 0.5 amps ($\sim 10\text{--}15$ V/cm) for between 2 and 20 h at 4°C . The precise length of the transfer is dependent upon a number of factors, including buffer composition, the percentage of acrylamide in the gel, and the molecular weight distribution of the proteins (high molecular weight proteins transferring more slowly). The optimum transfer time should therefore be determined by running a series of trial experiments, and monitoring the residual proteins in the gel as well as the proteins that have been transferred to the filter. (The transfer buffer can

be used for five transfers before it needs to be replaced.)

Passive-Diffusion Blotting

1. Proteins are separated by SDS PAGE: *see* Steps 1–4 above.
2. The gel is then placed in a large glass dish containing 500 mL of equilibration buffer and gently agitated for 3 h on a rotary shaker table.
3. The transfer cassette is assembled essentially as described above (Step 6) except that two nitrocellulose filters are used, one on each side of the gel (*see* Fig. 1).
4. The assembled transfer cassette is submerged in approximately 1.5 L of transfer buffer. After 24 h the buffer is replaced with fresh buffer and the transfer is allowed to continue for a further 2–4 d. (Again the optimum transfer time is dependent upon the composition of the gel and transfer buffer, as well as on the size and solubility of the proteins under investigation, and should ideally be established for each different class of experiments).

Protein Detection

1. Proteins can be detected by gently agitating the filter in the Amido black stain prepared previously (*see* Materials) for about 10 min.
2. The stain is then poured off and the filter is briefly rinsed with destain solution, followed by further washes in this solution, for approximately 1 h.
3. The filter is then dried between several sheets of filter paper, that are held flat with a heavy weight.

Notes

1. The potential applications of protein blotting are very wide-ranging and beyond the scope of this chapter. Examples of possible applications can be found in the following papers: immunoblotting (6,7), DNA binding

proteins (8,9), glycoproteins (10, 11), and receptor proteins (12).

2. The two methods that have been described in this chapter are both well-characterized and there are many references to their use in the literature. However, there are a number of alternative transfer techniques, such as Southern blotting (13) and vacuum blotting (14), but while these methods may be preferred for some applications they are not yet widely used.
3. Although nitrocellulose filters are the most commonly used filters in protein blotting experiments, several other types of transfer media are currently available.

(a) Nylon filters. These filters are reported to have similar binding properties to nitrocellulose, but they have the advantage of being considerably stronger. Although the fragility of nitrocellulose filters is seldom a serious problem, the nylon-based filters may be valuable where the filters have to be handled a lot, for example, in some immunoblotting experiments where the filters are recycled. However, for the majority of applications the extra expense of these filters cannot be justified.

(b) Diazo papers. A range of diazo papers, including both diazobenzoyloxymethyl (DBM) cellulose and diazophenylthioether (DPT) cellulose papers, are now commercially available in their more stable intermediate amino forms. Alternatively, these papers can be prepared in the laboratory (*see refs 15 and 16 for the preparation of DBM and DPT papers, respectively*). These papers are particularly useful where the transferred proteins need to be very tightly bound to the filter (as in some immunoblotting experiments) or possibly in facilitated active transfer experiments (17). Again few experiments currently justify the extra expense and time involved in preparing these filters.

Both ABM and APT papers have short half-lives and they should only be used in electroblotting experiments. Also because the reactive diazo groups will bind not only proteins, but free amino acids, the Tris/glycine/methanol electroblotting buffer

should be replaced with either a phosphate (5) or borate buffer (18).

- (c) Diethylaminoethyl (DEAE) anion exchange papers and membranes. These filters are available from a number of suppliers and have been successfully employed in DNA blotting experiments. The major advantage of these filters is that they allow the recovery of the transferred molecules (19).

4. It is important to note that, using the method described above, it is unlikely that all of the proteins will be transferred from the gel. The explanation for this would appear to be that there is preferential loss of SDS from the gel. As long as the SDS remains in the gel, the proteins are readily solubilized. However, owing to its small size and negative charge, the SDS migrates out of the gel more rapidly than the majority of the proteins, which are then trapped in the gel. The inclusion of SDS in the equilibration buffer considerably improves the transfer efficiency of this system. If SDS is also included in the transfer buffer, protein transfer reaches almost 100%. Unfortunately many proteins then fail to bind to the nitrocellulose filter, presumably because of the detergent action of SDS, which disrupts the hydrophobic interactions responsible for binding the proteins to the filter (*see* Fig. 2). Substituting non-ionic detergents only aggravates this situation, the uncharged detergent doing little to improve the migration of the proteins out of the gel while at the same time preventing many proteins from binding to the nitrocellulose.
5. Low molecular weight proteins (~60,000 daltons or less) are generally transferred far more efficiently than larger proteins, irrespective of the composition of the transfer buffer. Gibson has recently published a simple method designed to improve the transfer of high molecular weight proteins (20). Proteins are electrophoretically transferred essentially as described above. The major difference is that a filter paper soaked in a nonspecific protease is placed on the cathode side of the gel. Once the transfer starts, the protease migrates into the gel digesting any protein molecules it encounters. The resulting peptides are then easily transferred

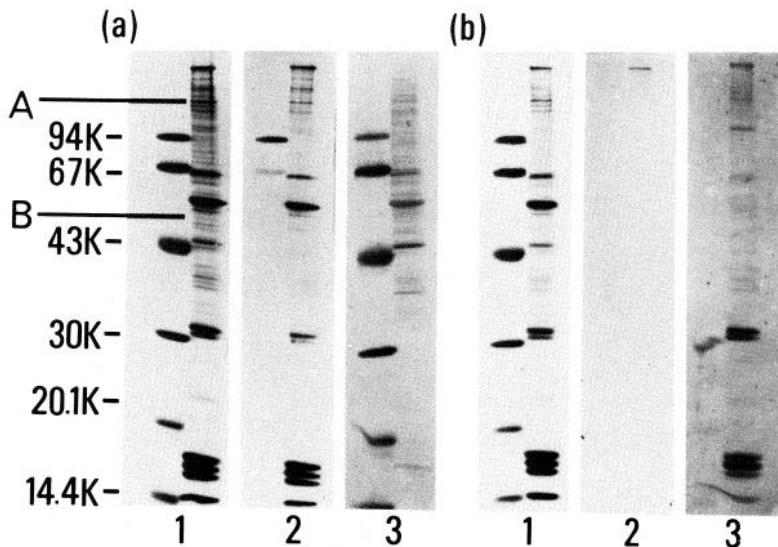


Fig. 2. Electrophoretic transfer efficiency. Pharmacia low molecular weight marker mix proteins and Chinese hamster nuclear proteins were separated by SDS polyacrylamide gel electrophoresis. The gels were then equilibrated in either (a) 20 mM Tris base, 192 mM glycine, 20% methanol or (b) the same buffer but containing 0.1% SDS, for 30 min. One gel from each buffer was then stained (1), while the remaining gels were electroblotted for 18.5 h at 0.5 A and 4°C, using the same buffers as used in the equilibration step. The post-transfer gels (2) and the filters (3) were then stained. [Polypeptides A and B, although minor components of the total nuclear protein sample, bind DNA probes with high affinity and in preference to all other nonhistone proteins (*see* Fig. 4, ref. 2).]

out of the gel. The principal drawback of this approach is that potentially important sites within the proteins may be destroyed and will not be recognized by probes specific to these sites.

6. Passive-diffusion blotting transfer efficiency is approximately one-tenth to one-hundredth of that seen in electroblot transfers. Two major factors are responsible for this: (i) The transfer is a passive process as opposed to the active transfer that occurs in electroblotting experiments. (ii) The long equilibration step prior to the transfer removes most of the SDS from the gel, leaving

the less soluble proteins trapped in the gel. If the length of the equilibration step is decreased, the transfer efficiency is improved. However, the rationale in using such a long equilibration step is that the proteins have sufficient time to renature. These renatured proteins may then be expected to be more efficient than electrophoretically transferred proteins in binding the probe molecules. This increased binding efficiency should therefore, in part at least, counterbalance the low transfer efficiency.

7. Proteins from both acid and isoelectric focusing (IEF) gels can be electrophoretically transferred to nitrocellulose filters in 0.7% (v/v) acetic acid (3). Proteins in this case are transferred towards the *cathode* and the position of the nitrocellulose filter should be reversed with respect to the gel therefore. Alternatively proteins can be transferred from these gels following equilibration in electroblot transfer buffer containing 2.0% (w/v) SDS (store at room temperature), or for IEF gels by capillary blotting, as described by Reinhart and Malamud (21).
8. The detection of transferred proteins is an important step in any protein blotting experiment, providing both a qualitative and a quantitative measure of the transfer efficiency. The staining of the transferred proteins also makes it easier to interpret the eventual probe binding pattern. A number of alternative stain recipes for both Amido black and Coomassie brilliant blue are to be found in the literature. However, we have been unable to find any method that produces results superior to those obtained using the method described above. Indeed some of these methods use very high concentrations of ethanol or methanol, which can severely distort or even dissolve the filter! When radioactively labeled proteins are used, these can be detected either by fluorography for ^3H , ^{14}C , and ^{35}S using Enhance spray (New England Nuclear), or in the case of ^{125}I by autoradiography. These methods have the considerable advantage of increasing the sensitivity by several orders of magnitude and consequently are very valuable when determining the optimum transfer conditions for a particular blotting experiment. However, care must be taken when using labeled proteins to ensure

that they will not interfere with the detection of the probe, which is also often radioactively labeled. As an alternative to using radiolabeled proteins, a high sensitivity silver staining method has recently been developed for use on protein blots (22). Although in principle this is a very useful technique, the method is unfortunately very time-consuming

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